

Continuation Under Constraint

Distinctions, Substrates, and Repair Across
Transformations

Flyxion

2026

Abstract

No finite substrate can be guaranteed sufficient for arbitrary future continuation across an open task space: this is the informal thesis this monograph makes precise and, in one central case, proves. Distinctions survive transformation only through renderings that discard some of what they carry; whether that discard is benign or corrupting depends on whether it is declared, not merely on its size. When renderings compose into a chain — a substrate chain, or budget relay — discard accumulates, and the Composition Failure Theorem shows that a sequence of locally rational transformations, each optimizing for its own task, can jointly produce a substrate insufficient for a later task none of them anticipated, without any single stage behaving negligently. The theorem is proved against an explicit numerical witness and shown to be independent of which local discard procedure a stage uses. A parallel result identifies the opposite case: a chain sharing one task throughout, adequately funded at every stage, cannot exhibit this failure, which is part of why formal proofs behave so differently from informal explanation under the same transmission process. The traditional distinction between preserving an artifact and reconstructing it from traces is replaced by a single continuous fidelity measure, with preservation recovered only as its high-fidelity limit; a companion chapter states plainly, rather than assumes away, the one dependency this correction cannot settle on its own — whether embodied perception

ever reaches that limit at all. Eight worked-domain chapters — chess, painting, maps, proofs, contracts, science, memory, and media pipelines — test the framework against cases chosen because they generated it, not because they illustrate it after the fact. A late chapter situates this monograph against the wider Distinguishability and Admissibility programs it draws on, verifying prior claims of informal inheritance against primary source text where that text was available, transcribing formulas directly where only the author could supply them, and retracting one citation to a text that does not appear to exist. Three findings survive that audit intact: the framework’s oldest working distinction already had a name in a program it had not engaged; its treatment of task-relative sufficiency was silently assuming an evaluability condition a related program makes explicit; and its central theorem occupies a setting a sibling theorem, proved for a single observer at one moment, does not cover and cannot be reduced to.

Note on Epistemic Status

This monograph distinguishes, throughout and without exception, between results that are proved, results that are conjectured, questions that are left explicitly open, and claims that are stated informally, ahead of formalization, as a working thesis rather than as any of the above. Five labels are used accordingly. A *Theorem* is a claim with a complete proof given in the chapter or in Appendix A. A *Proposition* is a claim with a complete, if sometimes short, proof, of lesser generality or centrality than a Theorem. A *Conjecture* is a claim believed true, motivated by argument or by worked examples, but not proved in this monograph. An *Open Question* is a question the monograph raises and explicitly declines to resolve, together with a statement of what resolving it would require. A *Principle* is a pre-formal statement of a thesis — most importantly the informal statement of global insufficiency in Chapter 1 and its restatement in Chapter 6 — introduced before the vocabulary needed to state it precisely has been built, and superseded once a formal Proposition or Theorem makes the same claim rigorously; it carries no claim to proof of its own.

The label Theorem is reserved for two kinds of claim: results proved new within this monograph, and results imported wholesale, with correct attribution, from an external source text that itself labels them Theorem. Exactly one result in this book is new in the first sense: the Composition Failure The-

orem (theorem 10.2), the diachronic, multi-agent result this monograph exists to establish. Nine results are imported in the second sense: the Redistribution Theorem from *Conservation of Ambiguity* (theorem 5.7); seven theorems from Distinguishability Geometry, Persistence Before Truth, and Observability Is Restorability imported in Chapter 3 – the Deficit Proxy Theorem, the Projection Deficit Bound, the Embedding Refinement theorem, Revision Monotonicity, Transport Stability, the theorem that persistence depends only on structure, and the Paradigm Shift Theorem; and the Domain Restriction Theorem from *Admissibility Before Truth*, imported at the transcribed-not-verified tier the front matter and Chapter 3 describe. Every other theorem-like claim in the book is labeled Proposition, Corollary, Conjecture, or Open Question according to the criteria above, and Appendix F consolidates every item labeled Conjecture or Open Question in one place, together with every substantive open item flagged in a Remark or Principle rather than a formal Theorem or Proposition environment, so that the book’s unproved commitments are never left implicit.

The single largest such commitment is the status of *MEM|8*’s relationship between memory and representation (chapter 12), which this monograph treats as a genuine, unresolved fork rather than resolving by assumption. Readers should treat Part III’s results as conditional on the narrower of the two available readings until that fork is settled by further work.

Relation to Prior Work

This monograph sits downstream of several prior works, verified to several different degrees of confidence, and this note distinguishes them explicitly rather than let a reader assume uniform rigor. Four sources have been checked by reading primary text directly: *Conservation of Ambiguity*, Distinguishability Geometry, Persistence Before Truth, and Observability Is Restorability. Two further sources, *Admissibility Before Truth* and *The Geometry of Correction*, are integrated in Chapter 3 on the strength of exact formulas and theorem statements transcribed directly from the source PDFs by the author, which is stronger evidence than paraphrase but is not the same as an independent reading of the primary file; every claim drawn from them is marked (*Transcribed from by the author; not independently verified against primary source text.*) rather than (*Imported from .*) throughout, and this distinction is maintained rather than allowed to blur after the first citation.

A retraction, since partially corrected. An earlier draft of this monograph cited *The Geometry of Reconstruction* as the source of the observation that repair operators require higher-order maintenance (proposition 9.8). No such text has been located, and that citation remains retracted. *The Geometry of Witnesses*, initially flagged as possibly forthcoming or nonexistent, is now confirmed by the author to exist and to have been read; it is not, however, transcribed or independently verified against primary source text, and what this monograph reports

from it (§3.8) rests on the author’s prose summary rather than exact formulas copied from source text, unlike *Admissibility Before Truth* and *The Geometry of Correction*. It is cited at this third, weaker tier accordingly, and the distinction is maintained rather than allowed to blur after the first citation.

From *Observability Is Restorability* this monograph imports the restorable-distinction set $\text{Rest}(\mathcal{R})$ and the basic notion of repair cost, both checked against primary text in Chapter 3 (§3.5). That chapter also reports a finding that bears on this monograph’s own repair-interference definition (definition 5.6): *Conservation of Ambiguity* attributes its derivative-form interference functional to Observability Is Restorability, but Observability Is Restorability’s own text defines interference algebraically instead. This monograph’s usage is correctly attributed to *Conservation of Ambiguity*, where the derivative form is actually stated; the discrepancy between the two source texts is recorded in remark 3.20 as a fact about the corpus, not a defect in this monograph’s citation.

From *Admissibility Before Truth*, transcribed rather than independently read, this monograph imports the Domain Restriction Theorem (§3.6) and connects it directly to this monograph’s own sufficiency apparatus (proposition 3.24). This text should not be confused with *The Admissibility Field*, the presumed foundational text of the wider Admissibility Program, which remains wholly unseen; *Admissibility Before Truth* is a distinct, narrower paper that happens to share the program’s name.

From *The Geometry of Correction*, likewise transcribed rather than independently read, this monograph imports the distinction between self-referential and reality-anchored evaluation and the coupling coefficient ρ (§3.7), connecting corruption (definition 2.8) to correction suppression (proposition 3.28).

From *Conservation of Ambiguity* this monograph imports the Redistribution Theorem in full and without modification

(theorem 5.7), together with its explicit corollary that ambiguity conservation is only a degenerate limiting case of the more general repair-budget conservation. This monograph does *not* claim to be a corollary of that theorem. The Redistribution Theorem governs a single observer’s competing distinctions at one moment; this monograph’s Composition Failure Theorem (theorem 10.2) governs a single distinction’s status as it moves through a sequence of independently-budgeted stages over time. Appendix C gives the side-by-side comparison in full, including a worked demonstration that one result is not derivable from the other without the additional budget-relay machinery developed in Part II.

Distinguishability Geometry and Persistence Before Truth are imported with verified source text in Chapter 3, which states precisely how far this monograph’s independently-built vocabulary – renderings, discard sets, root recovery, relay revision – corresponds to their actual formal apparatus, and where the correspondence is only partial or remains open. The Admissibility Program’s own foundational text, *The Admissibility Field*, has still not been consulted directly in any form; this monograph’s admissible continuation set (definition 4.1) is accordingly flagged in section 3.1 as an unresolved correspondence. Three further, distinct formal notions bearing on admissibility are now on record without being reconciled to it or to each other: Observability Is Restorability’s observer-relative admissibility manifold $\mathcal{A}_O$, the activation-manifolds application paper’s admissibility field $A : M \rightarrow \mathbb{R}_{\geq 0}$, and *Admissibility Before Truth*’s domain restriction $\text{dom}(\mathcal{T}) \subseteq \text{Rest}(\mathcal{R})$. Of the three, the last connects most directly to this monograph’s own apparatus, as §3.6 shows.

Contents

Abstract	iii
Note on Epistemic Status	v
Relation to Prior Work	vii

I Foundations: Distinctions, Substrates, and the Problem of Continuation **1**

1 The Continuation Problem	3
1.1 A Motivating Scenario	4
1.2 Informal Statement of the Thesis	4
1.3 Continuation, Not Representation	4
1.4 Three Motivating Domains	5
1.5 An Accurate Roadmap	6
1.6 A Note on How to Read the Labels	7
2 Distinctions and Distinguishability Loss	9
2.1 Distinguishability Geometry, Imported	9
2.2 Renderings as Projections	10
2.2.1 From Pairs to Named Distinctions	11
2.3 Underdetermination Versus Corruption	12
2.4 A Toy Example	13

3	The Four Operations of Distinction	15
3.1	Two Collisions in the Source Vocabulary	16
3.2	Two Deficits	17
3.3	The Four Operations	18
3.4	Persistence Sets	20
3.5	A Third Verified Source: Observability Is Restorability	21
3.6	A Fourth Source, Transcribed: Admissibility Before Truth	23
3.7	A Fifth Source, Transcribed: The Geometry of Correction	25
3.8	A Sixth Source, at a Weaker Tier: The Geometry of Witnesses	26
3.9	Where This Monograph's Apparatus Sits . . .	28
3.10	An Honest Scorecard	31
4	Admissibility and the Space of Continuations	33
4.1	Admissibility Theory, Imported and Specialized	34
4.2	What a Task Requires	35
4.2.1	Tasks as Filters on Distinction Space . .	35
4.3	Why Sufficiency Cannot Be Verified at Production Time	37
4.4	Closed and Open Hidden-State Domains . . .	38
5	Repair, Cost, and the Economics of Maintaining a Distinction	41
5.1	Repair Cost as the Cost of Preserving a Minimal Pair	42
5.2	Repair Cost, Sharpness, and Ambiguity	43
5.3	Repair Interference	44
5.4	The Redistribution Theorem	44
5.5	A Worked Instance, Before It Becomes a Relay .	46
6	From Synchronic Budgets to Diachronic Chains	49
6.1	The Mismatch, Stated Precisely	49

6.2	Why This Is Not a Trivial Relabeling	50
6.3	Statement of the Extension Task	52
II	The Budget Relay: A New Formal Object	53
7	Defining the Budget Relay	55
7.1	Why a Chain, and Not Just a Rendering	56
7.2	The Substrate Chain	57
7.2.1	Local Budgets as Stage-Local Repair Economies	58
7.3	Budget-Constrained Discard, and Its Underde- termination	58
7.4	The Budget Relay, Defined	60
7.4.1	Why Local Rationality Is Not Global Ra- tionality	61
7.5	The Root-Object Move	62
7.6	Cumulative Discard and Its Monotonicity	64
7.7	The Loss Ledger, Upgraded	65
7.7.1	The Loss Ledger as Institutional Memory	66
7.8	A Fully Worked Instance	67
7.8.1	Stage 0 $\rightarrow$ 1	67
7.8.2	Stage 1 $\rightarrow$ 2	68
7.8.3	A Task Neither Stage Anticipated	68
7.8.4	Root Recovery, Demonstrated	69
7.9	What This Definition Commits Us To	69
8	Coordination, Its Absence, and Partial Remedies	71
8.1	Why Full Coordination Is Usually Unavailable	72
8.2	Augmentation	72
8.3	Provenance and Loss-Ledger Sharing	73
8.4	Priors	74
8.5	Adjudication	75
8.5.1	Human Institutions as Adjudicators	76
8.6	Revision	76
8.7	Coordination Versus Pooling: A Boundary Case	77

8.8	The Limits of the Taxonomy	78
8.9	Coordination Versus Witness Reconstruction . .	78
9	Fan-Out: Parallel Budget Relays and Cross-Branch Disagreement	81
9.1	From Chains to Graphs	82
9.2	Branch Agreement and Silent Disagreement . .	83
9.2.1	A Minimal Two-Branch Witness	83
9.3	The Regress Inside the Confidence Profile . . .	85
9.4	Reconciliation	87
10	The Composition Failure Theorem	89
10.1	Making Local Admissibility Precise	90
10.2	Statement	91
10.2.1	Why No Stage Is Wrong	91
10.2.2	Relation to Arrow-Type Impossibility Results	92
10.3	Proof Strategy	93
10.4	Corollaries	94
10.5	What the Theorem Does Not Say	95
10.6	Relation to the Redistribution Theorem	95
III	Modes of Access: Preservation and Reconstitution	97
11	Preservation Mode, Reconstitution Mode, and the Fidelity Continuum	99
11.1	The Naive Binary	100
11.2	Fidelity, Defined	100
11.3	The Hard Test Case: A Chess Board in Front of You	102
11.4	Recursive Insufficiency	103

12 Memory as a Limiting Case: MEM 8 and the Ontology of Traces	105
12.1 Two Readings	106
12.2 Considerations on Each Side	107
12.3 Consequences of the Broad Reading	108
12.4 What Survives Regardless of the Fork	108
12.5 Worked Material Under the Narrow Reading	110
13 Modes of Access as Task-Relative, Not Object-Relative	111
13.1 The Criterion, Made Precise	112
13.2 Same Artifact, Different Modes	113
13.3 Why This Dissolves Rather Than Answers the Earlier Question	114
IV Worked Domains	117
14 Chess: The Closed Hidden-State Case	119
14.1 The Board as Substrate	119
14.2 Augmentation Worked in Full	120
14.3 The Simul Player as a Degenerate Relay	121
14.4 Revisiting the Hard Test Case	122
15 Painting: Open Causal Spaces and Benign Underdetermination	123
15.1 Spatial Compression	124
15.2 Temporal Compression and the Bent Branch	124
15.3 Why Painting Is the Canonical Open Domain	126
15.4 A Preview: Closedness and Declaration Are In- dependent	126
16 Maps: Scale as a Declared Discard Set	129
16.1 The Scale Bar as a Prehistoric Loss Ledger	129
16.2 Task-Relative Sufficiency, Concretely	130
16.3 Completing the Taxonomy	130

17 Proofs: Sufficiency Relative to a Closed Task and System	133
17.1 The Strongest Apparent Counterexample . . .	134
17.2 Why This Resolves Rather Than Merely Sidesteps the Objection	134
17.3 Why Proofs Do Not Exhibit Composition Failure	135
17.4 Verification as Preservation, Explanation as Reconstitution	136
18 Legal Contracts: Adjudication as Non-Informational Resolution	137
18.1 Contracts as Attempted Brute-Force Closure .	138
18.2 Adjudication, Worked	139
18.3 Litigating Intent as Reconstitution	139
19 Scientific Theories: Revision as Relay Replacement	141
19.1 A Theory as a Finite Substrate for an Open Observation Space	142
19.2 Revision, Not Augmentation	142
19.3 A Note on Adjacent Debates	143
19.4 Scientific Revolutions as Relay Replacement . .	144
20 Memory: Reconstitution and the Recursive Limit	147
20.1 Memory as a Three-Stage Relay	148
20.2 Schema-Driven Confident Error	149
21 Media Pipelines: Budget Relays Made Concrete	151
21.1 The Full Worked Chain, Named and Numbered	152
21.2 Fan-Out, Concretized	153
21.3 Practical Recommendations	154
V Synthesis	155
22 The Closing Statement	157
22.1 Restating the Arc	158

22.2	The Closing Statement	159
22.3	What Remains Open	159
22.4	What This Monograph Does Not Claim	160
22.5	The Strongest Objection	161
22.6	Future Research Directions	162
A	Full Proof of the Composition Failure Theorem	163
A.1	Recovery Tasks and the Witnessing Task Family	164
A.2	A Construction-Independent Exhaustion Lemma	165
A.3	Recalling Local Admissibility	166
A.4	The Canonical Witness	166
A.5	Extension to Arbitrary Relay Length	167
A.6	Proof of Theorem 10.2	168
A.7	Verifying the Corollaries	169
A.8	What This Proof Does Not Establish	169
B	Notation Reference	171
C	Relation to <i>Conservation of Ambiguity</i>: A Side-by-Side Comparison	175
C.1	Why One Does Not Reduce to the Other	176
D	Extended Worked Examples	179
D.1	Manuscript Transmission	179
D.2	Oral Tradition	180
D.3	Biological Inheritance	182
E	Toward a Simulation Framework	185
F	Open Questions Register	187
	Bibliography Note	195

CONTENTS

Part I

Foundations: Distinctions, Substrates, and the Problem of Continuation

Chapter 1

The Continuation Problem

Synopsis. This chapter introduces the question motivating the entire monograph: how can a future consumer continue a task when only a finite artifact remains available? Rather than asking whether a substrate represents its source faithfully in the abstract, the chapter shifts attention to continuation — the practical question of what actions remain possible given a surviving substrate. Chess, painting, and media pipelines are introduced as the recurring examples that generated the framework, before the chapter states its central thesis: no finite substrate can be guaranteed sufficient for arbitrary future continuation across an open task space, because future requirements cannot be fully anticipated at the time the substrate is produced.

1.1 A Motivating Scenario

An artifact is produced. A consumer arrives later, possibly much later, possibly through an intermediary the producer never anticipated, and must act on the basis of what is available to them. The question this monograph asks is simple to state and difficult to resolve: what must survive the interval between production and use for that later action to succeed?

1.2 Informal Statement of the Thesis

Principle 1.1 (Global Insufficiency, informal form). No finite substrate can be guaranteed sufficient for arbitrary future continuation across an open task space.

This statement is deliberately informal at this stage, and it is worth saying plainly what kind of book follows from taking it seriously. It is not a claim that information is destroyed, that communication is impossible, or that faithful transmission never occurs – Chapters 14 and 17 both exhibit clean cases where a substrate is fully adequate for its purpose. It is a narrower and more tractable claim: that adequacy is always relative to a task, that the relevant task cannot always be known in advance, and that this gap has a precise structure worth studying rather than a vague one worth lamenting. Chapter 4 gives the informal statement above a precise form as proposition 4.7, and Part II extends it from a single substrate to a chain of substrates, where the monograph’s central new theorem is proved (theorem 10.2).

1.3 Continuation, Not Representation

Continuation is defined operationally throughout this monograph: given only a substrate r , the set of actions a consumer

can correctly perform in pursuit of a task τ . This is deliberately narrower than asking whether r is a faithful, meaningful, or complete representation of some original object e in the abstract. Faithfulness-to-an-original is not directly measurable by a consumer who never had access to e ; sufficiency-for-a-task is.

1.4 Three Motivating Domains

Three domains recur throughout the monograph and are introduced here only informally, to be formalized in Part IV. A chess position illustrates a domain with closed, enumerable hidden state: what is missing from the visible board (castling rights, en passant eligibility, the repetition count) can in principle be listed exhaustively and restored. A painting illustrates the opposite case: a static configuration, such as a branch bent as if by wind, is compatible with an open-ended and non-enumerable set of causal histories, and no finite augmentation closes that space. A media pipeline – audio rendered into a transcript, a transcript compressed into a summary – illustrates a domain in which loss accumulates across a sequence of independently produced renderings, each locally reasonable, with no shared record of what any earlier stage discarded.

These three domains are not illustrations selected after the fact; they are the cases that generated the framework, and it is worth being explicit about what that means in practice, since it bears on how the rest of the book should be read. More than one definition in this monograph was revised after failing to survive contact with a worked numerical instance: the discard rule underlying the budget relay (definition 7.2) turned out to admit two genuinely different constructions only once a concrete example was pushed hard enough to expose the difference (remark 7.5), and a claimed theorem about cumulative loss (proposition 7.12) turned out, once stated against a concrete case, to be an immediate consequence of a definition rather than

an independent result requiring its original, stronger label. The worked examples of Part IV are load-bearing for this reason: they are not decoration added to a finished theory, they are part of how the theory was checked.

1.5 An Accurate Roadmap

Part I develops the vocabulary needed to state principle 1.1 precisely: distinguishability loss (chapter 2), then a chapter situating that vocabulary within the wider research program it descends from (chapter 3), checking prior claims of informal inheritance against primary source text where available and marking plainly where only weaker evidence could be had. Admissible continuation (chapter 4) and repair economics (chapter 5) follow, before Chapter 6 identifies exactly where this imported machinery stops applying and proves, rather than merely asserts, that no coordination mechanism salvages it without genuinely new construction (proposition 6.1). Part II introduces the budget relay, the monograph's one new formal object, and proves the Composition Failure Theorem (theorem 10.2) against an explicit, fully worked witness rather than only in the abstract. Part III resolves the preservation/reconstitution question raised by the informal motivating conversation behind this project, correcting its initial binary framing into a single continuous variable (definition 11.2), and states plainly, rather than papering over, the one dependency this correction cannot settle on its own: whether memory and representation stand in the relationship MEM|8 suggests (Chapter 12). Part IV returns to chess, painting, and the rest with the full apparatus available, closing with a numerically explicit pipeline witness (proposition 21.3) built from the same machinery as the book's central proof. Part V closes with a single statement tying distinguishability, admissibility, repair budgets, and continuation together, and an explicit accounting of what remains

unresolved.

Six appendices support this argument without carrying its weight. Appendix A gives the Composition Failure Theorem's full proof; Appendix B collects the book's notation in one place; Appendix C establishes, concretely rather than only structurally, that this monograph's central result is independent of *Conservation of Ambiguity's*; Appendix D extends the framework to domains not given full chapters; Appendix E sketches, without implementing, how the framework could be tested computationally; and Appendix F consolidates every conjecture and open question raised anywhere in the book into a single register, so that what remains unproved is never left for a reader to discover piecemeal.

1.6 A Note on How to Read the Labels

This monograph distinguishes proved results from conjectured ones throughout, and does so more insistently than is typical, because the central claim of the book is partly a claim about the limits of what any finite substrate – including this one – can establish once and for all. Exactly one result in the entire book is new: the Composition Failure Theorem this monograph exists to prove (theorem 10.2). Nine further results carry the label Theorem on the strength of being imported wholesale, with correct attribution, from a source text that itself labels them Theorem – including the Redistribution Theorem (theorem 5.7) and, at varying tiers of verification described in the front matter and in Chapter 3, seven theorems from Distinguishability Geometry, Persistence Before Truth, and Observability Is Restorability, plus one from Admissibility Before Truth transcribed rather than independently verified. Every other substantial claim is a Proposition, a Corollary, a Conjecture, or an Open Question, and the front matter's Note on Epistemic Status explains the difference precisely. A reader who wants to know, at any point,

exactly how much weight a given claim can bear should look at its label before its rhetoric.

Chapter 2

Distinctions and Distinguishability Loss

Synopsis. This chapter establishes the formal vocabulary used throughout the book. Drawing on Distinguishability Geometry, it treats information as a structured space of distinctions and describes renderings as projections that preserve some distinctions while collapsing others. The chapter’s central move is the distinction between benign underdetermination and corruption: not all loss is harmful, and corruption arises not from loss alone but from *undeclared* loss that obscures what has been discarded. This reframes information quality around disclosure rather than preservation alone.

2.1 Distinguishability Geometry, Imported

Distinguishability Geometry supplies this monograph’s most basic object: a space of distinctions $(X, \sim_X)$, together with an equivalence structure recording what is and is not told apart

within it. Nothing in this section modifies that prior formalism; it is specialized here to the case where X is the distinction-space of an object e under transformation.

Remark 2.1 (A Standing Simplification). Throughout this monograph, distinction spaces are treated as finite or countable, and the cardinality-based measures used later (definition 11.2 in particular) are stated only for that case. This is a simplification made for tractability, not a claim that every domain of interest is discrete in the relevant sense; a painting’s causal-history space, for instance, is arguably continuous rather than merely large. Where this matters, it is flagged again locally (chapter 15); it is not resolved here.

2.2 Renderings as Projections

Definition 2.2 (Rendering as Projection). A rendering r of an object e is a map $(X, \sim_X) \rightarrow (Y, \sim_Y)$ under which some distinctions separated in X become identified in Y .

This definition is stated at the level of Distinguishability Geometry’s own primitives: X and Y are sets of possible states, and $\sim_X, \sim_Y$ record which pairs of states are, and are not, told apart within each. This is the correct level of generality for the formalism, but it is not the level at which the rest of this monograph actually talks. From Chapter 4 onward, and especially throughout Chapter 7, this book speaks of *a* distinction d – “distinction d_3 ,” “the repair cost of d ,” “ d is dropped at stage i ” – as though a distinction were a single named thing, not a pair of states. Reconciling these two ways of talking, rather than quietly switching between them, is this chapter’s principal piece of technical work.

2.2.1 From Pairs to Named Distinctions

Definition 2.3 (Discarded Pair). Given a rendering $r : (X, \sim_X) \rightarrow (Y, \sim_Y)$ realized by an underlying map $\varphi : X \rightarrow Y$, an unordered pair $\{x, x'\} \subseteq X$ is a *discarded pair* of r if $x \not\sim_X x'$ (the pair is told apart in the source) but $\varphi(x) \sim_Y \varphi(x')$ (the pair becomes indistinguishable under r).

definition 2.3 is the pair-level primitive Distinguishability Geometry actually provides, and it is the correct formal object underlying everything that follows. It is, however, an awkward unit to reason about directly: nobody asks whether “the pair {recording with emphasis on the third syllable, recording without it}” was preserved; they ask whether *emphasis* was preserved. The bridge from pairs to named features is given by treating a feature as the thing that distinguishes a minimal pair.

Definition 2.4 (Minimal Pair for a Distinction). A named distinction d is represented, relative to a source $(X, \sim_X)$, by a minimal pair $\{x_d, x_{\neg d}\} \subseteq X$: two states differing exactly in whether the feature named by d holds, and told apart in the source ($x_d \not\sim_X x_{\neg d}$).

Definition 2.5 (Discarded Distinction Set, Revised). Given a rendering r and a fixed assignment of minimal pairs to named distinctions (definition 2.4), the discarded distinction set $D(r)$ is the set of named distinctions d whose minimal pair $\{x_d, x_{\neg d}\}$ is a discarded pair of r (definition 2.3):

$$D(r) = \{d : \{x_d, x_{\neg d}\} \text{ is discarded by } r\}.$$

definition 2.5 supersedes an earlier, looser formulation of the same definition, which described $D(r)$ directly as “the set of distinction-pairs separated in the source but identified in r ” and was consequently ambiguous between the pair-level and feature-level readings used elsewhere in the book. Every

later chapter’s usage – definition 2.7 and definition 2.8 below, Chapter 5’s treatment of $C_R(d)$ for a named d , and the entire apparatus of Chapter 7 – presumes the feature-level reading given here, and definition 2.5 is stated so as to make that presumption explicit and correct rather than convenient and unexamined.

Remark 2.6 (When the Bridge Is Clean). The identification in definition 2.5 between a discarded pair and a discarded feature is unproblematic exactly when X factors, at least locally, as a product of independent feature-dimensions – so that varying d alone, holding everything else fixed, is a meaningful operation on X and produces a genuine minimal pair rather than an artifact of an arbitrary choice of coordinates. This holds cleanly for the toy and worked examples used throughout this monograph (a transcript’s lexical content and prosodic emphasis are reasonably treated as independent dimensions of an audio recording) and is assumed as a standing simplification rather than argued for in general. Domains where features are not independently variable in this sense – where, for instance, no change to one property is possible without simultaneously changing another – are not treated here.

2.3 Underdetermination Versus Corruption

The distinction developed in this section is used, in one form or another, in every subsequent chapter of the monograph, and is the closest thing this book has to a single organizing move.

Definition 2.7 (Benign Loss). A discarded distinction $d \in D(r)$ is benign relative to a task τ if d is not required to perform τ correctly.

Definition 2.8 (Corrupting Loss). A discarded distinction $d \in D(r)$ is corrupting relative to τ if τ requires d and the rendering’s presentation obscures that d was discarded, as opposed to openly declaring the gap.

Proposition 2.9 (Corruption Requires Undeclared Loss). *Let r be a rendering with discard set $D(r)$. If every distinction in $D(r)$ is declared – that is, made available to the consumer as a known gap rather than presented as preserved – then loss alone does not imply corruption. Corruption requires both (i) the loss of a task-relevant distinction, and (ii) failure to disclose that loss.*

Proof Sketch. Corruption, per definition 2.8, is task-relative by construction. If $d \in D(r)$ is declared, a consumer pursuing a task requiring d can compensate for its absence, decline to rely on r for that task, or seek an alternative rendering; the distinction remains unavailable, but the consumer’s belief-state about r ’s sufficiency is not falsified by r itself. Undeclared loss is therefore necessary for corruption in the sense of definition 2.8, though not sufficient, since a task not requiring d is unaffected regardless of disclosure.

definition 2.7 and definition 2.8 are stated here in terms of “required to perform τ correctly” without yet saying precisely what correctness of performance means, since that requires the apparatus of admissible continuation sets developed in Chapter 4. proposition 4.6 there closes the loop, restating both definitions in the sharper, task-relative vocabulary Chapter 4 provides.

2.4 A Toy Example

Example 2.10. Lossy image compression discards high-frequency detail below a quantization threshold. When the compression ratio and quantization parameters are attached to the file as metadata, the loss is declared: a consumer needing pixel-exact reproduction can detect, before relying on the image, that it will not get it. When the same file is relabeled as a lossless master, an identical discard set becomes corrupting under definition 2.8, though nothing about the image data itself has

changed. In the vocabulary of definition 2.4, each discarded high-frequency component corresponds to a minimal pair of source images differing only in that component, rendered indistinguishable once quantized; the metadata either does or does not disclose that this minimal pair has been collapsed. The example is deliberately kept toy at this stage; it is developed properly, without simplification, in the media-pipeline case of Chapter 21.

Chapter 3

The Four Operations of Distinction

Synopsis. Chapter 2 built a self-contained apparatus for renderings and discard sets without situating it within the broader research program from which its vocabulary descends. This chapter does that situating, against six sources at three different levels of verification: Distinguishability Geometry and Persistence Before Truth, checked by reading primary text directly; Observability Is Restorability, likewise checked directly, which also surfaces an inconsistency between two of the other source texts; Admissibility Before Truth and The Geometry of Correction, integrated on the strength of formulas and theorem statements transcribed directly from source PDFs by the author rather than independently checked against primary source text; and The Geometry of Witnesses, reported only from the author's prose summary and cited at a correspondingly weaker tier still. Every claim is marked (*Imported from .*), (*Transcribed from by the author*;

not independently verified against primary source text.), or (Reported from a prose summary of by the author; neither transcribed formulas nor primary source text independently available.) according to which of these three tiers actually supports it, and the distinction is maintained rather than allowed to blur after the first citation. The honest finding is that this monograph built working analogues of three of Distinguishability Geometry's four primitive operations without recognizing them as such, resolved a fourth case only a later source could characterize, and discovered – by the accident of checking an unrelated source – that its oldest working distinction, benign versus corrupting loss, already had a name in a program it had not previously engaged.

3.1 Two Collisions in the Source Vocabulary

Before any importing can be done responsibly, two terminological collisions need to be named, because silently picking one convention over the other would misrepresent both source texts.

The first collision is internal to the wider research program this monograph draws on. Distinguishability Geometry's foundational object is the ontological triple $(X, \sim, T)$, in which T denotes an *ontology*: a description language or template hierarchy determining which descriptions of elements of X are reachable. Persistence Before Truth's foundational object is the structured domain $(X, \mathcal{T}, \mathcal{D}, \mathcal{R})$, in which $\mathcal{T}$ denotes a family of *admissible transformations* $\{T_\alpha : X \rightarrow X\}$. The same letter is doing incompatible work in two sibling foundational texts: one uses T for the language in which X is described, the other uses $\mathcal{T}$ for the maps that act on X . This monograph adopts the

merged notation

$$(X, \sim, T, \mathcal{T}),$$

with X the state space, $\sim$ the distinguishability relation, T the ontology in Distinguishability Geometry’s sense, and $\mathcal{T}$ the transformation family in Persistence Before Truth’s sense, and uses this notation for the remainder of the present chapter whenever both roles need to appear together.

The second collision is not resolved here, and it is important to say so plainly rather than let the first resolution create an impression of general tidiness. This monograph’s own admissible continuation set $\mathcal{A}_\tau(r)$ (definition 4.1) is an epistemic, consumer-relative notion: the continuations reachable given what a particular rendering discloses. The Admissibility Program’s own central object, so far as it can be reconstructed from a downstream application paper and a secondary summary rather than from the foundational text itself, is $\mathcal{A} \subseteq X$: a structural, non-epistemic subset of states whose continuation would not destroy the conditions for coherent further continuation, regardless of what any particular consumer can currently tell. These are not the same kind of object, and this monograph has not verified against the primary Admissibility Field source what the precise relationship between them is. This is recorded as an open item in Appendix F rather than papered over.

3.2 Two Deficits

(Imported from Distinguishability Geometry.)

Definition 3.1 (Ontological Triple). An ontological triple is $(X, \sim, T)$, where X is a set, $\sim$ is an equivalence relation on X (observational indistinguishability), and T is an ontology determining which descriptions of elements of X are reachable. The equivalence classes $[x]_\sim$ are called fibers.

Definition 3.2 (Coding Deficit). Let $L_T(x)$ be the shortest description of x reachable within T , and $L^*(x) = K(x)$ the Kolmogorov complexity of x . The coding deficit is $\delta_T(x) = L_T(x) - L^*(x)$.

Definition 3.3 (Distinguishability Deficit). Let $D_T(x)$ be the number of objects separable from x by descriptions reachable in T , and $D^*(x)$ the maximum such capacity attainable by any ontology. The distinguishability deficit is $\Delta_T(x) = D^*(x) - D_T(x)$.

Theorem 3.4 (Deficit Proxy). (*Imported from Distinguishability Geometry.*) Under faithful encoding (each expressible distinction requires at least one description bit), $\Delta_T(x) \leq C \delta_T(x)$ for a system-dependent constant $C > 0$. In particular, $\delta_T(x) = 0 \Rightarrow \Delta_T(x) = 0$.

The converse direction is stated in the source text as an open conjecture, not a theorem, and this monograph inherits that status rather than strengthening it: whether $\Delta_T(x) = 0 \Rightarrow \delta_T(x) = 0$ in general remains unproved.

This apparatus is considerably more developed than the loss vocabulary this monograph built independently in Chapter 2. Where $D(r)$ (definition 2.5) records only which named distinctions are present or absent, δ_T and Δ_T are quantitative, and the Deficit Proxy Theorem relates two genuinely different notions of loss – coding cost and distinguishing capacity – that this monograph never separated. section 3.9 states precisely how far the correspondence between the two vocabularies extends and where it does not.

3.3 The Four Operations

(*Imported from Distinguishability Geometry.*)

Distinguishability Geometry organizes representational change into two levels. Projection and embedding operate

at the distinction level, modifying $\sim$ within a fixed space. Revision and transport operate at the ontology level: revision changes T , and transport connects two distinct triples.

Definition 3.5 (Projection). A projection on $(X, \sim, T)$ is a surjective map $\sigma : X \rightarrow S$ inducing a coarsening $\sim'$ of $\sim$: $\sigma(x) = \sigma(y) \Rightarrow x \sim' y$.

Theorem 3.6 (Projection Deficit Bound). (*Imported from Distinguishability Geometry.*) Let $\sigma : X \rightarrow S$ be a projection inducing fiber partition $\{F_s\}_{s \in S}$, with X carrying a probability measure μ . Then $\Delta\delta(\sigma) \geq H(X \mid S)$, the conditional entropy of X given S .

Definition 3.7 (Embedding). An embedding of $(X, \sim, T)$ into $(Y, \sim', T')$ is an injective map $\phi : X \rightarrow Y$ such that $\phi(x) \sim' \phi(y) \Rightarrow x \sim y$. An embedding is faithful if the converse also holds.

Theorem 3.8 (Embedding Refinement). (*Imported from Distinguishability Geometry.*) If $\phi : (X, \sim, T) \rightarrow (Y, \sim', T')$ is a faithful embedding, then $\delta_{T'}(\phi(x)) \leq \delta_T(x)$: a faithful embedding never increases the coding deficit.

Definition 3.9 (Revision). An ontological revision is a transition $(X, \sim_T, T) \rightarrow (X, \sim_{T'}, T')$ where $T' \supset T$. A revision is productive if $\delta_{T'}(x) < \delta_T(x)$.

Theorem 3.10 (Revision Monotonicity). (*Imported from Distinguishability Geometry.*) For an expanding sequence of ontologies $T_0 \subset T_1 \subset \dots$, $\delta_{T_{n+1}}(x) \leq \delta_{T_n}(x)$ for all $n \geq 0$.

Corollary 3.11 (Asymptotic Deficit). (*Imported from Distinguishability Geometry.*) The sequence $(\delta_{T_n}(x))_{n \geq 0}$ is monotone non-increasing and bounded below by zero; hence $\lim_{n \rightarrow \infty} \delta_{T_n}(x) = \delta_\infty(x) \geq 0$ exists, with $\delta_\infty(x) = 0$ iff some reachable revision sequence converges to an optimal ontology for x .

Definition 3.12 (Transport). A transport map is $\tau : (X, \sim_X, T_X) \rightarrow (Y, \sim_Y, T_Y)$ with $x \sim_X x' \Rightarrow \tau(x) \sim_Y \tau(x')$, and distortion $\epsilon(\tau) \geq 0$ controlling the measure of fiber violations.

Theorem 3.13 (Transport Stability). (*Imported from Distinguishability Geometry.*) Let τ be a transport map with distortion at most ϵ , and let I_X, I_Y be invariants constant on fibers of $\sim_X, \sim_Y$ respectively, agreeing on perfectly transported fibers. Then $|I_X(x) - I_Y(\tau(x))| \leq C\epsilon$ for a constant C depending on the oscillation of the invariants across fiber boundaries.

Corollary 3.14 (Justification of Analogy). (*Imported from Distinguishability Geometry.*) An analogical inference transported by τ is valid up to error $C\epsilon(\tau)$; Transport Stability is the precise condition under which analogical reasoning is valid.

3.4 Persistence Sets

(*Imported from Persistence Before Truth.*)

Definition 3.15 (Structured Domain). A structured domain is a quadruple $(X, \mathcal{T}, \mathcal{D}, \mathcal{R})$, where X is a state space, $\mathcal{T} = \{T_\alpha : X \rightarrow X\}$ is a family of admissible transformations, $\mathcal{D} = \{d_i : X \rightarrow Y_i\}$ is a family of distinctions, and $\mathcal{R} = \{R_j : X \rightarrow X\}$ is a family of admissible reconstruction operators.

Definition 3.16 (Recoverability Relation). A distinction $d \in \mathcal{D}$ is recoverable under $T \in \mathcal{T}$ if there exists $R \in \mathcal{R}$ such that $d \circ R \circ T = d$.

Definition 3.17 (Persistence Set). The persistence set of a distinction d is $P(d) = \{T \in \mathcal{T} : \exists R \in \mathcal{R} \text{ with } d \circ R \circ T = d\}$.

Theorem 3.18 (Persistence Depends Only on Structure). (*Imported from Persistence Before Truth.*) Recoverability is invariant under any isomorphism of structured domains that preserves transformations, distinctions, and reconstruction operators; hence recoverable

persistence is a structural property rather than a property of particular representations.

This is the general, primitive object this monograph’s root-object apparatus (definition 7.8) turns out to be a narrow special case of. section 3.9 makes the correspondence precise.

3.5 A Third Verified Source: Observability Is Restorability

(Imported from Observability Is Restorability.)

A third source cited in this monograph’s front matter has since been checked against primary text, and the check surfaced something worth reporting precisely: not an error this monograph introduced, but an inconsistency already present between two source texts it draws on.

Definition 3.19 (Restorable Distinction Set). For a repair repertoire $\mathcal{R}$, $\text{Rest}(\mathcal{R}) = \{d \in \mathcal{D} : C_R(d) < +\infty\}$: the set of distinctions restorable at finite cost under every admissible perturbation.

$\text{Rest}(\mathcal{R})$, used throughout Chapters 5 onward, is this object, and the attribution to Observability Is Restorability was correct: $\text{Rest}(\mathcal{R})$ is exactly this source’s central construction, arrived at from repair paths, perturbations, and a substrate $(X_S, \mathcal{P}, \mathcal{D})$ considerably richer than anything this monograph builds on top of it. This monograph’s $C_R(d)$ is a bare scalar; Observability Is Restorability’s $C_R(d)$ is an infimum over actual repair paths through a topological state space, and the source text develops a full geometry on top of it – an interference matrix with spectral structure (bottleneck clusters, synergistic bundles), a repair pseudometric, an observer-relative admissibility manifold $\mathcal{A}_O$ characterized by a Repair Boundary Theorem,

and a Control-Geometry Collapse theorem identifying restorability with reachability under controllability hypotheses. None of this deeper apparatus is used in the present monograph; $\text{Rest}(\mathcal{R})$ is imported as a set, not as the geometry that generates it.

Remark 3.20 (An Inconsistency Between Two Source Texts, Not Introduced Here). *Conservation of Ambiguity* states that its repair interference functional $I(d_i, d_j)$ is “established in Observability Is Restorability,” and gives it in derivative form: the change in repair cost of d_j induced by an infinitesimal increase in repair investment devoted to d_i . This is exactly the form definition 5.6 uses, correctly attributed to *Conservation of Ambiguity*. But Observability Is Restorability’s own Definition 8.7 gives a different, algebraic form: $I(d_i, d_j) = C_R(d_i \wedge d_j) - C_R(d_i) - C_R(d_j)$, a superadditivity gap rather than a derivative. The sign conventions agree (positive: competition or conflict; negative: synergy; zero: independence), but the formulas are not the same object, and one text’s claim to inherit the other’s definition does not survive a direct check. This is recorded here because it bears on how much weight the phrase “established in Observability Is Restorability” can carry elsewhere in the corpus, not because it requires this monograph to choose between the two: everything in Chapters 5 onward uses only the derivative form, consistently, and nothing here depends on reconciling it with the algebraic one.

Theorem 3.21 (Paradigm Shift as Repair-Repertoire Discontinuity). (*Imported from Observability Is Restorability.*) Let $\mathcal{R}_{\text{old}}$ and $\mathcal{R}_{\text{new}}$ be repair repertoires with $\mathcal{R}_{\text{old}} \not\subseteq \mathcal{R}_{\text{new}}$ and $\mathcal{R}_{\text{new}} \not\subseteq \mathcal{R}_{\text{old}}$. Then $\text{Rest}(\mathcal{R}_{\text{old}}) \not\subseteq \text{Rest}(\mathcal{R}_{\text{new}})$ and conversely: the restorable-distinction sets are incomparable, with lost distinctions $\text{Rest}(\mathcal{R}_{\text{old}}) \setminus \text{Rest}(\mathcal{R}_{\text{new}})$ and gained distinctions the reverse difference, both generically nonempty.

This theorem is the resolution proposition 3.33 needed and did not have. That proposition showed relay revision matches

3.6. A FOURTH SOURCE, TRANSCRIBED: ADMISSIBILITY BEFORE TRUTH

Distinguishability Geometry’s revision operation only when the state space is held fixed and the ontology merely expands, and left the genuine paradigm-shift case – where the state space itself changes – uncharacterized. theorem 3.21 characterizes exactly that case, and does so within the repair-theoretic vocabulary this monograph already uses throughout Chapters 5 onward, rather than requiring an appeal to Distinguishability Geometry’s ontology-revision apparatus at all. section 3.9 restates this as a fifth correspondence.

3.6 A Fourth Source, Transcribed: Admissibility Before Truth

(Transcribed from Admissibility Before Truth by the author; not independently verified against primary source text.)

The following is imported on the strength of exact formulas and a proof sketch transcribed directly from the source PDF by the author, not from an independent reading of the primary file; per the front matter, this is marked throughout rather than allowed to blend into the fuller confidence of section 3.5’s verified import.

Definition 3.22 (Domain Restriction). A truth predicate $\mathcal{T}$ factors as $\mathcal{T} = \tilde{\mathcal{T}} \circ \iota$, where $\iota : \text{Rest}(\mathcal{R}) \hookrightarrow \mathcal{D}$ is the inclusion of restorable distinctions into the full distinction space.

Theorem 3.23 (Domain Restriction Theorem). *(Transcribed from Admissibility Before Truth by the author; not independently verified against primary source text.)* $\text{dom}(\mathcal{T}) \subseteq \text{Rest}(\mathcal{R})$: a truth predicate is defined only on distinctions restorable by the observer’s repair repertoire. Outside $\text{Rest}(\mathcal{R})$, a distinction is neither true nor false but unevaluable.

Proof Sketch. *(Transcribed from Admissibility Before Truth by the author; not independently verified against primary source text.)* Every

distinction in the domain of $\mathcal{T}$ must lie in $\mathcal{D}$ by well-typedness; $\mathcal{T}$ is then defined on $\text{Rest}(\mathcal{R})$ by restriction through ι , and outside $\text{Rest}(\mathcal{R})$ the predicate takes a third value marking non-evaluability, not a truth value of either polarity.

Proposition 3.24 (Sufficiency Presupposes Evaluability). *Task-relative sufficiency (definition 4.5) implicitly assumes $\mathcal{T}$'s domain restriction: r can be sufficient for τ only if every distinction τ requires already lies in $\text{Rest}(\mathcal{R})$ for the repair repertoire generating r .*

Proof Sketch. definition 4.5 asks whether $\mathcal{A}_\tau(r) \subseteq \mathcal{C}_\tau(e)$, which presupposes that membership in $\mathcal{C}_\tau(e)$ is a meaningful, evaluable question for the distinctions at stake. By theorem 3.23, that question is only meaningful for distinctions in $\text{Rest}(\mathcal{R})$; for a distinction outside it, asking whether a continuation is correct is type-confused rather than merely undetermined. This was never stated in Chapter 4, which built definition 4.5 without reference to any prior evaluability condition, silently assuming it. The assumption is almost always harmless in this monograph's worked domains, since the distinctions at stake (castling rights, causal history, lexical content) are restorable by construction whenever they are discussed at all – but it is an assumption, made visible only now.

proposition 3.24 is the closest this monograph comes to making contact with the Admissibility Program's actual formal content, as distinct from the still entirely unseen *Admissibility Field*. It does not resolve the three-way admissibility collision recorded in the front matter and in Appendix F, but it is the first of the four competing admissibility-adjacent notions now on record that connects to this monograph's own apparatus by a stated proof rather than by analogy alone.

3.7 A Fifth Source, Transcribed: The Geometry of Correction

(Transcribed from The Geometry of Correction by the author; not independently verified against primary source text.)

Definition 3.25 (Self-Referential and Reality-Anchored Evaluation). For an evaluation function $E : X \rightarrow \mathbb{R}$ and internal signal set S , a system is self-referential if $E(x) = f(S(x))$, depending only on internal signals. A system is reality-anchored if $E(x) = f(S(x), O(x))$ for an external channel O the system does not fully author or control.

Definition 3.26 (Coupling Coefficient). $\rho = \partial E / \partial O$. When $\rho \approx 0$, the external channel no longer moves the evaluation in practice, regardless of whether O remains present.

Principle 3.27 (Correction Suppression). *(Transcribed from The Geometry of Correction by the author; not independently verified against primary source text.)* Pathological self-reference does not usually mean no external corrective channel exists; it means the system has become insensitive to a channel that remains present.

This is stated as a Principle rather than a Theorem because the source text itself names it that way, and this monograph preserves that label rather than upgrading its apparent formality.

Proposition 3.28 (Corruption as Correction Suppression). *Corrupting loss (definition 2.8) is an instance of correction suppression: a rendering's presentation obscuring a discard is exactly a case where the rendering's apparent evaluation of its own sufficiency has decoupled from the external channel – the actual, undisclosed state of $D(r)$ – that would otherwise correct it.*

Proof Sketch. Let E be the confidence or sufficiency a rendering presents to a consumer, and let O be the true discard set $D(r)$,

a fact about the rendering the consumer does not directly observe. Declared loss (definition 2.7) keeps E coupled to O : the consumer's assessment of sufficiency tracks what was actually discarded, since it is disclosed. Undeclared loss (definition 2.8) is exactly the case where E no longer moves with O : the presented confidence is computed from S alone – the rendering's internal, self-referential account of itself – with $\rho \approx 0$ relative to the actual discard set. proposition 9.8's regress (Chapter 9) is the recursive version of the same failure: each confidence profile Q is itself a rendering whose own E may or may not track its own O , and the trust boundary (definition 9.9) is precisely the point at which this monograph stops asking whether $\rho \approx 0$ has crept in one level further down.

proposition 3.28 is offered as the tightest correspondence in this chapter, tighter even than the four/five-operations correspondences of section 3.9: it did not require reinterpreting either definition, only recognizing that definition 2.8's "obscures" and definition 3.26's " $\rho \approx 0$ " are the same phenomenon named twice, once in this monograph's discard-set vocabulary and once in the correction-theoretic vocabulary of a program this monograph had not previously engaged.

3.8 A Sixth Source, at a Weaker Tier: The Geometry of Witnesses

(Reported from a prose summary of The Geometry of Witnesses by the author; neither transcribed formulas nor primary source text independently available.)

What follows rests on the author's prose account of having read this text, not on transcribed formulas or primary source text, and is reported at that correspondingly weaker tier: informative enough to record and connect to this monograph's open questions, not solid enough to state as a Definition or Theorem the way §§3.6–3.7 do.

The reported central question is who maintains the repair repertoire $\mathcal{R}$ that Admissibility Before Truth (§3.6) treats as given. The proposed answer introduces a witness operator, reconstructing a repair operator from historical structure rather than repairing a distinction directly, and a reconstruction cover $C(r)$ – the set of histories from which a given repair operator can be reconstructed – with an associated redundancy, reported as $\rho(r) = |C(r)|$.

Remark 3.29 (A Notation Collision, Not Introduced Here). The Geometry of Witnesses’s reported redundancy and The Geometry of Correction’s coupling coefficient (definition 3.26) both use ρ for entirely unrelated quantities, apparently a collision between the two source texts’ own notational choices rather than an error introduced in this monograph. To avoid compounding the collision inside a single chapter, this monograph writes the witness redundancy as $\rho_W(r) = |C(r)|$ from this point on, reserving unadorned ρ for definition 3.26’s coupling coefficient throughout.

Zero, $\rho_W(r) = 0$, means the repair operator is permanently lost; one means a single point of failure; larger values mean resilience.

Remark 3.30 (A Candidate Answer to an Open Question, Unverified). Appendix F records, as unresolved, what $\mu(P(d))$ – the measure of a persistence set (definition 3.17) – might be in general. The reconstruction cover $C(r)$ and redundancy $\rho_W(r)$, as reported, are a plausible candidate operationalization of exactly this quantity for the specific case of root recovery (proposition 7.9): a root object with many independent reconstruction pathways is resilient in precisely the sense $\rho_W(r) \gg 1$ would formalize, and a root object with exactly one surviving copy in one archive is the $\rho_W(r) = 1$ case this monograph’s oral-tradition discussion (Appendix D) already worried about informally, without a name for it. This monograph does not adopt $\rho_W(r)$ as its own machinery, since it has not been verified;

it is recorded here as a specific, checkable candidate for future work rather than left as a vague gesture toward “some measure or other.”

3.9 Where This Monograph’s Apparatus Sits

Five correspondences concerning Distinguishability Geometry’s four operations are worth stating explicitly, because they were not recognized when the relevant chapters of this monograph were drafted, and because they differ sharply in how tight they are. Four are proved correspondences; one is an open, hedged analogy. Two further correspondences, concerning evaluability and correction rather than the four operations, were given already in §3.6 and §3.7 and are not repeated here.

Proposition 3.31 (Rendering Is Projection). *A rendering r in the sense of definition 2.2 is a projection in the sense of definition 3.5, and the discard set $D(r)$ of definition 2.5 corresponds to the fibers merged by that projection.*

Proof Sketch. Both are surjective maps from a source distinguishability structure to a coarser one, defined by which pairs of source states become identified. definition 2.2’s “some distinctions separated in X become identified in Y ” is definition 3.5’s coarsening condition stated in different words. What this monograph’s apparatus does not compute, and theorem 3.6 does, is a quantitative lower bound on how much is lost: $\Delta\delta(r) \geq H(X \mid Y)$ would give every discard set in this book a numeric floor it currently lacks. This is a genuine, available strengthening that has not been carried out.

Proposition 3.32 (Augmentation Is Embedding). *Augmentation with an enumerable hidden-state set H (proposition 4.9) is a faithful embedding in the sense of definition 3.7, and proposition 4.9 is an instance of theorem 3.8.*

Proof Sketch. The augmented rendering $r' = (r, H)$ is obtained from r by an injective map into a richer space that separates states r identified, exactly definition 3.7's shape; it is faithful precisely when H exhausts the task-relevant hidden state, which is proposition 4.9's hypothesis. theorem 3.8 then gives, for free and in full generality, the conclusion proposition 4.9 proved by a bespoke argument: a faithful embedding cannot increase the coding deficit, so augmentation cannot make sufficiency worse. This monograph's proof of proposition 4.9 is correct but is now known to be a special case of a general theorem it did not cite.

Proposition 3.33 (Relay Replacement Is Revision, Where the State Space Is Held Fixed). *Relay revision (definition 8.6) in which the underlying state space X is unchanged and only the descriptive ontology is expanded is an instance of definition 3.9, and theorem 3.10 guarantees the new theory's coding deficit is no worse than the old one's.*

Proof Sketch. When a superseded theory is replaced by one built on an expanded, but not qualitatively different, evidential ontology – more precise instruments within the same conceptual framework, say – this is exactly $T \subset T'$ on a fixed X , and theorem 3.10 applies directly. The correspondence is deliberately not claimed in full generality. Where a replacement changes what counts as a state at all – a genuine paradigm shift in the sense Chapter 19's note on adjacent debates explicitly declined to adjudicate – the transition is not pure revision in definition 3.9's sense, since that definition holds X fixed; such a case would need to be analyzed as some composition of revision and transport, and this monograph has not carried out that analysis.

Proposition 3.34 (Paradigm-Shift Relay Replacement Is Repair-Repertoire Discontinuity). *Relay revision in which the underlying production process changes so substantially that neither the old nor*

the new process's repertoire contains the other corresponds to theorem 3.21, with the substrate chain's producing repertoire at each era playing the role of $\mathcal{R}_{\text{old}}$ and $\mathcal{R}_{\text{new}}$.

Proof Sketch. This is exactly the case proposition 3.33 left uncharacterized. theorem 3.21 shows the resulting restorable-distinction sets are incomparable rather than one being a refinement of the other, which is a more decisive statement than “analysis not carried out”: it says the two eras' restorable distinctions are genuinely incommensurable in the technical sense of neither set containing the other, with concrete lost and gained distinctions on both sides. Chapter 19's example of a theory's evidential basis being enlarged rather than replaced falls under proposition 3.33 instead; the paradigm-shift case this proposition covers is the sharper one, where old and new genuinely talk past each other on some distinctions in both directions.

Remark 3.35 (Root Recovery as Persistence-Set Membership). proposition 7.9 is naturally read as the claim that a specific transformation – the composite production process from A_0 to A_n – lies in the persistence set $P(d)$ (definition 3.17) for a discarded distinction d , exactly when the reconstruction operator “recompute from the preserved A_0 under a revised policy” is available in $\mathcal{R}$. This reframes Appendix F's root-preservation bootstrapping question sharply: rather than asking qualitatively whether recovery is possible, one can ask what $\mu(P(d))$ is – what fraction of the relevant transformation family still admits recovery – and this monograph has not attempted that computation.

Remark 3.36 (Fidelity and Transport: An Open Correspondence). Chapter 11's fidelity $F(r, \tau)$ (definition 11.2) and theorem 3.13's invariant-oscillation bound both measure how well structure survives a transformation, but they are different mathematical objects – a cardinality ratio of admissible continuation sets against a Lipschitz-style bound on a fiber-constant invariant – and this monograph has not shown one reduces to the

other. This is the weakest of the five correspondences in this section, offered as an open question rather than a result, and recorded as such in Appendix F.

3.10 An Honest Scorecard

Before this chapter, a reader could reasonably have concluded that this monograph recognized only one of Distinguishability Geometry's four primitive operations – projection – as load-bearing, with embedding, revision, and transport absent. That conclusion would have been too pessimistic and not quite accurate. proposition 3.31 confirms projection was there from Chapter 2 onward. proposition 3.32 shows embedding was present, unnamed, in the augmentation mechanism of Chapters 4 and 14. proposition 3.33 and proposition 3.34 together show revision was present, unnamed, in Chapter 19 for both the case Distinguishability Geometry's revision operation covers and the sharper paradigm-shift case it does not, with Observability Is Restorability supplying the missing half. Only transport remains without a demonstrated instance in this monograph rather than merely an unnamed one: remark 3.36's candidate is an analogy, not yet a correspondence. Three operations out of four fully accounted for, and the fourth genuinely open rather than merely unexamined, is a materially different and more encouraging finding than one operation out of four – and it is the finding this monograph is prepared to defend, rather than a stronger claim it has not earned.

Two further findings sit outside the four-operations scorecard entirely. proposition 3.24 shows this monograph's sufficiency apparatus was silently assuming an evaluability condition Admissibility Before Truth makes explicit, discovered only once that source was transcribed rather than paraphrased. proposition 3.28 shows this monograph's oldest and most-used distinction – benign versus corrupting loss, present since Chap-

ter 2 – is the same phenomenon The Geometry of Correction names correction suppression, recognized only by accident of the author happening to transcribe a second, unrelated-seeming source. Neither finding would have been visible from the four-operations framing alone, which is itself a reason to keep checking new sources against this monograph's apparatus rather than treating the reconciliation work in this chapter as finished.

Chapter 4

Admissibility and the Space of Continuations

Synopsis. Having defined what can be lost, this chapter turns to what remains possible. Using concepts imported from Admissibility Theory, it defines admissible continuation sets and task-relative sufficiency: a substrate is never sufficient in the abstract, only relative to a particular task. The chapter argues that sufficiency cannot generally be certified at creation time, since future tasks are often unknown then, and introduces the distinction between closed hidden-state domains, where omitted information can in principle be enumerated and restored, and open domains, where no finite augmentation can exhaust future informational needs. This distinction becomes one of the framework's central organizing principles.

4.1 Admissibility Theory, Imported and Specialized

Admissibility Theory supplies the second basic object this monograph needs: given a substrate, the set of continuations still reachable from it. As with Distinguishability Geometry in Chapter 2, nothing in the prior formalism is altered here; it is specialized to the continuation problem of chapter 1.

Definition 4.1 (Admissible Continuation Set). $\mathcal{A}_\tau(r)$, the admissible continuation set of task τ given rendering r , is the set of continuations of τ that remain reachable given only r .

This definition leaves the internal structure of a “continuation” deliberately unspecified, and deliberately so: a continuation is a legal move in Chapter 14’s setting, a plausible causal narrative in Chapter 15’s, a verified-or-rejected verdict in Chapter 17’s, and a binding ruling in Chapter 18’s. The formalism is domain-agnostic on purpose; each worked domain in Part IV supplies its own concrete continuation space rather than inheriting one from this chapter.

Proposition 4.2 (Admissibility Bounded by Full Access). *For any rendering r of an object e and any task τ , $\mathcal{A}_\tau(r) \subseteq \mathcal{A}_\tau(e)$.*

Proof Sketch. A rendering (definition 2.2) can only identify distinctions present in the source, never introduce new ones; any continuation reachable using only r is therefore reachable using the full source e as well, since e determines everything r does and possibly more. The reverse containment need not hold, which is exactly what makes discard consequential.

Proposition 4.3 (Admissibility Is Monotone Under Further Rendering). *If r' is itself a rendering of r – that is, r' is obtained from r by a further projection in the sense of definition 2.2 – then $\mathcal{A}_\tau(r') \subseteq \mathcal{A}_\tau(r)$.*

Proof Sketch. Immediate from proposition 4.2 applied with r playing the role of the source and r' the rendering of it: r' can only identify distinctions already present or absent in r , and cannot recover any continuation r itself could not reach.

proposition 4.3 is the admissibility-theoretic counterpart of Chapter 7's proposition 7.12: continuation possibilities can only shrink or stay level as a substrate is further rendered, never expand, in the absence of a preserved earlier stage to re-derive from (proposition 7.9).

4.2 What a Task Requires

4.2.1 Tasks as Filters on Distinction Space

It helps to have a single image in mind before the formal machinery arrives. A task does not need everything a substrate might carry; it needs a specific subset of the available distinctions, and is indifferent to the rest. Verifying a proof needs the logical content and nothing about the author's handwriting; reading a map for a driving route needs road connectivity and is indifferent to elevation contours a hiking map would need instead. A task, in this sense, acts as a filter over distinction space: it partitions what a substrate carries into what matters for this purpose and what does not, and Chapter 2's benign/corrupting distinction (definitions 2.7 and 2.8) is already implicitly using this filter, since "not needed for the task" is exactly "outside the filter." The rest of this chapter makes the filter precise enough to compute with, but the filter is the idea; the correctness set below is its formal expression.

Chapter 2 spoke of a distinction being "required" by a task τ without saying precisely what that means. The remainder of this chapter makes that precise, and in doing so corrects an oversimplification present in an earlier formulation of task-relative sufficiency: that formulation spoke of "the true or intended

continuation” of a task, in the singular, which is adequate for tasks like verification that admit exactly one correct outcome but is not adequate for tasks like Chapter 15’s evocative reconstruction, which by design tolerates more than one acceptable completion.

Definition 4.4 (Correctness Set). A task τ , evaluated against an object e , determines a nonempty correctness set $\mathcal{C}_\tau(e)$: the set of continuations that count as successfully performing τ . $\mathcal{C}_\tau(e)$ is a singleton for tasks demanding a unique correct outcome (verification against a fixed proof, say) and may contain many continuations for tasks tolerating multiple acceptable completions (evocative reconstruction from a painting, say, where several causal histories are equally acceptable).

Definition 4.5 (Task-Relative Sufficiency, Revised). r is sufficient for τ (evaluated against the true object e) iff

$$\mathcal{A}_\tau(r) \subseteq \mathcal{C}_\tau(e).$$

This revises, and is intended to supersede, the earlier singular-outcome phrasing. The subset formulation in definition 4.5 is chosen because it correctly unifies the two cases an adequate definition needs to cover simultaneously. For a verification-type task, $\mathcal{C}_\tau(e)$ is a singleton, and definition 4.5 demands that r pin down exactly that outcome: $\mathcal{A}_\tau(r)$ must not merely intersect the correct answer but be contained in it, ruling out any admissible but wrong continuation being reachable alongside the right one. For a task tolerating multiple acceptable completions, $\mathcal{C}_\tau(e)$ is larger, and definition 4.5 only requires that every continuation r makes reachable falls inside that larger acceptable set – which is satisfied even when several distinct, mutually incompatible continuations remain open, exactly the situation Chapter 15 calls benign underdetermination.

Proposition 4.6 (Benign Loss Preserves Sufficiency). *If every distinction in $D(r)$ is benign relative to τ (definition 2.7), and e itself would be sufficient for τ , then r is sufficient for τ .*

4.3. WHY SUFFICIENCY CANNOT BE VERIFIED AT PRODUCTION TIME

Proof Sketch. By definition 2.7, no distinction required to determine membership in $\mathcal{C}_\tau(e)$ is among those r discards; hence every continuation in $\mathcal{A}_\tau(r)$ is also determinable from e as belonging to $\mathcal{C}_\tau(e)$, since the discarded distinctions play no role in that determination. By proposition 4.2, $\mathcal{A}_\tau(r) \subseteq \mathcal{A}_\tau(e)$ always, and whenever e itself is sufficient for τ , every continuation in $\mathcal{A}_\tau(e)$ already lies in $\mathcal{C}_\tau(e)$; combining the two containments gives $\mathcal{A}_\tau(r) \subseteq \mathcal{C}_\tau(e)$, which is definition 4.5 directly.

proposition 4.6 is the formal statement of the coherence this monograph’s two central chapters are supposed to have with each other: Chapter 2’s loss-level vocabulary (benign versus corrupting) and this chapter’s continuation-level vocabulary (admissible versus sufficient) are not two independent frameworks that happen to agree on examples; benign loss is, by construction, exactly the kind of loss that cannot violate sufficiency.

4.3 Why Sufficiency Cannot Be Verified at Production Time

Proposition 4.7 (Task Unavailability at Ingestion). *If the future task space T relevant to a substrate is open – unbounded, not fixed at the time the substrate is produced – then sufficiency of a rendering r relative to T cannot be certified when r is produced.*

Proof Sketch. Suppose otherwise: suppose r could be certified sufficient for every $\tau \in T$ at production time. Then $\mathcal{A}_\tau(r) \subseteq \mathcal{C}_\tau(e)$ for every $\tau \in T$, which requires r to have preserved every distinction any such τ might need to rule out an incorrect continuation. Since T is open, a task τ^* can always be constructed, after the fact, requiring a distinction not represented in r – for instance, a distinction r ’s producer had no reason to anticipate – for which $\mathcal{A}_{\tau^*}(r) \not\subseteq \mathcal{C}_{\tau^*}(e)$. This contradicts the certification

assumed above. Hence no such certification is available in general.

This result is the formal version of principle 1.1 for a single substrate. It motivates the shift, completed in definition 7.8, from optimizing a rendering for sufficiency – which proposition 4.7 shows to be generally impossible to verify – to optimizing it for re-derivability from a preserved root object.

4.4 Closed and Open Hidden-State Domains

Definition 4.8 (Closed versus Open Hidden-State Domain). A domain has *closed* hidden state if the distinctions omitted from a given state are drawn from a finite, enumerable list – for instance, a chess position’s castling rights, en passant eligibility, and repetition count. A domain is *open* otherwise – for instance, the causal history of a depicted configuration in a painting, which is compatible with an unbounded and non-enumerable set of candidate causes.

Proposition 4.9 (Closed Hidden-State Sufficiency). *In a domain with a finite hidden-state set H , the augmented rendering $r' = (r, H)$ is sufficient for any task τ whose correctness set $\mathcal{C}_\tau(e)$ is determined by distinctions drawn from H , provided all task-relevant elements of H are included.*

Proof Sketch. Since H is finite, exhaustive enumeration eliminates ambiguity about which hidden variables are missing: every continuation’s membership in $\mathcal{C}_\tau(e)$ becomes computable from r' once the enumerated variables are fixed, so $\mathcal{A}_\tau(r') \subseteq \mathcal{C}_\tau(e)$ as required by definition 4.5. Chess serves as a witness (Chapter 14): a board position augmented with castling rights, en passant status, and repetition count is sufficient for determining every legal continuation of the game, whereas the visible board alone is not.

Proposition 4.10 (Sufficiency Achievable Only in the Closed Case). *Sufficiency in the sense of proposition 4.9 – achieved by finite augmentation of the hidden state – is available only in closed hidden-state domains.*

Proof Sketch. In an open hidden-state domain (definition 4.8), no finite set H exhausts the candidate causes or hidden variables relevant to a given task, by definition of openness; hence no finite H makes proposition 4.9’s hypothesis satisfiable. Painting (Chapter 15) is the canonical witness to this proposition’s converse case, and there $\mathcal{C}_\tau(e)$ is typically large rather than singleton, which is exactly what allows sufficiency to be achieved by other means (proposition 4.6) even though augmentation specifically is unavailable.

CHAPTER 4. ADMISSIBILITY AND THE SPACE OF CONTINUATIONS

Chapter 5

Repair, Cost, and the Economics of Maintaining a Distinction

Synopsis. This chapter imports the repair-theoretic machinery of *Conservation of Ambiguity*: repair cost, ambiguity, and interference as quantitative descriptions of how finite budgets constrain distinguishability. The Redistribution Theorem is restated and carefully scoped as a description of how a single observer reallocates finite resources across distinctions within one repair economy at one moment. The chapter's key emphasis, on which the rest of the monograph depends, is that ambiguity conservation is not that theorem's central result; the deeper conserved quantity is the repair budget itself, and ambiguity conservation is recovered only as a restrictive limiting case.

5.1 Repair Cost as the Cost of Preserving a Minimal Pair

Chapter 2 defined a named distinction d via a minimal pair $\{x_d, x_{-d}\}$ (definition 2.4) and defined d as discarded by a rendering precisely when that pair collapses (definition 2.5). This chapter’s imported apparatus speaks instead of the *repair cost* of d , and it is worth stating plainly, before using the two vocabularies interchangeably as every later chapter does, exactly how they connect.

Definition 5.1 (Repair Cost, Specialized). For a named distinction d with minimal pair $\{x_d, x_{-d}\}$ subject to a perturbation process (a transformation, a transmission channel, a compression step), the repair cost $C_R(d)$ is the cost of maintaining $x_d \not\sim x_{-d}$ – that is, of keeping the pair told apart – under that perturbation.

definition 5.1 is a specialization of Chapter 5’s imported notion of repair cost to the minimal-pair vocabulary of Chapter 2, and it is what licenses Chapter 7’s otherwise unexplained move of treating $C_R(d)$ and $D(r)$ as talking about the same underlying phenomenon: a distinction is discarded by a rendering exactly when insufficient repair investment was made, relative to that rendering’s perturbation, to keep its minimal pair apart.

Remark 5.2 (Repair Cost as Deficit-Closing Investment). Chapter 3 imports Distinguishability Geometry’s coding deficit $\delta_T(x)$ (definition 3.2), and the connection to repair cost is close enough to be worth stating even before that chapter is reached: funding $C_R(d)$ is, in the vocabulary of that later chapter, precisely the investment required to keep d ’s minimal pair distinguishable under an ontology T that would otherwise identify them – that is, to hold $\delta_T(d)$ at zero rather than letting it rise. This monograph’s repair-economic vocabulary and Distinguishability Geometry’s deficit vocabulary are describing the same underlying phenomenon from two directions: one asks what it

costs to prevent a collapse, the other asks how much description length is lost once the collapse has already happened.

5.2 Repair Cost, Sharpness, and Ambiguity

(Imported from *Conservation of Ambiguity*.)

Definition 5.3 (Distinction Sharpness). For $d \in \text{Rest}(\mathcal{R})$ a repair-stable distinction, the sharpness of d is $\sigma(d) = C_R(d)$, the repair cost required to maintain d under admissible perturbation.

Definition 5.4 (Operational Ambiguity). The ambiguity of a distinction is a monotone decreasing function of its repair cost; the simplest choice is $A(d) = 1/C_R(d)$.

Remark 5.5 (Ambiguity’s Functional Form Is a Convention, Not a Law). *Conservation of Ambiguity* states this explicitly, and it is worth repeating rather than letting the simplest choice harden into an assumption: any strictly decreasing monotone transformation of $C_R(d)$ produces an equivalent *ordering* of distinctions by ambiguity, so $A(d) = 1/C_R(d)$ is adopted here for concreteness and ease of computation in worked examples, not because it is uniquely forced by anything proved. Nothing in this monograph depends on the specific functional form; every result stated in terms of $A(d)$ could be restated, without loss, in terms of the ordering it induces.

Ambiguity, under definition 5.4, is not ignorance, incomplete information, or subjective uncertainty. It measures the degree to which a distinction has been left under-maintained: a distinction is ambiguous because insufficient repair capacity has been committed to stabilizing it against perturbation, not because information about it is unknowable in principle.

5.3 Repair Interference

Definition 5.6 (Repair Interference). For $d_i, d_j \in \text{Rest}(\mathcal{R})$, the repair interference $I(d_i, d_j)$ is the change in repair cost of d_j induced by an infinitesimal increase in repair investment devoted to d_i . Positive interference indicates repair competition; negative interference indicates repair synergy; zero interference indicates operational independence.

This definition is stated exactly as *Conservation of Ambiguity* gives it, correctly attributed. It is worth flagging, rather than discovering by accident later, that the same source text attributes this formula to Observability Is Restorability, and Observability Is Restorability’s own verified text defines repair interference differently – algebraically rather than as a derivative. remark 3.20 (Chapter 3) gives the precise comparison; nothing in this chapter or those that follow depends on resolving it, since this monograph uses only the derivative form, consistently, throughout.

5.4 The Redistribution Theorem

Theorem 5.7 (Redistribution Theorem). (*Imported from Conservation of Ambiguity.*) Let $\Phi = A(\cdot) + \lambda C'_R(\cdot) + \mu I(\cdot, \cdot)$ be the repair budget functional of a finite observer, with total budget Φ_{tot} held fixed. Then every admissible variation satisfying $\delta C_R(d_1) > 0$ for some distinction d_1 induces compensating variations throughout the remaining distinction space such that $\delta \Phi = 0$. Consequently, no local increase in distinction sharpness can occur without global redistribution of ambiguity, repair cost, or repair interference.

Proof Sketch. (*Imported from Conservation of Ambiguity.*) The result follows from constrained variation of the repair budget functional Φ under fixed total budget. Since the observer possesses no additional maintenance capacity, every increase in

repair investment devoted to one region of distinction space must appear elsewhere as a corresponding change in ambiguity, repair cost, or repair interference. The full proof, including the three interference regimes (competition, synergy, independence), is given in *Conservation of Ambiguity* and is not repeated here.

Scope note. theorem 5.7 governs a single observer’s competing distinctions at a single moment: a *synchronic* result. It says nothing, on its own, about what happens to a single distinction’s status as it is carried across a sequence of independently-budgeted stages over time. That extension is the subject of Chapter 6 and Part II.

Corollary 5.8 (Ambiguity Conservation as a Limiting Case). *(Imported from Conservation of Ambiguity.) If repair costs and interference remain fixed ($\delta C_R(\cdot) = \delta I(\cdot, \cdot) = 0$), then $\delta A(\cdot) = 0$: ambiguity conservation is recovered as a special case of repair-budget conservation. In realistic systems, sharpening one distinction almost always changes both repair costs and interference relationships, so ambiguity alone is generally not conserved; the conserved quantity is the complete repair budget Φ .*

This corollary is the reason this monograph does not treat itself as a corollary of theorem 5.7: the earlier framing of the research program that produced this book – “ambiguity is conserved, merely relocated” – is precisely the special case corollary 5.8 shows to be exceptionally restrictive and generally false. The new work required here (Part II) does not inherit its central claim from this theorem; it must be argued independently.

5.5 A Worked Instance, Before It Becomes a Relay

Chapter 7 introduces a running numerical example: five distinctions $d_1, \dots, d_5$ with repair costs $C_R(d_1) = 1$, $C_R(d_2) = 2$, $C_R(d_3) = 3$, $C_R(d_4) = 2$, $C_R(d_5) = 4$. There, the costs are held fixed and a *budget* β_i varies from stage to stage of a relay. Here, before any relay exists, it is worth seeing what theorem 5.7 says about the same five distinctions considered as the simultaneous concern of a *single* observer with one fixed total budget Φ_{tot} – the case the theorem actually governs.

Suppose the observer initially funds all five distinctions exactly at their stated costs, for a total commitment of $1 + 2 + 3 + 2 + 4 = 12$, and suppose $\Phi_{\text{tot}} = 12$ exactly, leaving no slack. Now suppose the observer wishes to sharpen d_3 – to invest further in it beyond its baseline cost, driving $C_R(d_3)$ up and $A(d_3)$ down, perhaps because d_3 has become newly important. theorem 5.7 states that this cannot happen for free: with Φ_{tot} fixed at 12 and already fully committed, $\delta C_R(d_3) > 0$ forces a compensating $\delta \Phi = 0$ elsewhere in the functional. If interference is negligible ($I \approx 0$, as in the worked relay of section 7.8), the compensation must come directly out of the repair cost or ambiguity terms of the remaining four distinctions – concretely, some combination of d_1, d_2, d_4, d_5 must have its own funding reduced by exactly the amount d_3 's sharpening consumed, raising their ambiguity in turn.

This is the single-observer, single-moment phenomenon theorem 5.7 describes, and it is worth contrasting directly with what happens to the *same five numbers* once Chapter 7 turns them into a two-stage relay with $\beta_0 = 4$ and $\beta_1 = 3$. There, d_3 's low priority at stage 0 does not trigger any compensating adjustment to d_1 or d_2 's funding elsewhere in a shared Φ_{tot} , because no such shared quantity exists across stages (definition 7.6): d_3, d_4 , and d_5 are simply dropped, permanently, with

nothing anywhere in the system compensating for their loss. The redistribution theorem 5.7 guarantees within one economy has no counterpart at all across the boundary between β_0 and β_1 . Chapter 6 argues for this contrast in prose; proposition 6.1 there states it as a precise claim about why no functional analogous to Φ can be defined across a relay's stages in the first place.

CHAPTER 5. REPAIR, COST, AND THE ECONOMICS OF MAINTAINING A DISTINCTION

Chapter 6

From Synchronic Budgets to Diachronic Chains

Synopsis. This chapter identifies the conceptual gap that motivates the monograph's primary contribution. Existing repair-budget theory describes trade-offs among distinctions maintained by one observer at one moment; real information systems instead consist of chains of transformations performed by many independent agents across time. Local repair-budget reasoning cannot simply be applied stage by stage, because no global coordination mechanism exists and each stage acts with incomplete knowledge of future tasks and other stages' decisions. This motivates the new formal object introduced in Part II.

6.1 The Mismatch, Stated Precisely

theorem 5.7 governs competition among distinctions maintained by one observer at one time. The continuation problem of Chapter 1 concerns a single distinction, or a bounded clus-

ter of distinctions, whose status must survive transmission across a sequence of observers or processes over time. These are different objects, and the difference is not cosmetic.

6.2 Why This Is Not a Trivial Relabeling

Three features of the continuation problem have no counterpart in the Redistribution Theorem’s setting. First, no single global budget functional constrains an entire chain of independently operating stages; each stage’s budget is its own. Second, each stage’s repair decisions are typically made without visibility into any other stage’s decisions, past or future. Third, the chain is directional: a later stage cannot un-discard a distinction an earlier stage has already collapsed, whereas theorem 5.7’s variational argument implicitly allows reallocation in any direction within a single observer’s simultaneous budget.

The temptation to treat these as differences of degree rather than of kind is worth resisting directly, since a natural first response to the continuation problem is to try applying theorem 5.7 stage-by-stage – to ask what a “local Redistribution Theorem” at each B_i would say, and to hope that composing n local instances yields something like a global conservation law across the whole chain. The remainder of this section shows why that hope fails, and why it fails for a structural reason rather than merely a technical one.

Proposition 6.1 (No Global Conserved Quantity Across a Relay). *There is, in general, no functional $\Phi_{\text{relay}} = f(\beta_0, \dots, \beta_{n-1})$ of a budget relay’s stage budgets satisfying an analogue of theorem 5.7’s conservation law under variation of any single β_i .*

Proof Sketch. theorem 5.7’s conservation law depends essentially on the interference functional $I(\cdot, \cdot)$ being defined over pairs of distinctions competing within a *single* repair-stable

domain $\text{Rest}(\mathcal{R})$, so that a variation in the repair investment devoted to one distinction has a well-defined, computable effect on the repair cost of another via $I(\cdot, \cdot)$ (definition 5.6). Under definition 7.6, however, stage i 's interference function I_i is defined only over distinctions present at stage i ; no interference term $I(d, d')$ is defined, or is even well-typed, for d a distinction competing for β_i and d' a distinction competing for β_j at a different stage j , since the two distinctions need not both survive to be simultaneously present at any single stage for such a term to be evaluated. A variation $\delta\beta_i$ therefore has no defined effect on β_j for $j \neq i$ within this formalism, not merely an uncoordinated or unobserved one: the mathematical object *theorem 5.7's* proof varies – a single Φ built from terms sharing one common domain $\text{Rest}(\mathcal{R})$ – has no counterpart spanning stages, since $\text{Rest}(\mathcal{R})$ itself is stage-local by definition 7.1. No conservation law can be stated for a quantity that has not been defined, and Φ_{relay} has not been defined for exactly this reason.

proposition 6.1 sharpens the informal claim made in earlier drafts of this argument – that “no global coordination mechanism exists” – into a specific mathematical reason rather than a description of a missing institutional arrangement. It is not merely that no one has coordinated the stages of a typical relay; it is that the formal object a coordination claim would need to be about does not exist under definition 7.6's stage-local interference structure. This is also why Chapter 8's partial-coordination mechanisms cannot simply consist of “computing Φ_{relay} and applying theorem 5.7 to it”: that theorem is unavailable in this setting not because the chain is uncoordinated but because the relevant conserved quantity has no definition to conserve. Coordination mechanisms must therefore do something other than extend Chapter 5's apparatus outward; they must introduce new machinery, which is exactly what Chapter 8's taxonomy does.

6.3 Statement of the Extension Task

The remainder of this monograph is organized around building a formal object capable of expressing “locally rational, globally uncoordinated” loss accumulation – the budget relay of Chapter 7 – and proving a composition result analogous in spirit to theorem 5.7 but diachronic in structure: the Composition Failure Theorem of Chapter 10. Given proposition 6.1, that theorem cannot be obtained by generalizing theorem 5.7’s variational argument to a larger domain; it must be proved directly against definition 7.1 and definition 7.2’s irreversibility, as it is in Appendix A.

Principle 6.2 (Global Insufficiency, restated). A substrate chain in which each transformation is locally budget-rational can nonetheless become globally insufficient for an open task space, without any single stage being negligent.

This sentence is, in an important sense, the actual center of the monograph – more so than any single theorem statement that follows from it. It is stated here informally, and precisely, because everything in Part II exists to make it rigorous, and proposition 6.1 is the first indication that making it rigorous will require genuinely new machinery rather than a larger application of what Chapter 5 already provides.

Part II

The Budget Relay: A New Formal Object

Chapter 7

Defining the Budget Relay

Synopsis. This chapter introduces the budget relay, the monograph's primary new theoretical object: a chain of renderings connected by independent local repair economies, each governed by its own costs, priorities, and constraints. The chapter formalizes how distinctions are discarded under local budget pressure and why such losses become irreversible once no record of them remains. Root objects and budget-aware loss ledgers are introduced as the mechanisms for measuring, and in some cases recovering from, downstream insufficiency; preserving a root object is shown to be fundamentally different from preserving every downstream rendering, since it alone permits re-derivation under newly discovered tasks.

This chapter carries disproportionate weight in the monograph relative to its length. The object defined in definition 7.6 is the only genuinely new mathematical object this book introduces; every later theorem is, in an important sense, a property of this definition rather than an independent contribution. If this definition is not judged inevitable and useful on its own

terms, no proof in Chapter 10 will rescue it. The chapter is therefore written at a slower pace than its material would otherwise require, and it deliberately stress-tests the definition against a fully worked numerical instance before the rest of the monograph is permitted to build on it. Two genuine gaps surface in the course of this stress-testing — the underdetermination of the local discard rule, and the bootstrapping cost of preserving a root object — and both are carried forward explicitly into Appendix F rather than quietly absorbed into the definition.

7.1 Why a Chain, and Not Just a Rendering

Chapter 2 defined a rendering as a single projection from a source’s distinction space to a smaller one, and Chapter 4 asked, of a single such rendering, whether it is sufficient for a given task. Neither chapter had reason to ask what happens when a rendering is itself produced from an earlier rendering, and that one from a still earlier one, with a different party responsible for each step and no party responsible for the sequence as a whole. This is not a generalization added for its own sake. It is the ordinary condition of almost every substrate a real consumer actually encounters: a transcript is produced from a recording, not from the recorded event itself; a textbook’s account of a proof is produced from the proof, not independently derived; a legal memorandum summarizing a contract is produced from the contract, which was itself drafted from negotiation notes that no longer exist. Treating each of these as an isolated rendering of some single, fixed original – as Chapters 2 and 4 implicitly do when they speak of “the” discard set of “a” rendering r of “an” object e – silently assumes that e itself, and not some intermediate rendering of it, is always available to compare against. That assumption is false for essentially every substrate chain of practical interest, and its falsity is precisely

what Chapter 6 identified as the reason the repair-budget apparatus of Chapter 5 does not, by itself, resolve the continuation problem of Chapter 1.

7.2 The Substrate Chain

Definition 7.1 (Substrate Chain). A substrate chain is a sequence

$$A_0 \xrightarrow{B_0} A_1 \xrightarrow{B_1} \dots \xrightarrow{B_{n-1}} A_n$$

where each A_i is a rendering (definition 2.2) and each transition is governed by a local budget triple $B_i = (C_i, I_i, \beta_i)$: a local repair cost function, a local interference function, and the budget available at that stage.

Two restrictions are built into definition 7.1 and are worth stating openly rather than leaving implicit. First, the chain is linear and finite: each A_i has exactly one predecessor and one successor, and there are finitely many stages. The branching generalization, in which a single A_i gives rise to several independent successors, is deferred to Chapter 9, where it is shown to introduce a genuinely new failure mode (silent cross-branch disagreement) rather than being a straightforward extension of what follows here. Whether a substrate chain with infinitely many stages behaves differently in kind, and not merely in degree, from the finite case treated throughout this monograph is left unaddressed and is recorded as an open question in Appendix F. Second, A_0 is not assumed to be the original object e of Chapter 2; it is only assumed to be the earliest rendering the chain's analysis takes as given. A substrate chain is therefore always relative to a choice of starting point, and nothing in this chapter's machinery requires that starting point to be loss-free relative to whatever preceded it.

7.2.1 Local Budgets as Stage-Local Repair Economies

The triple $B_i = (C_i, I_i, \beta_i)$ is not an unrelated new primitive. It is Chapter 5's repair-economic apparatus, restricted to a single stage and to that stage's own repair-stable domain. Concretely, C_i is the restriction of the repair cost function $C_R(\cdot)$ of definition 5.3 to the distinctions present at stage i ; I_i is the restriction of the interference functional $I(\cdot, \cdot)$ of definition 5.6 to pairs of distinctions competing for stage i 's budget; and β_i plays the role, at stage i alone, that the total budget Φ_{tot} plays for a single observer in theorem 5.7. The relationship between β_i and Φ_{tot} is best understood by contrast rather than by analogy: Φ_{tot} is conserved across an observer's entire, simultaneous distinction space, so that theorem 5.7 can speak of resources being redistributed *within* it. No such global quantity is conserved across a substrate chain's stages. $\beta_0, \beta_1, \dots, \beta_{n-1}$ are, in general, entirely independent numbers set by independent parties for independent reasons, and nothing in definition 7.1 relates β_i to β_j for $i \neq j$. This is the formal expression of the distinction drawn informally in Chapter 6: the Redistribution Theorem governs one Φ_{tot} shared by many distinctions; a substrate chain has many β_i , each shared by nothing beyond its own stage.

7.3 Budget-Constrained Discard, and Its Underdetermination

Definition 7.2 (Budget-Constrained Discard). A distinction d present in A_i is dropped in A_{i+1} if maintaining d exceeds β_i , or if maintaining d interferes too strongly, per I_i , with a higher-priority distinction competing for the same stage's budget.

definition 7.2 is deliberately schematic, and it is worth being honest, before building anything further on it, about exactly what the schema leaves open. The phrase "higher-priority distinction" presupposes a priority ordering over the distinctions

7.3. BUDGET-CONSTRAINED DISCARD, AND ITS UNDERDETERMINATION

present at stage i , and definition 7.2 does not say where that ordering comes from or what happens when two admissible orderings would license different discards under the same budget β_i . This is not a minor omission to be patched in passing. Two natural instantiations of definition 7.2 exist, they are not equivalent, and the monograph's later theorems need to be checked against both rather than silently assumed to hold for whichever one is easiest to compute with.

Construction 7.3 (Greedy Priority Discard). Fix a priority ordering $d_{(1)} \succ d_{(2)} \succ \cdots \succ d_{(m)}$ over the distinctions present at stage i , reflecting their importance to the stage's own task τ_i . Process the distinctions in this order, maintaining a running total of committed budget. A distinction is funded, and hence preserved into A_{i+1} , if the remaining budget after funding every higher-priority distinction is at least $C_i(d)$; otherwise it is dropped, and processing continues to the next distinction in the ordering without returning to reconsider a dropped distinction later.

Construction 7.4 (Cost-Optimal Discard). Given the same priority ordering interpreted now as a value assignment $v(d_{(k)})$ decreasing in k , select the subset S of stage i 's distinctions maximizing $\sum_{d \in S} v(d)$ subject to $\sum_{d \in S} C_i(d) \leq \beta_i$, breaking ties in favor of higher-priority distinctions.

Remark 7.5 (The Two Constructions Diverge). construction 7.3 and construction 7.4 are the discard rule and the knapsack-optimal packing of the same underlying priorities and costs, respectively, and they can select different subsets for the same (C_i, I_i, β_i) and the same priority ordering. A single low-priority, low-cost distinction that the greedy rule would skip in order to fund several higher-priority items may be exactly the distinction a global optimization would sacrifice one mid-priority item to include, whenever doing so raises the total value funded. Neither construction is privileged by definition 7.2 as stated;

the definition is a constraint on the *outcome* of a stage’s discard decision, not a specification of the *procedure* that produces it.

This divergence is not a defect to be resolved before the theory can proceed. It is the correct level of generality for a schema meant to cover discard decisions made by a codec’s rate-control algorithm, a human editor’s judgment, a court’s interpretation of relevance, and an institution’s retention policy, none of which can be assumed to implement the same procedure. What it does mean, and what is recorded explicitly here so that it cannot be forgotten by the time Appendix A is drafted, is that any theorem claiming to hold for *every* budget relay — including the Composition Failure Theorem of Chapter 10 — must hold under both construction 7.3 and construction 7.4, and ideally under any construction satisfying only the qualitative constraint of definition 7.2. This robustness requirement is added to Appendix F as a condition Appendix A’s eventual proof must be checked against, not assumed to satisfy by construction.

7.4 The Budget Relay, Defined

Definition 7.6 (Budget Relay). A substrate chain (definition 7.1) is a *budget relay* if, for every stage i , the transition B_i is selected using only the current rendering A_i and the stage’s own local task τ_i — that is, without reference to the loss ledgers $L_0, \dots, L_{i-1}$ of any earlier stage, without reference to the budget triple B_j of any other stage $j \neq i$, and without appeal to any interference term spanning distinctions held at different stages.

This restates, and sharpens, an earlier and vaguer formulation of the same condition. The earlier formulation spoke of stages having no “shared interference term or shared discard record...by default,” which left unclear exactly what a stage is

and is not permitted to consult when setting its own budget. definition 7.6 as now stated answers that question directly: a stage may see the rendering handed to it and may know its own task, and nothing else about the chain’s history or future. This is doing the real work of the definition, and it is worth dwelling on why. A substrate chain in which every stage consults a common ledger and a common, chain-wide interference functional before setting its own budget is, for the purposes of this monograph’s apparatus, indistinguishable from a single observer’s economy distributed across time: its total behavior is governed by theorem 5.7 rather than by anything new. That case is not excluded from consideration — Chapter 8 treats exactly this kind of shared-ledger access as one point on a spectrum of partial-coordination mechanisms — but it is not the default case, and definition 7.6 is written to isolate the default case cleanly: a relay in which coordination, if it exists at all, must be added back in deliberately and explicitly, rather than assumed from the outset.

Remark 7.7 (Degenerate Cases). A substrate chain with $n = 1$ satisfies definition 7.6 vacuously, since there is only one stage and no other stage’s budget or ledger to consult; this case reduces to the single rendering already treated in Chapters 2–4 and adds nothing new. The genuinely new content of definition 7.6 appears only once $n \geq 2$, and the sharpest illustrations of that content, developed in section 7.8 below, use $n = 3$.

7.4.1 Why Local Rationality Is Not Global Rationality

It is worth pausing on a distinction definition 7.6 makes available but does not itself name: a stage can be perfectly rational relative to everything it can see, and the chain it belongs to can still be irrational relative to everything no single stage could see. This is not a paradox and not a failure of individual reasoning. Each stage’s optimization problem is well-posed and correctly solved: given A_i , τ_i , and β_i , find what τ_i requires as

well as the budget allows. Nothing about that problem asks the stage to consider τ_{i+1} , let alone τ^* , the eventually-relevant task Chapter 10 constructs. Local rationality is rationality relative to a local objective function; global rationality, if the phrase means anything at all here, would have to be rationality relative to every future task the chain might eventually serve — and proposition 4.7 already showed that objective is not well-posed at any single stage, since the relevant future tasks are not yet known when that stage acts. The chain cannot be irrational for failing an optimization problem it was never given, and it cannot be rational for solving one either; “global rationality” is simply not an available property for any individual stage to have or lack. What survives is only the composition-level question Chapter 10 asks: given that every stage solved its own well-posed local problem, was the sequence of solutions jointly sufficient for a task none of them was solving for? The answer can be no, and that is the entire content of this monograph’s central theorem — not a claim that someone reasoned badly, but a claim that a well-posed collection of local optima need not compose into a global one.

7.5 The Root-Object Move

proposition 4.7 showed that sufficiency cannot be certified at the time a substrate is produced, because the relevant future task space is generally unknown then. Within a budget relay this observation sharpens into a design question rather than remaining a mere impossibility result: since no stage can know which of its discards a future, unanticipated task will need reversed, is there anything a relay’s earliest stage can do that does not require anticipating the future task space at all?

Definition 7.8 (Root Object). A distinguished rendering A_0 in a substrate chain, treated as the reference point against which

all downstream loss is measured. Where possible, ingestion-time policy should optimize for re-derivability of downstream renderings from A_0 , rather than for sufficiency relative to an as-yet-unknown τ .

Proposition 7.9 (Root Recovery). *If the root object A_0 survives unaltered, any downstream rendering A_i can be regenerated under an alternative budget allocation B'_i .*

Proof Sketch. By definition 7.1, every A_i for $i > 0$ is generated from A_0 via some composition of transitions. If A_0 persists, the transition B_i can be replaced with B'_i and A'_i recomputed directly from A_0 ; the loss incurred under B_i is therefore reversible whenever A_0 is available. Without a preserved A_0 , distinctions discarded at any stage are permanently unavailable to every later stage, since definition 7.2 provides no mechanism for a distinction to reappear once dropped.

remark 3.35 (Chapter 3) situates this proposition within the more general persistence-set apparatus of Persistence Before Truth, and reframes the bootstrapping problem below in terms of the measure of a specific persistence set rather than only qualitatively.

proposition 7.9 is easy to overstate, and it is worth being precise about what it does not claim. It does not claim that preserving A_0 is free. Preserving a root object is itself a repair decision – A_0 must be maintained against whatever perturbations threaten it (storage degradation, format obsolescence, simple deletion under retention-policy pressure) – and that maintenance itself has a cost that must be funded by *some* budget, decided by *someone*, before any downstream task requiring it is known to exist.

Remark 7.10 (The Root-Preservation Bootstrapping Problem). Preserving A_0 so that proposition 7.9 remains available is itself an instance of proposition 4.7: the decision to fund A_0 's preservation must be made before the future task space that would

justify the expense is known, by definition of that task space being open. An institution cannot, even in principle, calculate the return on archiving a root object against tasks that do not yet exist. This does not undermine definition 7.8's recommendation so much as clarify its status: root preservation is not a solution to the sufficiency problem of chapter 4, but a wager that re-derivability will be worth more, in expectation, than whatever the preservation itself costs. Whether this wager can be given a more precise decision-theoretic treatment, rather than being left at the level of a design recommendation, is left open in Appendix F.

7.6 Cumulative Discard and Its Monotonicity

The claim that loss compounds along an uncoordinated chain has been used informally since Chapter 6. Making it precise requires first defining what “cumulative” discard means for a chain of more than one transition, since definition 2.5 was stated only for a single rendering.

Definition 7.11 (Cumulative Discard Set). For a substrate chain and indices $i \leq j$, write $D_k := D(A_k \rightarrow A_{k+1})$ for the discard set of the single transition B_k . The cumulative discard set between stages i and j is

$$D(A_i \Rightarrow A_j) := \bigcup_{k=i}^{j-1} D_k.$$

Proposition 7.12 (Monotonicity of Cumulative Discard). *For any substrate chain in which the root object is not available for recovery (proposition 7.9 does not apply), and for any $i \leq j \leq l$,*

$$D(A_i \Rightarrow A_j) \subseteq D(A_i \Rightarrow A_l).$$

Proof Sketch. Immediate from definition 7.11: $D(A_i \Rightarrow A_l)$ is a union over a strictly larger index range than $D(A_i \Rightarrow A_j)$, and a union over a superset of index terms contains the union over the subset. No further argument is required once cumulative discard is defined as a union of per-transition discard sets, since definition 7.2 supplies no operation by which a distinction can be re-added to a later A_l once absent from an intermediate A_k for $k < l$.

This result was labeled a Theorem in an earlier draft of this chapter, and it is worth recording explicitly why that label has been withdrawn. proposition 7.12’s inequality is, once definition 7.11 is stated correctly, an immediate consequence of set union rather than a substantive derivation, and the monograph’s own epistemic-labeling discipline reserves the word “Theorem” for results whose proof carries independent weight. The genuine content of the claim that loss compounds along a chain is not the inclusion itself but the prior, harder-won observation that definition 7.2 provides no mechanism for reversal absent proposition 7.9 – and that observation is a modeling commitment built into definition 7.1, not a theorem about it.

7.7 The Loss Ledger, Upgraded

Definition 7.13 (Budget-Aware Loss Ledger). The loss ledger L_i attached to transition B_i records not merely the discard set $D_i = D(A_i \rightarrow A_{i+1})$, but the pair $L_i = (D_i, B_i)$: the discarded distinctions *together with* the local budget that produced the discard. This allows a downstream consumer to distinguish a distinction dropped because it was irrelevant to that stage’s own purpose from one dropped because that stage’s budget was simply small – a distinction with different implications for whether the loss might be recoverable elsewhere in the relay.

definition 7.13 is what separates a budget relay’s provenance machinery from the plain loss ledger of Chapter 2: it is a

coordination device, not merely a passive record, and its role as such is developed fully in Chapter 8. It is worth noting immediately, however, a limitation that definition 7.6 itself imposes on L_i 's usefulness: because a budget relay's stage $i + 1$ does not consult L_j for $j < i$ when setting B_{i+1} , the mere existence of a loss ledger at every stage does not, by itself, prevent the composition failures analyzed in Chapter 10. A loss ledger is a necessary condition for a downstream consumer, arriving after the fact and inspecting the whole chain, to diagnose what happened; it is not, on its own, a mechanism by which an intermediate stage avoids causing the problem in the first place.

7.7.1 The Loss Ledger as Institutional Memory

The loss ledger is worth a second look through the vocabulary §3.8 reported, at its own weaker tier, from *The Geometry of Witnesses*. That source's reported central question — who maintains the repair operator, once repair operators themselves can be lost — has a direct analogue here: who maintains the loss ledger, once ledgers themselves can be discarded, corrupted, or never consulted? L_i is not self-maintaining. It is itself a rendering (definition 2.2, applied to the ledger rather than to A_i), with its own discard set, and an institution that stops funding its own record-keeping loses the ledger before it loses the substance the ledger was tracking. In the reported vocabulary of §3.8, a well-archived loss ledger — copied across independent systems, cited in multiple downstream audits, backed up in more than one location — has high reconstruction redundancy ρ_W ; a ledger kept in one place by one administrator, discarded the moment that role changes hands, has $\rho_W = 1$ and is one retirement away from $\rho_W = 0$. This is not a new mechanism this monograph is adopting, since §3.8's apparatus remains unverified; it is a naming of a real, practical vulnerability that the loss ledger, as defined here, does nothing to prevent on its

own – a ledger is only as durable as whatever maintains it, and definition 7.13 is silent on what that is.

7.8 A Fully Worked Instance

The definitions above are abstract enough that their consequences are easy to state and easy to misjudge. This section fixes one small, fully explicit numerical relay and works every definition above against it by hand, so that later chapters – particularly the proof of Chapter 10 and its constructive witness in Chapter 21 – have a concrete instance to generalize from rather than only an abstract schema.

Consider a source rendering A_0 carrying five distinctions, $d_1, \dots, d_5$, with repair costs $C(d_1) = 1$, $C(d_2) = 2$, $C(d_3) = 3$, $C(d_4) = 2$, and $C(d_5) = 4$, and suppose interference among them is negligible, so that $I \approx 0$ throughout and the greedy construction (construction 7.3) and the optimal construction (construction 7.4) can be compared directly against the same numbers.

7.8.1 Stage $0 \rightarrow 1$

Suppose stage 0's task τ_0 ranks the distinctions by priority $d_1 \succ d_2 \succ d_4 \succ d_5 \succ d_3$, and that $\beta_0 = 4$. Applying construction 7.3: d_1 is funded (cost 1, remaining budget 3); d_2 is funded (cost 2, remaining budget 1); d_4 cannot be funded (cost 2 > 1 remaining) and is dropped; d_5 cannot be funded (cost 4 > 1 remaining) and is dropped; d_3 cannot be funded (cost 3 > 1 remaining) and is dropped. Thus A_1 retains $\{d_1, d_2\}$ and $D_0 = \{d_3, d_4, d_5\}$. Applying construction 7.4 to the same priorities and budget yields the identical subset in this instance, since $\{d_1, d_2\}$ is in fact the value-maximizing feasible set here; remark 7.5's divergence is real in general but does not arise for every choice of numbers, and this instance was chosen partly to

establish a clean baseline before a divergent case is considered elsewhere.

The loss ledger at this stage is $L_0 = (\{d_3, d_4, d_5\}, B_0)$, with B_0 recording $\beta_0 = 4$ against the priority ordering above – so that a downstream consumer inspecting L_0 can see that d_3 , d_4 , and d_5 were dropped because τ_0 ranked them low and β_0 was small relative to their combined cost, not because they were judged worthless in any absolute sense.

7.8.2 Stage 1 $\rightarrow$ 2

Suppose stage 1's task τ_1 requires only $\{d_1, d_2\}$, exactly what survived into A_1 , and that $\beta_1 = 3$ – exactly sufficient to fund both at cost $1 + 2 = 3$. Then A_2 retains $\{d_1, d_2\}$ and $D_1 = \emptyset$. By definition 7.11, $D(A_0 \Rightarrow A_2) = D_0 \cup D_1 = \{d_3, d_4, d_5\}$, unchanged from $D(A_0 \Rightarrow A_1)$ – an explicit instance of proposition 7.12's inclusion, here realized as equality rather than strict containment, since stage 1 happened to discard nothing further.

7.8.3 A Task Neither Stage Anticipated

Suppose a task τ^* is later discovered, requiring recovery of d_3 – some distinction, present in A_0 , that neither τ_0 nor τ_1 had reason to prioritize. Since $d_3 \notin A_1$ and A_2 is derived only from A_1 , d_3 is unrecoverable from A_2 by definition 7.2's irreversibility and confirmed unrecoverable in general by proposition 7.12. Neither B_0 nor B_1 was irrational relative to its own task: β_0 was spent optimally against τ_0 's stated priorities, and β_1 was spent optimally, indeed exactly, against τ_1 's. A_2 is nonetheless insufficient (definition 4.5) for τ^* . This four-paragraph construction is already, in miniature, the mechanism that Chapter 10 formalizes as Theorem 10.2 and that Chapter 21 exhibits again at full scale against a real media pipeline; nothing about the general

theorem's proof will do more than what has already happened here with five distinctions and two stages.

7.8.4 Root Recovery, Demonstrated

Suppose, finally, that A_0 itself was preserved as a root object (definition 7.8) rather than discarded once A_1 was produced. Upon discovering τ^* , a new transition B'_0 can be computed directly from A_0 using a revised priority ordering that ranks d_3 above d_4 , yielding a new rendering A'_1 retaining $\{d_1, d_2, d_3\}$ and satisfying τ^* . This is proposition 7.9 made concrete: the insufficiency exhibited above was permanent for the original chain $A_0 \rightarrow A_1 \rightarrow A_2$, but not permanent for the underlying situation, precisely because A_0 , and not merely A_1 or A_2 , remained available to be re-derived from.

7.9 What This Definition Commits Us To

It is worth closing this chapter with an explicit account of exactly what has been assumed, so that later chapters relaxing any one of these assumptions can be read as deliberate extensions rather than silent changes of subject. A substrate chain, as defined here, is linear rather than branching (relaxed in Chapter 9); finite rather than unbounded (left open in Appendix F); governed at each stage by a discard rule whose outcome, but not procedure, is constrained by definition 7.2 (with the resulting underdetermination flagged in remark 7.5 as a robustness condition the proofs of Chapter 10 must satisfy); irreversible in its losses except through the specific, costly mechanism of root recovery (proposition 7.9, itself subject to the bootstrapping problem of remark 7.10); and uncoordinated by default, in the precise sense given in definition 7.6, with coordination treated in Chapter 8 as something that must be added back in deliberately rather than assumed. Every one of these five com-

mitments is a modeling choice, not a discovered fact about the world, and the worked instance of section 7.8 was constructed specifically to make each commitment's consequences visible in a case small enough to check by hand before trusting the general theory built on top of it.

Chapter 8

Coordination, Its Absence, and Partial Remedies

Synopsis. This chapter investigates what happens when stages within a relay attempt to compensate for the absence of global coordination. Augmentation, provenance, priors, adjudication, and revision are reinterpreted not as an unrelated list of techniques but as distinct strategies for partially coordinating otherwise independent repair economies, each addressing a different failure mode: augmentation where hidden state is enumerable, provenance where loss can at least be recorded, priors where a consumer can plausibly fill a benign gap, adjudication where authority can terminate ambiguity without recovering information, and revision where the relay itself must be replaced. None fully solves the coordination problem; each narrows a specific failure mode.

8.1 Why Full Coordination Is Usually Unavailable

A field-recording engineer, the vendor of a transcription model, and a human editor who later summarizes the transcript typically share no protocol, no common budget, and often no direct communication at all. definition 7.6's uncoordinated case is not a pathological edge case to be assumed away; for most substrate chains of practical interest, it is the default condition. This chapter asks what a stage *can* do to compensate for that default, given that Chapter 6 already showed (proposition 6.1) that no extension of Chapter 5's single-observer apparatus is available to fall back on.

Definition 8.1 (Coordination Mechanism). A coordination mechanism is any deliberate modification to definition 7.6's default independence condition, permitting some stage or external party access to information — a loss ledger, a hidden-state enumeration, an authoritative ruling, or the raw root object itself — that a fully uncoordinated relay would not provide.

Five such mechanisms recur across the domains treated in Part IV, and it is worth resisting the temptation to treat them as an unrelated list of ad hoc devices collected because they happen to be useful. Each addresses a specific, identifiable gap in definition 7.6's independence condition, and organizing them by which gap they close is more informative than organizing them alphabetically or by domain.

8.2 Augmentation

Augmentation adds an enumerated hidden-state set H directly to a rendering, pre-committing budget to distinctions a stage's own task does not itself require, on the strength of knowing in advance that some later consumer will. By proposition 4.10,

this strategy is viable only in closed hidden-state domains (definition 4.8): Chapter 14 develops the case in full, where H — castling rights, en passant status, repetition count — is finite and known at the time a position is recorded, so that funding it requires no prediction about which future task will need it, only the (comparatively modest) foresight that *some* task among a known, finite list might. Chapter 3 (proposition 3.32) identifies augmentation as an instance of Distinguishability Geometry’s embedding operation, which supplies a more general justification for why it cannot hurt sufficiency than the bespoke argument given here and in proposition 4.9.

Remark 8.2 (Augmentation Does Not Solve the Open Case). Augmentation is unavailable, not merely difficult, in open hidden-state domains (definition 4.8): there is no finite H to pre-commit budget to, since proposition 4.10 shows no finite augmentation closes an open domain’s hidden-state space. Painting (Chapter 15) is not a case where augmentation would help if only more budget were available; it is a case where augmentation, as a strategy, does not apply regardless of budget.

8.3 Provenance and Loss-Ledger Sharing

Definition 8.3 (Provenance-Coordinated Relay). A substrate chain is provenance-coordinated if definition 7.6’s independence condition is relaxed to permit stage i ’s transition B_i to be selected with reference to the loss ledgers $L_0, \dots, L_{i-1}$ of every earlier stage, while still forbidding reference to any later stage’s budget or ledger.

This is the weakest relaxation of definition 7.6 that still qualifies as coordination at all: it requires no advance agreement between stages, only that each ledger be produced and made accessible to whatever comes after it, and it preserves the chain’s directionality — a stage still cannot see the future, only more of the past than definition 7.6 otherwise allows.

Proposition 8.4 (Provenance Sharing Does Not Restore Sufficiency Alone). *A provenance-coordinated relay (definition 8.3) can still exhibit the failure mode of theorem 10.2.*

Proof Sketch. Reading $L_0, \dots, L_{i-1}$ tells stage i , and eventually a downstream consumer, *what* was discarded and under what budget — but not what a future, unknown τ^* will require, since proposition 4.7 applies to the ledger’s reader exactly as it applies to the stage that produced the ledger. Knowing that d^* was dropped at stage 0 does not, by itself, supply d^* ; recovering it requires either proposition 7.9’s root access, which definition 8.3 does not grant, or advance knowledge of τ^* , which proposition 4.7 rules out in general. Provenance sharing converts an undetectable insufficiency into a diagnosable one; it does not convert it into an avoided one.

proposition 8.4 is why Chapter 7 described the budget-aware loss ledger (definition 7.13) as a coordination device rather than a solution: it is necessary for a downstream party to *diagnose* what happened, which matters a great deal for the reasons Chapter 21 develops, but it is not sufficient to *prevent* what happened, which requires the stronger and costlier machinery of root preservation.

8.4 Priors

A consumer-side substitute for coordination: when a relay supplies no interference information, the consumer supplies plausible reconstructions of the missing distinction from general knowledge. Painting (chapter 15) is the paradigm case, and it is available precisely because, and only because, the discarded distinction’s absence is benign (definition 2.7) relative to the consumer’s task: the correctness set $\mathcal{C}_\tau(e)$ of definition 4.4 is large enough that a plausible guess, rather than the exact discarded distinction, is sufficient to land inside it. Priors are unavailable, or worse than unavailable, exactly where

definition 2.8's conditions hold: a confidently wrong prior substituted for an undisclosed loss is corruption compounded by false confidence, not a benign completion.

8.5 Adjudication

Proposition 8.5 (Adjudication Terminates Ambiguity). *Let $\mathcal{J}$ be an authority external to a substrate chain. Adjudication by $\mathcal{J}$ produces a unique continuation even when the distinguishability information available in the chain is incomplete.*

Proof Sketch. Under adjudication, the continuation is selected by an authoritative decision procedure rather than inferred from available evidence. The underlying information deficit — the discarded distinctions — is unaffected by the ruling; what changes is that ambiguity is terminated procedurally rather than resolved informationally.

It is worth stating precisely, in the vocabulary of Chapter 4, what adjudication actually does to definition 4.4's correctness set. Adjudication does not discover which continuation truly belongs to $\mathcal{C}_\tau(e)$; it substitutes $\mathcal{J}$'s ruling as the operative correctness set going forward, regardless of whether that ruling coincides with whatever the undiscoverable true $\mathcal{C}_\tau(e)$ would have contained. Chapter 18 treats this substitution as the defining feature of legal continuation under irreducible uncertainty: courts do not typically claim to have recovered the parties' true intent so much as to have settled, bindingly, what will be treated as though it were. Adjudication is, in this taxonomy, the only mechanism that terminates ambiguity without requiring any additional distinguishability information to be recovered or supplied. It trades informational completeness for a binding decision.

8.5.1 Human Institutions as Adjudicators

Courts are the paradigm case, but far from the only one, and it is worth naming a few others so that adjudication does not read as a narrowly legal mechanism dressed in general vocabulary. Peer review adjudicates whether a manuscript’s claims are adequately supported when reviewers cannot fully recover the researcher’s actual experimental process, substituting a binding accept-or-reject decision for the unrecoverable question of whether the work is, in some deeper sense, correct. Standards committees adjudicate which of several competing technical proposals becomes the specification, terminating a design ambiguity that further technical argument alone might never resolve, since the disagreement is often about priorities rather than facts. Editorial boards for critical editions of ancient or damaged texts adjudicate which reading enters the authoritative version when manuscript branches disagree (proposition D.1, Appendix D), knowingly binding future readers to a decision rather than to a recovered certainty. In each case, what makes the mechanism adjudication rather than augmentation, revision, or priors is the same structural fact proposition 8.5 identifies: the deciding body’s ruling becomes the operative correctness set by authority, not by having closed the information gap that made a ruling necessary in the first place.

8.6 Revision

Definition 8.6 (Relay Revision). Wholesale replacement of an entire substrate chain $A_0 \rightarrow \dots \rightarrow A_n$ by a new chain built from new production, rather than re-derived from the original chain’s root object.

Revision is easy to confuse with root recovery (proposition 7.9), and it is worth being precise about the difference. Root recovery re-derives a new downstream rendering A'_i from

the *same* preserved A_0 , under a revised discard policy; it presupposes A_0 survived and is available. Revision, as defined here, does not presuppose this: it discards the entire existing chain, root included, and begins again from new evidence, a new theory, or a new production process entirely. Chapter 19 develops this case in full: a superseded scientific theory is not re-derived from some preserved “root” observation set in the way proposition 7.9 re-derives a rendering; the theory itself, and often the observational framework that produced the evidence for it, is replaced.

8.7 Coordination Versus Pooling: A Boundary Case

The five mechanisms above all operate *within* the assumption that stages remain separate — each retains its own local budget, and coordination consists of granting limited, structured access across that separation (a ledger here, an authority there), not erasing it. Appendix C’s demonstration that pooling the witness’s stage budgets into one shared total can restore sufficiency (proposition C.1) is a more radical move than any mechanism in this chapter’s taxonomy: pooling does not coordinate separate local economies, it eliminates the separation itself, collapsing the relay back into the single-observer setting Chapter 5 already covers. None of augmentation, provenance-sharing, priors, adjudication, or revision does this; each leaves definition 7.1’s stage structure intact and works within it. This is worth flagging explicitly because pooling is, in a precise sense, strictly more powerful than anything on this chapter’s list — it is what makes theorem 5.7 available again — and its absence from the taxonomy is not an oversight but a reflection of how rarely it is actually achievable: pooling requires exactly the kind of unified, advance-negotiated authority over an entire chain’s resources that Section 8.2’s discussion of augmentation

already identified as the exception rather than the rule.

8.8 The Limits of the Taxonomy

Proposition 8.7 (No Coordination Mechanism Restores Universal Sufficiency). *No coordination mechanism drawn from Sections 8.2–8.6 can guarantee sufficiency of a substrate chain over an open task space T .*

Proof Sketch. This follows directly from proposition 4.7: each coordination mechanism above operates by preserving, recovering, or authoritatively resolving *some additional* set of distinctions, but T 's openness means a task τ^* can always be constructed requiring a distinction outside whatever finite additional set any given mechanism supplies. Coordination narrows the gap identified in principle 6.2; it does not close it. This result was labeled a Theorem in an earlier draft; the label has been corrected to Proposition here, consistent with the monograph's epistemic-labeling discipline, since the argument is a direct application of proposition 4.7 rather than an independent derivation.

Chapter 10 is, in effect, the sharper and more constructive version of proposition 8.7: it does not merely state that no mechanism guarantees universal sufficiency, but exhibits the mechanism by which locally rational, uncoordinated stages compose into global insufficiency in the first place, and Appendix A exhibits it with a construction robust to the choice of local discard rule.

8.9 Coordination Versus Witness Reconstruction

A terminological note before this final connection, since the word “witness” is already doing separate work in this mono-

8.9. COORDINATION VERSUS WITNESS RECONSTRUCTION

graph: a *witness* in Appendix A's sense is a constructed example proving an existential claim (the canonical witness of construction A.5); a *witness operator* in §3.8's reported sense is something else entirely, an object that reconstructs a repair capability from historical structure. The two senses do not interact, and this section concerns only the second.

Every mechanism in this chapter's taxonomy coordinates *distinctions* across separated stages – augmentation pre-funds them, provenance records their loss, priors guess at them, adjudication rules on them, revision replaces the chain that lost them. None of them coordinates the *capacity to coordinate* itself: nothing in this chapter's apparatus asks what happens when the mechanism doing the coordinating – the shared ledger, the standing court, the augmentation policy – is itself lost. That is precisely the question §3.8 reports The Geometry of Witnesses as asking one level up: not which distinctions survive, but which repair capacities survive, and through what historical structure they could be reconstructed if they did not. Coordination, as this chapter defines it, presupposes a stable coordinating party; witness reconstruction, at its reported weaker tier, is what would be needed if that presupposition failed. This monograph does not develop that further layer, since §3.8's apparatus remains unverified, but the shape of the gap is worth naming: this chapter's taxonomy is complete for coordinating distinctions and silent about coordinating coordinators. Appendix F records this gap explicitly rather than leave it implicit in this paragraph alone.

CHAPTER 8. COORDINATION, ITS ABSENCE, AND PARTIAL REMEDIES

Chapter 9

Fan-Out: Parallel Budget Relays and Cross-Branch Disagreement

Synopsis. The framework is extended here from linear chains to branching structures, where a single source generates multiple downstream renderings optimized for different purposes. Branch agreement and silent disagreement are introduced to describe situations in which branches appear compatible while relying on incompatible assumptions about discarded information. Confidence and quality estimates are shown to be renderings in their own right, subject to the same limitations as the artifacts they describe, motivating a trust boundary as the practical stopping point that prevents an infinite regress in evaluating confidence and provenance systems. The chapter closes by asking what happens when branches are brought back together rather than left to diverge: reconciliation can recover distinctions no single branch preserved

alone, though it does not resolve genuine conflicts between branches that disagree about the same distinction rather than merely discarding different ones.

9.1 From Chains to Graphs

Definition 9.1 (Fan-Out Relay). A fan-out relay is a directed graph of renderings with a single source A_0 and two or more independent successor branches $A_0 \xrightarrow{B^{(1)}} A^{(1)}$, $A_0 \xrightarrow{B^{(2)}} A^{(2)}$, ..., each governed by its own local budget triple and each satisfying definition 7.6’s independence condition with respect to every other branch: branch k ’s transition is selected without reference to any other branch’s budget or ledger.

definition 9.1 generalizes definition 7.1 directly: audio rendered in parallel into a transcript, an acoustic-feature analysis, and an emotion analysis is the paradigm case, and each branch may itself continue as a linear substrate chain of arbitrary length beyond the point of divergence. What is new here is not the branching structure by itself, but what it makes possible: two renderings that both trace back to the same source, and that a downstream party might reasonably expect to agree with each other, can instead diverge in ways that neither branch’s own local admissibility (definition 10.1) does anything to prevent.

Example 9.2 (Version Control as a Fan-Out Relay). A version-control branch structure is a fan-out relay in essentially undisguised form: a shared commit A_0 gives rise to independent branches, each accumulating its own history of transitions governed by whichever contributor is working on it, with no requirement that any branch consult another’s changes before committing its own. Two branches independently editing the same function for different purposes – one optimizing

it for speed, one adding a safety check – are exactly definition 9.1’s two-branch case, and a downstream party merging both branches later is performing, in effect, an ad hoc instance of the reconciliation problem taken up in §9.4.

9.2 Branch Agreement and Silent Disagreement

Definition 9.3 (Branch Agreement). Two renderings $A^{(i)}$, $A^{(j)}$ derived from a common ancestor are in agreement on a distinction d if neither branch’s discard set contradicts the other’s claims about d .

Definition 9.4 (Silent Disagreement). Branches disagree silently when each branch’s confidence or quality profile Q fails to flag that another branch depended on a distinction that the first branch discarded or rendered unstable.

The definitions above are easy to accept in the abstract and easy to under-examine in practice, so this section works a single construction in full, in the same style as Appendix A’s canonical witness, rather than resting on a description of the transcript/acoustic case alone.

9.2.1 A Minimal Two-Branch Witness

Construction 9.5 (Two-Branch Witness). Let A_0 carry two distinctions: d , the resolved identity of an ambiguous phoneme, with cost $C(d) = 2$; and q_d , a meta-distinction recording whether the source evidence for d is stable or unstable, with cost $C(q_d) = 1$. Branch X (a transcript) has task τ_X requiring exactly $\{d\}$, with budget $\beta_X = 2$. Branch Y (an acoustic-stability analysis) has task τ_Y requiring exactly $\{q_d\}$, with budget $\beta_Y = 1$.

Proposition 9.6 (The Witness Produces Silent Disagreement). *In construction 9.5, branch X is locally admissible with respect to τ_X and discards q_d ; branch Y is locally admissible with respect to τ_Y and discards d ; and a consumer with access only to branch X cannot detect that q_d – and hence the instability branch Y would have reported – was ever at issue.*

Proof. For branch X : the required distinction $\{d\}$ has cost $R_X = C(d) = 2 = \beta_X$ exactly. The only non-required distinction present, q_d , has cost $C(q_d) = 1 > \beta_X - R_X = 0$. By lemma A.4, q_d is dropped under any construction satisfying definition 7.2 with d ranked above q_d , so X is locally admissible (definition 10.1) and $D_X = \{q_d\}$.

For branch Y : symmetric. The required distinction $\{q_d\}$ has cost $R_Y = C(q_d) = 1 = \beta_Y$ exactly, and the non-required d has cost $C(d) = 2 > \beta_Y - R_Y = 0$, so by lemma A.4, d is dropped under any construction, Y is locally admissible, and $D_Y = \{d\}$.

By proposition 9.8, X 's confidence profile Q_X is itself a rendering, produced from whatever X retains; since $q_d \notin A^{(X)}$, Q_X has no distinction available from which to construct an instability flag, and X 's output presents d 's resolved value without qualification. A consumer relying on branch X alone therefore receives no signal — not a low-confidence flag, not a declared gap, nothing — indicating that q_d was ever a live consideration, even though branch Y , derived from the identical source A_0 , would have reported it as unstable. This is exactly definition 9.4's condition: Q_X fails to flag that another branch's finding (instability, via q_d) bears on distinction d , which X itself depended on. $\square$

This construction is deliberately built to reuse Appendix A's exhaustion lemma rather than argue the point freshly, and the reuse is informative in its own right: the same mechanism that forces a construction-independent discard within a single linear chain (lemma A.4) forces construction-independent,

asymmetric discard across two branches of a fan-out, and it is precisely this asymmetry — X drops what Y keeps, and vice versa — that produces silent disagreement rather than mere independent loss.

Proposition 9.7 (Silent Disagreement Exists in General). *Given two branches produced independently (definition 9.1) from a common source, there exist source distinctions whose instability is reported by one branch and concealed by the other.*

Proof Sketch. construction 9.5 together with proposition 9.6 constitutes an existence proof: at least one pair of branches, with at least one such distinction, is exhibited explicitly and shown to satisfy definition 9.4 under a construction-independent argument. This result was labeled a Theorem in an earlier draft; the label has been corrected to Proposition, consistent with the monograph’s epistemic-labeling discipline, since an existence claim witnessed by one explicit construction is not of the same weight as theorem 10.2’s general composition result.

9.3 The Regress Inside the Confidence Profile

Proposition 9.8 (Confidence Profiles Are Renderings). *A confidence or quality profile Q attached to a rendering A is itself produced by a projection, and therefore has its own discard set $D(Q)$.*

Proof Sketch. Confidence estimation maps source evidence – acoustic likelihoods, model logits, annotator judgments – into a finite label or score. Any such finite mapping collapses distinctions in the sense of definition 2.2; Q therefore satisfies the definition of a rendering and inherits a discard set like any other.

proposition 9.8 produces a regress: trusting Q requires trusting the process that produced Q , which has its own confidence measure Q' , and so on. This regress does not resolve

within the formalism; it is cut off by convention rather than by argument, and it is worth being explicit about the cost of extending it even one level further. Verifying Q against a further Q' is itself a repair decision in the sense of Chapter 5: it consumes budget that could otherwise be spent on the object-level distinctions Q was meant to qualify, and proposition 6.1's argument applies just as well to a chain of nested confidence measures as to a chain of substantive renderings. Real systems stop extending this chain not because the regress terminates on its own but because someone decides, by convention, where it is no longer worth the expenditure.

Definition 9.9 (Trust Boundary). A designated layer – for instance, a fixed, versioned confidence model – beyond which a relay does not recursively question its own loss-accounting. The trust boundary is a declared convention, not a theorem; proposition 9.8's regress is halted by stipulation at this layer, and an auditor of the relay is, in effect, trusting the boundary itself rather than anything beyond it.

Definition 9.10 (Boundary-Relative Sufficiency). A rendering is boundary-sufficient for a task if it is sufficient (definition 4.5) conditional on every confidence and quality profile at or beyond the declared trust boundary (definition 9.9) being accepted without further verification.

definition 9.10 names what practitioners actually rely on, as distinct from the stronger and generally unattainable notion of unconditional sufficiency: a system is almost never verified all the way down to its source distinctions, and boundary-sufficiency makes explicit exactly how much trust that verification is quietly resting on.

9.4 Reconciliation

Everything so far in this chapter has treated branches as diverging and stopping there. Branches are also, routinely, brought back together, and it is worth asking what a fan-out relay's apparatus says about that.

Definition 9.11 (Reconciliation). Given branches $A^{(1)}$, $A^{(2)}$ derived independently from a common source A_0 , a reconciliation is a rendering A^{merge} derived from both, attempting $D(A^{\text{merge}}) = D(A^{(1)}) \cap D(A^{(2)})$: only what *both* branches lost is treated as genuinely lost, and anything either branch preserved is carried into the merge.

Proposition 9.12 (Reconciliation Can Recover What No Branch Preserved Alone). *If d is preserved by $A^{(1)}$ but not $A^{(2)}$, and d' is preserved by $A^{(2)}$ but not $A^{(1)}$, a reconciliation combining both branches can preserve both d and d' , which neither branch achieves alone.*

Proof Sketch. By definition 9.11, A^{merge} 's discard set is the *intersection* of the two branches' discard sets, not the union of their preserved sets alone but a genuine combination: $d \notin D(A^{(1)})$ and $d' \notin D(A^{(2)})$ together give $d, d' \notin D(A^{(1)}) \cap D(A^{(2)}) = D(A^{\text{merge}})$, so both survive the merge even though each was absent from one of the two source branches.

This is the positive counterpart to definition 9.4's negative result, and it is worth being precise about why the two do not contradict each other. Silent disagreement is what happens when a downstream consumer relies on *one* branch alone; reconciliation is available only to a party with access to *both*, and proposition 9.12 presupposes exactly the access Chapter 9's worked witness (construction 9.5) showed a single-branch consumer lacks.

Remark 9.13 (Genuine Conflicts Are Not Solved by Reconciliation). definition 9.11 presupposes the two branches' discard

sets are the only source of disagreement between them. This fails whenever both branches preserve the *same* distinction d but disagree about its value – branch X from example 9.2 optimizing a function one way, branch Y optimizing it incompatibly – since $d \notin D(A^{(1)}) \cup D(A^{(2)})$ in this case, and definition 9.11’s intersection rule gives no guidance about which branch’s value of d the merge should adopt. This is a merge conflict in the ordinary sense, and definition 9.11 does not resolve it automatically; resolving it requires falling back on one of Chapter 8’s mechanisms – adjudication by whichever party owns the merge, or a prior favoring one branch’s convention over the other’s – rather than treating reconciliation as a fully general, conflict-free operation. Reconciliation recovers what non-overlapping loss destroyed; it does not adjudicate what overlapping, incompatible preservation produced. Whether the boundary between these two cases can be characterized in advance is recorded as open in Appendix F rather than left implicit here.

Chapter 10

The Composition Failure Theorem

Synopsis. This chapter presents the monograph's central new result. The Composition Failure Theorem demonstrates that a sequence of locally rational transformations can collectively produce a globally insufficient outcome, with no stage behaving negligently, incompetently, or irrationally: failure emerges because each stage optimizes for its own local task while future tasks remain unknown. This formalizes a pattern familiar from archival systems, institutions, scientific communication, and information-processing pipelines, in which a distinction that seemed irrelevant at one stage becomes essential later.

(New to this monograph.)

10.1 Making Local Admissibility Precise

Before the theorem can be stated, the phrase it turns on — a transition being “locally admissible with respect to its own stage’s task” — needs a definition sharper than definition 7.2’s general schema alone provides. definition 7.2 says a distinction is dropped when funding it exceeds budget or when it interferes too strongly with a higher-priority distinction; it does not say how priority itself is supposed to track a stage’s task. The following definition closes that gap, and everything downstream — the theorem, its corollaries, and the proof of Appendix A — depends on exactly this closure.

Definition 10.1 (Locally Admissible Transition). A transition B_i is locally admissible with respect to τ_i if its discard set D_i is produced by a construction satisfying definition 7.2 in which every distinction required by τ_i (definition 2.7) is ranked strictly above every distinction not required by τ_i .

This definition is deliberately weak in one respect and strong in another. It is weak in that it does not choose between construction 7.3 and construction 7.4, or any other construction satisfying definition 7.2: any of them counts as locally admissible provided the priority ordering they operate on is correctly aligned with τ_i . It is strong in that this alignment is not optional — a transition that funds a distinction τ_i does not need ahead of one it does need fails to be locally admissible, regardless of how sophisticated the underlying discard construction is. A transition satisfying definition 10.1 cannot be accused of irrationality relative to τ_i : it funds everything τ_i needs, in full, before spending anything on distinctions τ_i does not need. This is exactly the standard the theorem needs every stage to meet, since the entire point of the result is that failure can occur even when no stage falls short of it.

10.2 Statement

Theorem 10.2 (Composition Failure). *Let $A_0 \xrightarrow{B_0} \dots \xrightarrow{B_{n-1}} A_n$ be a budget relay (definition 7.6) in which each transition B_i is locally admissible (definition 10.1) with respect to its own stage's task τ_i . Then there exist open task spaces T (proposition 4.7) containing some $\tau^* \notin \{\tau_0, \dots, \tau_{n-1}\}$ such that A_n is insufficient (definition 4.5) for τ^* , even though no individual transition was locally irrational relative to its own τ_i .*

Each clause of this statement is load-bearing, and it is worth checking what happens if any one is dropped. Without definition 7.6's independence condition, the chain reduces to a single observer's economy and theorem 5.7 governs it instead, as Appendix C shows concretely. Without local admissibility, the theorem would be uninteresting: of course an incompetently-run relay can end up insufficient. Without T being open, τ^* could in principle have been anticipated and funded for, and the failure would trace back to ordinary negligence rather than to the structure of uncoordinated composition itself. The theorem's content is precisely that none of these outs are available: the relay is uncoordinated by definition, every stage did what it was supposed to do, and the task that breaks it was not available to be planned for.

10.2.1 Why No Stage Is Wrong

It is worth stating this as plainly as possible, since it is the single most commonly missed reading of the theorem. "No stage is wrong" does not mean the outcome is somehow acceptable, or that no one should have done anything differently in hindsight. It means that identifying which stage to blame is a type error: blame presupposes a standard the blamed party fell short of, and every stage in theorem 10.2's hypothesis met its own standard exactly. Stage i 's standard is τ_i ; stage i 's transition B_i

is locally admissible with respect to τ_i by hypothesis; there is no further standard stage i could have been held to without already knowing τ^* , which proposition 4.7 rules out. Asking “which stage failed” is asking a question with a false presupposition, in the same way that asking which digit of a correctly computed sum is wrong is a false presupposition when the sum is correct. The theorem’s discomfort, for readers who feel there ought to be someone to hold responsible, is not a flaw in the theorem; it is the theorem’s content, restated as a feeling.

10.2.2 Relation to Arrow-Type Impossibility Results

Readers familiar with social choice theory may notice a family resemblance to Arrow’s impossibility theorem, which shows that no voting rule aggregating individual preferences can simultaneously satisfy a short list of independently reasonable fairness conditions. The resemblance is worth naming and then bounding carefully, since the two results are not the same theorem in different clothing. Both share the structure of showing that a collection of locally reasonable requirements — individually rational voters, individually rational relay stages — cannot be jointly satisfied by any aggregation procedure meeting a further, seemingly modest condition (independence of irrelevant alternatives, in Arrow’s case; uncoordinated composition, in this monograph’s). Where they differ is in what is being aggregated and against what standard: Arrow’s theorem is about preference aggregation under a fixed, closed set of alternatives and voters; theorem 10.2 is about distinction preservation across an open task space, and its failure mode is diachronic in a way Arrow’s is not, since it depends on a later task appearing that no stage could have voted on in the first place because it did not yet exist. The comparison is offered as an aid to intuition for readers who already know Arrow’s theorem, not as a claim that one result specializes to or generalizes the other; no such reduction is attempted here.

10.3 Proof Strategy

The full proof is given in Appendix A; this section previews its shape so the construction does not appear unmotivated when read there.

Proof Sketch. The proof constructs an explicit witness relay using the five-distinction instance introduced in section 7.8 and reused in section 5.5, together with a family of *recovery tasks* τ_d — one for each named distinction d , asking only whether d holds in the source (definition A.1) — which serves as the open task space T . A stage whose required distinctions exactly, or near-exactly, exhaust its local budget is shown (lemma A.4) to drop every non-required distinction *regardless of which construction satisfying definition 7.2 is used*, which is what allows the witness to be locally admissible in the strong sense of definition 10.1 without leaving room for a more clever discard policy to have rescued the dropped distinction. Some distinction d^* — concretely, d_5 in the worked witness — is required by neither stage’s task and is dropped at the earliest stage as a direct consequence. By proposition 7.12, d^* remains unavailable at every later stage absent root recovery (proposition 7.9), which the witness does not invoke. Taking $\tau^* = \tau_{d^*}$ then gives insufficiency directly, by lemma A.2: a rendering is sufficient for a recovery task exactly when it has not discarded the distinction in question, and A_n has.

This preview differs in one respect from an earlier, less precise sketch of the same argument, which described d^* as dropped because it “interferes with a distinction τ_i does require.” The actual mechanism proved in Appendix A is sharper and does not rely on interference at all: d^* is dropped because, after funding every required distinction in full, no budget capable of funding d^* remains — a conclusion that holds regardless of interference structure and regardless of which construction satisfying definition 7.2 produced the allocation. The

interference-based mechanism remains available in principle for other witnesses, but it is not what the canonical witness of this monograph uses, and the sketch above is corrected accordingly.

10.4 Corollaries

Corollary 10.3 (Local Improvement Does Not Guarantee Global Sufficiency). *Improving any single stage's local budget cannot, by itself, eliminate all instances of the failure mode exhibited in theorem 10.2.*

Proof Sketch. Verified directly against the witness in Appendix A: increasing β_1 cannot restore a distinction already absent from A_1 by the time stage 1 acts, since proposition 7.12 forbids recovery absent root access. Only an improvement at the specific stage where the loss occurred could have prevented it, and even that requires knowing, at that stage, that the eventually-relevant task would need the distinction in question – precisely the knowledge proposition 4.7 shows is generally unavailable.

Corollary 10.4 (No Free Lunch for Relays). *A budget relay with n uncoordinated stages has cumulative discard (definition 7.11) that is non-decreasing in n (proposition 7.12), realized with equality across arbitrarily many additional pass-through stages (proposition A.8); the coordination mechanisms of Chapter 8 can narrow, but not eliminate, the resulting insufficiency (proposition 8.7).*

Proof Sketch. The non-decrease is proposition 7.12 directly, and Appendix A shows this bound is achieved with equality rather than merely respected, since pass-through stages contribute no further discard. This corollary should not be read as a claim that cumulative discard *risk* grows in some stronger, quantitative sense across relays in general; that stronger claim is not established here and is recorded as open in Appendix F. What

is established is the existence half only: discard does not shrink, and can accumulate to the full extent this chapter's witness demonstrates.

10.5 What the Theorem Does Not Say

It is worth listing plainly what theorem 10.2 does not claim, since a result this general invites over-reading in several directions at once. It does not say that every budget relay fails; the homogeneous-task chains of proposition 17.3 (Chapter 17) are relays that provably do not, and closed hidden-state domains under augmentation (proposition 4.9) are a further case where sufficiency is achieved outright. It does not say that coordination is worthless; Chapter 8's mechanisms measurably narrow the failure mode even though proposition 8.7 shows none of them close it entirely. It does not identify a rate at which insufficiency grows, only that it does not shrink (corollary 10.4), and the sharper, quantitative version of that claim is explicitly left open rather than implied. It does not say anything at all about whether a given real institution, pipeline, or memory system is currently suffering from this failure mode; the theorem establishes that the failure mode is possible under stated hypotheses, not that it is occurring in any particular case, which would require checking those hypotheses against the case in question. And it does not require malice, incompetence, or even error at any stage, which is the point the preceding two sections existed to make and which this list exists to keep from being forgotten once the theorem is put to use.

10.6 Relation to the Redistribution Theorem

theorem 10.2 is not a special case of theorem 5.7, and is not derivable from it without the budget-relay machinery of Chapter 7. The Redistribution Theorem concerns simultaneous com-

petition for one observer's budget; the Composition Failure Theorem concerns a single distinction's fate across a sequence of independent, non-communicating budgets. Appendix C develops this contrast concretely rather than only structurally: pooling the witness's two stage budgets into a single shared total is shown there (proposition C.1) not merely to make theorem 5.7 applicable again, but to actually succeed at funding the distinction the separated relay was forced to drop, using the identical total nominal resource. The insufficiency this chapter's theorem exhibits is therefore a consequence specifically of definition 7.6's prohibition on transferring unspent capacity across stages, not of any absolute scarcity of resource – confirming that the two theorems occupy genuinely disjoint settings rather than one subsuming the other.

Part III

Modes of Access: Preservation and Reconstitution

Chapter 11

Preservation Mode, Reconstitution Mode, and the Fidelity Continuum

Synopsis. This chapter revisits the traditional distinction between direct access and reconstruction and replaces it with a single fidelity continuum. Every consumer interacts with some rendering, and every rendering preserves only part of the original distinction structure, so preservation becomes a limiting case of reconstitution rather than a separate ontology. Fidelity is introduced as a measure of how closely a rendering preserves an original object's admissible continuation space, and the chapter argues that reconstitution does not escape the Composition Failure Theorem: a reconstructed trace is simply a new relay stage, subject to the same constraints as any other.

11.1 The Naive Binary

An initial formulation of access modes distinguishes preservation mode – the consumer has direct causal access to the artifact itself – from reconstitution mode – the consumer has access only to traces from which structure must be regenerated. This chapter shows that the binary does not survive scrutiny and replaces it with a single continuous variable.

Remark 11.1. Nearly every case initially classified as preservation is in fact mediated by some rendering: a photocopy, a scan, a decoded audio file. A photograph of a painting is not the painting; a scan of a manuscript is not the manuscript; a decoded audio stream is not the acoustic event it encodes. Such cases are already points on the loss spectrum of definition 2.2, not a qualitatively separate category.

11.2 Fidelity, Defined

Definition 11.2 (Fidelity). For a rendering r of an object e and a task τ ,

$$F(r, \tau) = \frac{|\mathcal{A}_\tau(r) \cap \mathcal{A}_\tau(e)|}{|\mathcal{A}_\tau(e)|},$$

a measure of how closely the admissible continuation set available from r approximates the admissible continuation set available from the original object e itself, for the task τ in question. Per remark 2.1, this is stated only for the finite or countable case.

Remark 11.3 (Relation to Transport Stability). Fidelity is not the only available measure of how well structure survives a transformation. Distinguishability Geometry’s Transport Stability theorem bounds a different but related quantity, and Chapter 3 (remark 3.36) states plainly that this monograph has not shown

one reduces to the other; the relationship is recorded as open rather than assumed.

Remark 11.4 (The Intersection Simplifies). By proposition 4.2, $\mathcal{A}_\tau(r) \subseteq \mathcal{A}_\tau(e)$ always, so the intersection in definition 11.2 is simply $\mathcal{A}_\tau(r)$ itself: $F(r, \tau) = |\mathcal{A}_\tau(r)|/|\mathcal{A}_\tau(e)|$. The intersection form is retained in the definition because it makes explicit, without relying on a separately proved proposition, exactly what is being compared; the simplification is used freely from here on.

Remark 11.5 (A Standing Assumption: Ground-Truth Sufficiency). Throughout this chapter it is assumed that $\mathcal{A}_\tau(e) \subseteq \mathcal{C}_\tau(e)$: the true object e is itself always sufficient for any task evaluated against it. This is close to definitional given how $\mathcal{C}_\tau(e)$ was fixed relative to e in definition 4.4, but it is worth stating openly, since corollary 11.7 below depends on it.

Proposition 11.6 (Preservation as a Limit Case). *“Preservation mode” is the limit of reconstitution mode as $F(r, \tau) \rightarrow 1$ for the tasks τ in question; it is not a distinct ontological category but a point on the continuum defined by definition 11.2.*

Proof Sketch. As $F(r, \tau) \rightarrow 1$, $\mathcal{A}_\tau(r)$ converges to $\mathcal{A}_\tau(e)$: the continuations available from the rendering become indistinguishable, for the purposes of τ , from those available from the original. Reconstitution from a sufficiently high-fidelity trace is then operationally indistinguishable from direct access, which is what “preservation” was intended to capture.

Corollary 11.7 (Fidelity One Implies Sufficiency). *If $F(r, \tau) = 1$, then r is sufficient for τ (definition 4.5).*

Proof Sketch. $F(r, \tau) = 1$ gives $\mathcal{A}_\tau(r) = \mathcal{A}_\tau(e)$ by remark 11.4, and $\mathcal{A}_\tau(e) \subseteq \mathcal{C}_\tau(e)$ by remark 11.5, so $\mathcal{A}_\tau(r) \subseteq \mathcal{C}_\tau(e)$, which is exactly definition 4.5.

This corollary sharpens proposition 11.6’s informal “operationally indistinguishable” into a precise claim: the fidelity

continuum's upper limit is not merely subjectively hard to tell apart from direct access, it is provably sufficient in the sense Chapter 4 already uses throughout the rest of the monograph.

Proposition 11.8 (Benign Loss Preserves Perfect Fidelity). *If every distinction in $D(r)$ is benign relative to τ (definition 2.7), then $F(r, \tau) = 1$.*

Proof Sketch. By the same argument as proposition 4.6: no distinction discarded by r is required to determine any continuation's membership in $\mathcal{A}_\tau(e)$, so no continuation reachable from e becomes unreachable from r for the purposes of τ , giving $\mathcal{A}_\tau(r) = \mathcal{A}_\tau(e)$ and hence $F(r, \tau) = 1$ by remark 11.4.

proposition 11.8 is the fidelity-continuum counterpart of proposition 4.6, and the two propositions together are why this chapter's continuous measure and Chapter 4's binary sufficiency criterion cohere rather than compete: perfect fidelity for a task and sufficiency for that task are the same claim, viewed through cardinality in one case and through set containment in the other.

11.3 The Hard Test Case: A Chess Board in Front of You

A chess board physically present to a player is the strongest available candidate for genuine, unmediated preservation. Yet perceiving the board is itself a rendering process – a visual system producing a percept from ambient light – and is therefore, by remark 11.1, already a point on the fidelity continuum rather than a case falling outside it. Whether *any* case achieves fidelity exactly equal to 1, or whether the entire category of strictly unmediated access is an idealization available only to formal or machine systems (a verifier checking a proof directly against explicit axioms, per Chapter 17) and never to embodied perception, is left open here.

Open Question 11.9. Does any instance of “direct access” to an artifact involve strictly zero mediation, or is unmediated access an idealization that applies cleanly only to formal/machine verification and never to embodied perception? Resolving this requires engaging directly with the question raised in full in Chapter 12.

11.4 Recursive Insufficiency

Proposition 11.10 (Recursive Insufficiency). *If A_i is a trace consumed in reconstitution mode, the act of reconstitution – schema plus trace producing a usable object – is itself a transition B_i in a budget relay in the sense of definition 7.1, and is therefore subject to theorem 10.2.*

Proof Sketch. Reconstitution takes a trace A_i and a schema and produces a usable object A_{i+1} for continuation. This is not merely analogous to a relay transition; it can be given the same triple structure directly. The schema’s capacity to correctly fill a given gap plays the role of C_i (some gaps are cheaply and reliably filled by strong priors, others are not); competition among candidate fillings for the same gap, or among several gaps drawing on the same limited attention or working memory, plays the role of I_i ; and whatever bounded capacity the reconstitution process has available – time, attention, cognitive resources – plays the role of β_i . Under this identification, $A_i \rightarrow A_{i+1}$ satisfies definition 7.1 exactly, not by loose analogy, and lemma A.4’s mechanism applies to reconstitution precisely as it applies to any other transition: a required feature of the reconstructed object that costs more than the reconstitution process’s available capacity is not recovered, regardless of how the schema is implemented.

This corrects a claim made informally in the research program that generated this monograph, in which reconstitution appeared briefly to “escape” the insufficiency problem

CHAPTER 11. PRESERVATION MODE, RECONSTITUTION MODE, AND THE FIDELITY CONTINUUM

by changing register from information recovery to regeneration. proposition 11.10 shows this is not the case, and shows it by identification rather than metaphor: reconstitution relocates the insufficiency problem to the trace and the schema, and theorem 10.2 applies to that relocation exactly as it applies to any other stage of a relay, because it *is* another stage of a relay.

Chapter 12

Memory as a Limiting Case: MEM|8 and the Ontology of Traces

Synopsis. This chapter treats memory as both an example and a challenge to the framework. Two readings of MEM|8 are set out: a narrow reading, under which memory is simply another reconstructive process operating over traces, and a broad reading, under which memory or process is ontologically prior to representation itself, implying that all access to artifacts is already reconstructive. The chapter does not attempt to resolve this dispute. It instead analyzes the consequences of each reading and identifies which parts of the framework remain stable regardless of the outcome, treating the fork as an explicit, acknowledged dependency rather than concealing it behind a premature resolution.

This chapter is a friction chapter, not a worked example, and is written to be read as such. It does not resolve the question

it raises; it states the question precisely enough that the rest of the monograph's dependence on one answer or the other is visible, and it closes with a table stating exactly which results survive regardless of the answer and which do not.

12.1 Two Readings

MEM|8's core claim — that memory is prior to representation rather than a degraded copy of it — admits at least two readings with very different consequences for this monograph, and the difference between them is worth setting out in full before either is evaluated.

Under the narrow reading, biological recall specifically is reconstructive: a trace-based regeneration process rather than retrieval of a stored copy. This reading is well supported by the eyewitness-memory literature and is compatible with treating memory as one more point on the fidelity continuum of Chapter 11, alongside photocopies, scans, and decoded audio. On this reading, memory is unusual among renderings only in degree — its fidelity tends to be low and its schema-driven gap-filling tends to be confident about its own accuracy in a way a photocopier is not — not in kind.

Under the broad reading, representation *in general* is a stabilization of prior memory or process, not the reverse: perception, recognition, and even the sense of directly encountering an external object are themselves outputs of an ongoing memory-like process, rather than a separate, more basic capacity that memory merely draws on afterward. Under this reading, no consumer — including a chess player looking at a physical board — ever has strictly unmediated access to an external artifact, since the very act of perceiving the board is already a memory-like construction rather than a transparent window onto it. This would undermine the very idea of a fixed root object (definition 7.8) for any embodied consumer, since a root

object's usefulness (proposition 7.9) depends on there being a stable thing to re-derive from, and the broad reading denies that anything external is ever encountered other than through the same reconstructive process that governs recall.

12.2 Considerations on Each Side

It would be premature, and beyond this monograph's scope, to adjudicate between these readings, but it is worth surveying briefly what would need to be true for each to hold, so that the fork is a substantive question rather than an empty one. The narrow reading is the more conservative position and requires only what is already well established: reconstructive-memory research shows that recall is schema-driven and error-prone in characteristic, well-documented ways, without requiring any claim about perception or representation more broadly. The broad reading is a stronger, more general claim, and its most natural allies are views in cognitive science that treat perception itself as a constructive or predictive process rather than a passive registration of external stimuli — the general family of ideas sometimes described as treating perception as a controlled or constrained hallucination, or as governed by predictive processing in which incoming signal is used to correct an ongoing internally generated model rather than read off directly. If some view in this family is correct, the boundary between "perceiving the board" and "remembering the board" may not be a boundary between two different kinds of access at all, but two points on one continuous process — which is exactly what the broad reading claims and what this monograph's own Part III argued for the boundary between preservation and reconstitution more narrowly (proposition 11.6). This is a genuine structural echo, and it is tempting to treat it as evidence for the broad reading; this monograph resists that temptation, since an echo between two frameworks is not the same as a

proof that one licenses the other, and the question of whether perception and memory share a single underlying mechanism is an empirical matter this book is not positioned to settle.

12.3 Consequences of the Broad Reading

If the broad reading is correct, the admissibility and distinguishability apparatus of Part I would need to be built on top of a process-first ontology, rather than assuming a stable artifact that a consumer either does or does not have access to. This monograph explicitly declines to resolve this question here. It is stated as the single largest open dependency of the entire framework.

Open Question 12.1. Does MEM|8's claim generalize beyond biological recall (the narrow reading) to representation as such (the broad reading)? If the broad reading holds, Part I's treatment of artifacts as stable objects available for direct or indirect access requires revision from the ground up, rather than the more modest correction made in proposition 11.6.

12.4 What Survives Regardless of the Fork

Both readings agree that a trace or a percept is a finite, lossy substrate; they disagree only about whether any *non*-trace substrate exists at all for an embodied consumer. This distinction is sharp enough to sort the monograph's results cleanly into those that depend on it and those that do not.

12.4. WHAT SURVIVES REGARDLESS OF THE FORK

Result	Narrow read- ing	Broad reading
proposition 11.10	holds	holds
theorem 10.2	holds	holds
proposition 17.1 (proofs)	holds	holds
proposition 11.6	holds as stated	upper limit never reached
open Question 11.9	resolved: no	open in stronger form
Ch. 14's board case	fidelity $\rightarrow$ 1 unavailable	fidelity $\rightarrow$ 1 un- available
Embodied cases, Part IV	conditional	structurally re- vised

The pattern in this table is not accidental. Results stated about *formal or machine* substrates — a proof checked by an algorithm against explicit axioms, a substrate chain defined over named distinctions without reference to how those distinctions come to be perceived — do not mention embodied access at all, and so cannot be disturbed by a dispute about what embodied access amounts to. theorem 10.2 in particular is stated entirely in terms of renderings, budgets, and discard sets; nothing in its proof (Appendix A) invokes a consumer's perceptual access to anything, so it is simply not the kind of claim the MEM|8 fork bears on. Results that *do* mention embodied access — most sharply, open Question 11.9 and every worked-domain discussion of a physical object sitting in front of an observer — are exactly the ones the fork can move. The table's purpose is to make that boundary explicit rather than leave it to be discovered piecemeal across later chapters.

Remark 12.2 (The Fork's Scope, Precisely). A further distinction is worth making explicit, since it is easy to conflate in practice and doing so caused an error corrected during the drafting of this monograph: the fork bears on whether a consumer's perceptual access to a *given* rendering can reach fidelity 1 relative

to that rendering. It does not bear on whether that rendering was already low-fidelity for independent, domain-structural reasons upstream of any perceptual question. Chapter 15's painting case illustrates the distinction cleanly: the rendering from event to canvas is low-fidelity for causal-inference tasks by proposition 15.2 alone, entirely independent of the fork; a further rendering from canvas to a specific viewer's percept is where the fork actually applies. By proposition 4.3, fidelity cannot increase under further rendering, so the fork can only leave an already-low fidelity the same or lower – it can never resolve an upstream ambiguity into certainty. Chess's board case (Chapter 14) is fork-relevant precisely because it lacks this confound: the board's own fidelity for determining legal continuations is high once augmented (proposition 4.9), so whether a player's perceptual access reaches that same fidelity is the fork's question in comparatively pure form.

12.5 Worked Material Under the Narrow Reading

The remainder of the monograph's treatment of memory (Chapter 20) proceeds under the narrow reading, since it is the reading supported by existing empirical literature independent of MEM|8's broader claim, and flags explicitly, wherever embodied perception is discussed in Part IV, that the broad reading remains an open alternative per open Question 12.1.

Chapter 13

Modes of Access as Task-Relative, Not Object-Relative

Synopsis. This chapter synthesizes the preceding discussion by arguing that access mode is determined by the relationship among consumer, task, and time, not by an intrinsic property of an artifact. The same object can function as a preserved source for one task while acting as an incomplete trace for another; proofs, paintings, contracts, and memories are revisited to show how a single artifact occupies different points on the fidelity continuum depending on what continuation is being attempted. This dissolves the earlier preservation/reconstitution binary into a more general relational framework.

13.1 The Criterion, Made Precise

Chapter 11 defined fidelity for a rendering r of an object e relative to a task τ . What was left implicit there is that r itself depends on who is asking and when: the substrate a particular consumer actually has in hand is not a fixed fact about the world, it is a fact about that consumer at that moment.

Definition 13.1 (Substrate Available to a Consumer). For a consumer c at time t , $r_{c,t}$ denotes the rendering (definition 2.2) c has causal access to at t .

Definition 13.2 (Access Mode). The access mode of consumer c to a task τ at time t is $M(c, \tau, t) := F(r_{c,t}, \tau)$.

Under this formalization, definition 13.2 is not an independent postulate about the nature of access; it is fidelity, already defined, simply evaluated at whichever substrate a specific consumer’s specific circumstances make available. This is a stronger and more useful move than describing mode as “a relation among consumer, task, and time” in the abstract, because it says exactly what that relation *is*: not a new primitive, but an old one indexed correctly.

Proposition 13.3 (Mode as a Relation). *Access mode is a relation $M(c, \tau, t)$ over consumer c , task τ , and time t – not a property $M(A)$ of an artifact A in isolation. The same artifact can occupy different points on the fidelity continuum of definition 11.2 simultaneously for different consumer-task pairs.*

Proof Sketch. Immediate from definition 13.2: $M(c, \tau, t)$ is defined as a function of $r_{c,t}$ and τ jointly, and $r_{c,t}$ is itself consumer- and time-indexed by definition 13.1. M collapses to a property of a single artifact A alone only in the special, non-generic case where $r_{c,t} = A$ for every consumer and every time – which is exactly what fails whenever different consumers have access to different renderings of the same underlying object, or the same consumer’s access changes over time.

13.2 Same Artifact, Different Modes

A formal proof is high-fidelity (definition 11.2) for the task of verification, since $r_{c,t}$ for a verifier with the text in hand is close to the proof itself; the same proof is lower-fidelity for the task of explanation or generalization, since $r_{c,t}$ for a reader attempting those tasks is typically their memory or partial recollection of the argument, not the text. A painting is, for its painter during production, close to direct access; for a later viewer inferring causal history from static cues (Chapter 15), two separate rendering steps are at issue and are worth keeping apart. The step from event to canvas is already low-fidelity for the causal-inference task, by proposition 15.2 alone, regardless of who views the canvas or how directly they perceive it. The step from canvas to a specific viewer's percept is the one Chapter 12's fork actually bears on (open Question 12.1): how close $r_{c,t}$ sits to the canvas itself. By proposition 4.3, the viewer's fidelity for the causal-inference task is bounded above by the canvas's own already-low fidelity for that task, so resolving the fork can only leave the viewer's fidelity the same or lower – it cannot resolve the ambiguity proposition 15.2 already established into certainty. The benign-ambiguity conclusion of proposition 15.3 is therefore fork-independent, even though the raw perceptual fidelity of viewing the canvas is not: the proof of proposition 15.3 requires only that the task tolerate multiple continuations, not that the viewer's perception of the canvas be unmediated. A contract clause is high-fidelity for the task of reading its literal text; the same contract is low-fidelity for the task of establishing original intent, which is typically litigated by reconstructing a trace-based approximation of a past state of mind (Chapter 18) – $r_{c,t}$ for a court asking about intent is correspondence, drafting history, and testimony, not the contract text itself.

CHAPTER 13. MODES OF ACCESS AS TASK-RELATIVE,
NOT OBJECT-RELATIVE

Artifact	Task	$r_{c,t}$	Fidelity
Proof	Verification	The proof text it- self	High
Proof	Explanation	Reader’s recollec- tion of the argu- ment	Low
Painting	Production	Canvas under the painter’s hand	High
Painting	Causal infer- ence	The canvas itself	Low, fork- independent
Painting	Perceiving the canvas	Viewer’s percept of the canvas	Open (open Question 12.1)
Contract	Literal read- ing	The contract text itself	High
Contract	Establishing intent	Correspondence, drafting history, testimony	Low

The table’s third column is doing the real work: in every row, what changes across tasks for the same artifact is not the artifact but which substrate a consumer pursuing that particular task actually has access to, exactly as definition 13.2 formalizes.

13.3 Why This Dissolves Rather Than Answers the Earlier Question

It is worth being precise about what this chapter, and Part III as a whole, has and has not accomplished. It has not settled whether embodied perception ever achieves fidelity exactly 1; that question is left explicitly to Chapter 12’s unresolved fork. What it has done is show that the original question motivating this part of the monograph — is a given artifact accessed in preservation mode or reconstitution mode? — was not well posed to begin with, because it asked for a property of the arti-

13.3. WHY THIS DISSOLVES RATHER THAN ANSWERS THE EARLIER QUESTION

fact when the answer was always a property of the artifact together with a consumer, a task, and a time. Once definition 13.2 makes that dependency explicit, the preservation/reconstitution distinction does not need to be defended as a real boundary somewhere on the fidelity continuum; it can simply be retired in favor of the continuum itself, with “preservation” surviving only as a convenient name for the high-fidelity end of a scale that has no other privileged points. Part IV proceeds on this basis: rather than asking of chess, painting, maps, proofs, contracts, science, memory, and media pipelines which mode of access each one exemplifies, it asks, for each domain and each task, what $r_{c,t}$ actually is and how close it sits to the source.

Part IV

Worked Domains

Chapter 14

Chess: The Closed Hidden-State Case

Synopsis. Chess serves as the canonical closed hidden-state domain. This chapter examines the gap between the visible board and the additional state required for correct continuation, such as castling rights and the repetition count, and shows that because this hidden state is finite and enumerable, augmentation restores sufficiency in principle. Chess offers a clean illustration of the framework's core concepts without the complications introduced by open-ended real-world domains, and the chapter closes by revisiting, through the case of a physical board, the question of what direct access actually means.

14.1 The Board as Substrate

A chess position's visible state – piece placement – omits castling rights, en passant eligibility, and the repetition count, each of which is required to determine legal continuations in some po-

sitions. This is the canonical witness for definition 4.8’s closed case: the omitted variables are drawn from a short, enumerable list, in sharp contrast to painting’s open causal space (Chapter 15).

Example 14.1. Two positions can have identical piece placement – the same pieces on the same squares – while differing in whether either player retains the right to castle, if one position arose from a king or rook having moved and moved back while the other did not. A rendering that records only piece placement cannot distinguish these; a rendering that also records castling rights can. This is a minimal pair in the exact sense of definition 2.4: two source states differing in one named distinction, told apart in the source, and identified by a rendering that omits it.

14.2 Augmentation Worked in Full

Augmenting the visible board with the hidden-state set H – castling rights, en passant status, and repetition count – satisfies proposition 4.9: the augmented rendering $r' = (r, H)$ is sufficient for determining every legal continuation, since H exhausts the game’s hidden state relevant to legality. This can be stated even more precisely using Appendix A’s recovery-task machinery: each element of H corresponds to a recovery task τ_h (definition A.1) – “does this side retain castling rights,” say – and lemma A.2 gives directly that r' is sufficient for τ_h if and only if $h \notin D(r')$. Since H is included in full, every such τ_h is satisfied simultaneously, which is exactly what proposition 4.9’s conclusion amounts to once stated at the level of individual recovery tasks rather than only at the level of the augmented rendering as a whole. Chess is, in this sense, the domain where the recovery-task family T_{rec} introduced to construct Appendix A’s witness happens to be finite and known

in advance, rather than open – the one condition under which proposition 4.7’s impossibility result does not bind.

14.3 The Simul Player as a Degenerate Relay

A simultaneous-exhibition player moving between boards is formalized as a degenerate one-stage relay in which B_0 is deliberately reapplied fresh at each encounter with a given board, rather than carried forward across encounters: the player reads the current position cold and acts on it, with no continuity of an internal working state between visits.

Proposition 14.2 (Stateless Re-Entry Is Sufficient When the Board Is). *If the augmented rendering $r' = (r, H)$ of a chess position is sufficient for the task of determining the best legal move (proposition 4.9), then a player re-entering the game statelessly at that position – with no memory of how it arose – can determine the same best move as a player with full continuous memory of the game.*

Proof Sketch. By proposition 4.9, r' is sufficient for the task, so $\mathcal{A}_\tau(r') \subseteq \mathcal{C}_\tau(e)$ regardless of which consumer is evaluating it or what that consumer’s history has been. A continuously-present player’s continuation is computed from $\mathcal{A}_\tau(r')$ exactly as a freshly re-entering player’s is, since neither player has access to information beyond r' once castling rights, en passant status, and repetition count are included in it. The game’s history matters only insofar as it determines H ’s current values, and H is, by hypothesis, already reflected in r' .

proposition 14.2 is what licenses treating the simul player as a legitimate limiting case rather than a curiosity: the entire reason walking a circle of boards and reading each one cold works at all is that chess’s hidden state is closed and, when augmented, fully captured in the position itself, exactly as section 8.2’s discussion of augmentation as a coordination mechanism requires. This contrasts sharply with the genuine

multi-stage relays of Chapter 21, where state *is* intended to carry forward across stages that do not share a single fixed task, and where theorem 10.2's failure mode becomes possible precisely because that carried-forward state is uncoordinated across stages with differing priorities – a condition that does not arise in the simul case because every visit to a given board shares exactly one task.

14.4 Revisiting the Hard Test Case

The physical board in front of a player is, per open Question 11.9, the strongest available candidate for fidelity $F(r, \tau) = 1$ under definition 11.2. In the vocabulary of Chapter 13, the question is precisely what $r_{c,t}$ is for a player c at the moment of moving: is it the board itself, or is it c 's visual percept of the board, already processed by whatever mechanism governs perception? If the narrow reading of chapter 12 is correct, this distinction can be drawn cleanly and $r_{c,t}$ can be treated as the board itself for practical purposes, recovering the naive intuition that a player has direct access to the game. If the broad reading is correct, $r_{c,t}$ is already a reconstructive product of the player's perceptual system, and the question of whether fidelity ever actually reaches 1 here — rather than merely approaching it closely enough for chess's purposes — is left open pending resolution of open Question 12.1. Nothing in this chapter's treatment of augmentation and stateless re-entry depends on settling that question: proposition 14.2 holds given whatever $r_{c,t}$ turns out to be, provided it includes H , and the fork bears only on how literally “the board itself” can be taken as that substrate.

Chapter 15

Painting: Open Causal Spaces and Benign Underdetermination

Synopsis. This chapter examines painting as the canonical open hidden-state domain. Paintings routinely omit vast amounts of information while remaining meaningful and useful; spatial compression, implied motion, and inferred causal history are analyzed as cases where a viewer reconstructs a rich world from sparse cues. Unlike chess, no finite augmentation can enumerate every causal history consistent with a painted configuration. The chapter argues that ambiguity here is not merely tolerated but often serves the intended task directly, making painting the clearest illustration in the monograph of the difference between corruption and benign underdetermination.

15.1 Spatial Compression

A small number of brushstrokes suggesting a cluster of leaves, and a larger arrangement of such clusters suggesting a forest, exploit a viewer's priors under definition 2.7: the omitted detail is redundant given what the visual system already supplies, not required for the viewer's task of recognizing foliage.

Remark 15.1 (Priors as Repair Cost Reduction). It is worth stating this connection to Chapter 5 explicitly rather than leaving spatial compression and repair economics as two separate vocabularies that happen to apply to the same example. A painter can afford to discard a leaf-level distinction d at negligible $C_R(d)$ precisely because the viewer's own perceptual system performs, for free and at the moment of viewing, whatever reconstruction would otherwise have been needed to supply d . The painter's low repair investment is not recklessness; it is an accurate assessment that the *effective* cost of maintaining d for this consumer's task is already near zero, since the consumer's prior does the work the painter would otherwise have had to fund.

15.2 Temporal Compression and the Bent Branch

A branch bent as if by wind is a static configuration that is low-probability under a viewer's prior model of objects at rest, forcing an inference of cause. Informally,

$$P(\text{cause} \mid \text{state}) \propto P(\text{state} \mid \text{cause}) P(\text{cause}),$$

used here descriptively, to fix intuition, rather than as a proved result of this monograph.

Proposition 15.2 (Open Causal Space). *No finite augmentation of a painting's visible configuration determines a unique causal history for that configuration.*

Proof Sketch. For any finite augmentation set H added to the image, multiple distinct causal histories – wind, a bird’s departure, a hand’s release, accumulated then-melted snow – remain consistent with the image together with H . The causal space is therefore open in the sense of definition 4.8, and no finite H closes it.

Proposition 15.3 (Benign Ambiguity). *Multiple admissible causal histories for a painted configuration need not reduce success at the task of evocative reconstruction.*

Proof Sketch. The task of evocative reconstruction (definition 4.5 applied with $\tau =$ “evoke a plausible dynamic scene”) does not require a unique history; any history in $\mathcal{A}_\tau(r)$ satisfies τ equally well. Ambiguity here is benign in the sense of definition 2.7 precisely because the task does not discriminate among the remaining admissible continuations. By proposition 11.8, this ambiguity is also compatible with $F(r, \tau) = 1$ for this task, even though the same rendering would have far lower fidelity for a task demanding the branch’s one true cause.

Remark 15.4 (This Conclusion Does Not Depend on Chapter 12’s Fork). This chapter’s argument concerns the rendering from event to canvas: the painter’s projection, which is low-fidelity for causal-inference tasks by proposition 15.2 regardless of who later views the canvas or how directly they perceive it. A further rendering step – from canvas to a specific viewer’s percept – is where Chapter 12’s unresolved fork actually enters (open Question 12.1), and Chapter 13 treats it separately for exactly this reason. Because fidelity cannot increase under further rendering (proposition 4.3), proposition 15.3’s conclusion is unaffected by how that further question is eventually resolved.

15.3 Why Painting Is the Canonical Open Domain

proposition 15.2 is the direct converse witness to proposition 4.10: painting demonstrates that sufficiency by augmentation is unavailable precisely where the hidden state is open, exactly as chess (chapter 14) demonstrates its availability where the hidden state is closed.

15.4 A Preview: Closedness and Declaration Are Independent

It is tempting, having seen chess (closed, and its hidden state straightforwardly enumerable) and painting (open, and its causal space irreducibly ambiguous), to treat closedness and transparency as the same property. Chapter 16 shows they are not: a map depicts a domain (terrain) that is, in its full physical complexity, at least as rich as anything a painting depicts, yet a map’s discard set is closed and declared by its scale bar. The relevant two-by-two structure is worth stating here in preview, since painting and chess between them supply only two of its four cells.

		Declared discard	Undeclared discard
Closed state	hidden	Maps (chapter 16)	Chess without <i>H</i>
Open state	hidden	A destroyed-records archive	Painting (this chapter)

The upper-right cell is a chess position reported without its hidden state – closed, but undeclared, and hence merely incomplete rather than irreducibly ambiguous, since *H* could in principle be supplied. The lower-left cell is populated by a case not otherwise treated in this monograph: an archive that

15.4. A PREVIEW: CLOSEDNESS AND DECLARATION ARE INDEPENDENT

explicitly records that certain materials were destroyed, without being able to enumerate what they contained – open, since the destroyed content’s specifics are gone rather than merely omitted from a known list, but declared, since the fact and approximate scope of the loss is a matter of record. Painting occupies the remaining cell: open, and undeclared in the specific sense that a painting does not carry an explicit ledger of which causal histories it fails to distinguish, relying instead on the viewer’s tolerance for ambiguity (proposition 15.3) rather than on disclosure. Chapter 16 develops the closed-and-declared cell in full.

CHAPTER 15. PAINTING: OPEN CAUSAL SPACES AND BENIGN UNDERDETERMINATION

Chapter 16

Maps: Scale as a Declared Discard Set

Synopsis. Maps are examined here as one of the oldest and most successful examples of explicit, declared information reduction. A map’s scale functions as a declared discard set, informing the user in advance which distinctions have been intentionally omitted. The chapter shows that different scales support different continuation tasks, that insufficiency is not equivalent to error, and that a map can be perfectly adequate for one purpose while being wholly inadequate for another — underscoring how transparency alone can turn what would otherwise be corrupting loss into benign simplification.

16.1 The Scale Bar as a Prehistoric Loss Ledger

Definition 16.1 (Scale-Induced Discard). A map at scale $1:s$ discards every distinction d whose minimal pair (definition 2.4) corresponds to a real-world separation below a resolution threshold $\theta(s)$, where θ is increasing in s .

Proposition 16.2 (Scale as Declared Discard). *A map's scale declaration is an instance of the loss ledger (definition 7.13).*

Proof Sketch. By definition 16.1, a stated scale specifies, implicitly, the class of spatial distinctions below threshold $\theta(s)$ that the map does not represent. This is functionally equivalent to a declared discard set (definition 2.5) attached openly to the rendering, satisfying definition 2.7 rather than definition 2.8 for any task not requiring sub-threshold detail.

16.2 Task-Relative Sufficiency, Concretely

Example 16.3. A 1:50,000 map has $\theta(50,000)$ on the order of tens of meters: two buildings on the same block collapse to the same mark, but two distinct towns do not. This map is sufficient (definition 4.5) for route planning between towns and insufficient for locating a specific building on a specific block, without being incorrect at either task – the discarded distinctions are declared by the scale bar itself, and proposition 2.9 confirms this declared loss does not constitute corruption regardless of which task a given user brings to the map. A 1:5,000 map lowers θ by an order of magnitude, becoming sufficient for the building-level task at the cost of no longer fitting an entire province on one page – a different, but equally declared, discard set.

16.3 Completing the Taxonomy

Chapter 15 previewed a two-by-two structure crossing closed against open hidden state with declared against undeclared discard, and placed maps in the closed-and-declared cell. It is worth saying precisely why maps belong there and not in the open cell painting occupies, since terrain is, considered in the

abstract, an enormously complex physical domain – arguably as complex as anything a painting depicts.

Proposition 16.4 (Map Closedness Is a Property of the Discard Function, Not the Domain). *A map’s hidden-state space is closed (definition 4.8) regardless of the physical complexity of the terrain it depicts.*

Proof Sketch. By definition 16.1, everything a map discards is characterized by a single, known threshold $\theta(s)$ applied uniformly: the discarded distinction set is exactly $\{d : d\text{'s separation} < \theta(s)\}$, which is a well-defined, if large, set fixed by s alone. This is an enumerable characterization of what is missing, satisfying definition 4.8’s closed case, regardless of how much real-world detail falls below $\theta(s)$. Painting’s causal-history space (proposition 15.2) is open not because it is *large* but because no threshold-based or otherwise finite characterization of the missing causes exists at all; terrain’s complexity, in contrast, is fully characterized by $\theta(s)$ once the scale is fixed.

proposition 16.4 is the precise statement behind Chapter 15’s claim that closedness is a property of the *discard declaration*, not of the depicted domain’s complexity: a map’s terrain could in principle be as intricate as a painted scene, and the map would still be closed, because what makes a domain open in this monograph’s sense is the absence of any finite characterization of what is missing, not the sheer quantity of what is missing.

Chapter 17

Proofs: Sufficiency Relative to a Closed Task and System

Synopsis. This chapter addresses the strongest apparent counterexample to the monograph's thesis: a formal proof, once written, appears complete. The chapter shows this is correct precisely for verification relative to a fixed axiom system, and precisely why it stops being correct the moment explanation, intuition, generalization, pedagogy, or future theoretical change become the task instead. Proofs therefore demonstrate that task-relative sufficiency is genuinely achievable in the closed case, while simultaneously reinforcing why task boundaries matter as much as they do throughout the rest of the book.

17.1 The Strongest Apparent Counterexample

Any framework claiming that no finite substrate is generally sufficient invites the objection: but a machine-checkable proof, once written, is complete for verification. This chapter states precisely why that objection is correct as far as it goes, and precisely where it stops applying – a role this chapter is written to make explicit, since the objection would otherwise hang unanswered over the rest of the monograph.

Proposition 17.1 (Verification Sufficiency). *For a fixed axiom system S , a machine-checkable proof p is sufficient (definition 4.5) for the task of verification relative to S .*

Proof Sketch. The verification algorithm for S is finite and decidable by construction; every distinction it requires to certify p appears explicitly in p together with S 's axioms. No distinction required by the verification task is discarded, so definition 4.5 is satisfied.

Corollary 17.2 (Explanation Is Not Guaranteed). *proposition 17.1 does not extend to the tasks of explanation, generalization, or adaptation to a changed axiom system.*

Proof Sketch. These tasks lie outside the closed task space $\{\tau_{\text{verify}}\}$ for which S and p were held fixed in proposition 17.1's hypothesis; by proposition 4.7, sufficiency for tasks outside a closed, anticipated set is not guaranteed by a finite substrate.

17.2 Why This Resolves Rather Than Merely Sidesteps the Objection

proposition 17.1 shows the impossibility claim of principle 1.1 was never intended to apply where both the task and the formal

system are closed and fixed in advance; corollary 17.2 shows that the moment either is allowed to vary, the same proof reverts to an ordinary instance of proposition 4.7. Sufficiency, throughout this monograph, is always sufficiency relative to a declared task space, and a proof's celebrated reliability is precisely a case where that task space has been closed by convention.

17.3 Why Proofs Do Not Exhibit Composition Failure

Chapter 7 onward has treated failure as arising when a chain's stages pursue *different* local tasks $\tau_0, \dots, \tau_{n-1}$, each locally reasonable, that jointly fail to anticipate a later τ^* . A proof's transmission – through typesetting, copying, publication, translation of notation – is also a chain of renderings in the sense of definition 7.1. It is worth stating precisely why this particular chain typically avoids theorem 10.2's failure mode for the verification task, even though it is exactly the kind of chain the theorem is about in general.

Proposition 17.3 (Homogeneous-Task Chains Preserve What They Share). *If every stage of a substrate chain shares the same task, $\tau_0 = \tau_1 = \dots = \tau_{n-1} = \tau$, and every stage's budget satisfies $\beta_i \geq R_\tau$ (the total cost of τ 's required distinctions), then A_n is sufficient for τ .*

Proof Sketch. By definition 10.1, each stage ranks τ 's required distinctions above all others, and by hypothesis $\beta_i \geq R_\tau$ at every stage, so every required distinction is funded at every stage: no required distinction is ever dropped, by definition 7.2. Since this holds for every i , τ 's required distinctions survive intact into A_n , and A_n is sufficient for τ by definition 4.5.

proposition 17.3's hypothesis is exactly what typesetting and transmission processes for mathematical proofs typically

satisfy for $\tau = \tau_{\text{verify}}$: every party handling the text – a typesetter, a journal, a translator of notation – treats preserving the logical content as the overriding priority, well-resourced enough that $\beta_i \geq R_{\tau_{\text{verify}}}$ at every stage. This is why proofs do not feel like an exception requiring special pleading: proposition 17.3 shows that a chain avoids composition failure for a given task exactly when every stage treats that task as priority throughout, which is the normal condition for a proof’s transmission and verification, but conspicuously is *not* the normal condition for a proof’s transmission with respect to explanation. Nothing about the transmission chain prioritizes preserving the motivating intuition, the historical context of the discovery, or the informal picture that made the result plausible in the first place; those distinctions are exactly the ones a typesetting or publication process has no particular reason to fund, and corollary 17.2 is the consequence.

17.4 Verification as Preservation, Explanation as Reconstitution

A proof text is high-fidelity (definition 11.2) for verification, since $r_{c,t}$ for a verifier with the text in hand is close to the proof itself (definition 13.2); understanding *why* the proof works is typically reconstitution-mode, built from a reader’s trace-based reconstruction of the argument’s structure, and is therefore subject to proposition 11.10 in exactly the way memory is (Chapter 20).

Chapter 18

Legal Contracts: Adjudication as Non-Informational Resolution

Synopsis. Contracts attempt to regulate future contingencies through finite language, making them a natural example of continuation under uncertainty. This chapter examines how legal systems respond when a contract encounters a situation its authors did not anticipate, and presents adjudication as a distinctive mechanism: a court resolves ambiguity through authoritative decision rather than by recovering additional distinctions about the past. Law is, on this framework, a domain in which continuation proceeds despite persistent, unresolved informational insufficiency.

18.1 Contracts as Attempted Brute-Force Closure

A contract's enumerated contingencies ("in the event of X , Y , or Z ") attempt to close what is, in fact, an open contingency space by brute enumeration, resembling augmentation (section 8.2) in strategy while targeting a domain that does not satisfy definition 4.8's closed case.

Example 18.1. A force majeure clause enumerating war, natural disaster, and government action as excusing conditions is an augmentation set $H_{\text{contract}} = \{\text{war, natural disaster, government action}\}$ in exactly the sense of proposition 4.9: it pre-commits the parties' agreement to cover these named contingencies without needing to anticipate, at the time of drafting, which one might occur. The clause is sufficient for any dispute falling within H_{contract} . A disruption outside this list – one neither party had reason to name at drafting time – is not covered, not because the clause was drafted carelessly, but because H_{contract} , like any finite augmentation, cannot exhaust an open space of future circumstances.

Proposition 18.2 (Contractual Augmentation Targets the Wrong Kind of Domain). *A contract's enumerated-contingency strategy achieves proposition 4.9's sufficiency only for contingencies within its enumerated set H_{contract} ; by proposition 4.10, no finite H_{contract} achieves sufficiency for the full space of future circumstances, since that space is open in the sense of definition 4.8.*

Proof Sketch. Immediate from proposition 4.10 applied with the future-circumstance space playing the role of the open hidden-state space: augmentation is the correct strategy in form, exactly as it is for chess (chapter 14), but the domain it is applied to lacks the closure property chess's hidden state has, so the strategy's guarantee does not transfer.

This mismatch between strategy and domain is, on the framework developed here, a structural explanation for the persistence of contractual litigation: no finite enumeration of contingencies exhausts an open space of future circumstances, and every contract that enumerates contingencies is, in this precise sense, applying chess's solution to painting's problem.

18.2 Adjudication, Worked

proposition 8.5 is developed here in full: a court's ruling on a disputed clause selects a binding continuation without recovering any additional distinguishability information about the parties' original intent. The information deficit – what the parties actually meant at the time of drafting – remains after the ruling exactly as before it; what has changed is that further continuation is now permitted to proceed on the ruling's basis. In the vocabulary of Section 8.5, the court substitutes its ruling as the operative correctness set $\mathcal{C}_\tau(e)$ going forward, regardless of whether that ruling coincides with whatever the undiscoverable true correctness set would have contained.

18.3 Litigating Intent as Reconstitution

Establishing original intent is a reconstitution-mode task (chapter 13): for a court asking about intent, $r_{c,t}$ is correspondence, drafting history, and testimony, none of which constitutes direct access to the intent itself (definition 13.1). This is a further instance of proposition 11.10: the traces used to reconstruct intent are themselves finite substrates subject to theorem 10.2, which is part of why intent disputes resist final resolution by evidence alone and are ultimately settled, per proposition 8.5, by authority rather than by information recovery.

Chapter 19

Scientific Theories: Revision as Relay Replacement

Synopsis. Scientific theories are treated here as finite renderings of an open empirical world. The chapter argues that a theory inevitably encounters observations it did not anticipate and therefore cannot remain sufficient indefinitely; scientific progress is reframed as relay replacement rather than local correction, with a new theory replacing an old relay once accumulated insufficiency becomes too great to ignore. Unlike most domains treated in the monograph, science actively depends on discovering its own insufficiency, making it a particularly revealing test of the framework.

19.1 A Theory as a Finite Substrate for an Open Observation Space

Proposition 19.1 (Theories Are Renderings of an Open Observation Space). *A scientific theory is a finite rendering of a potentially unbounded space of future observations.*

Proof Sketch. The observation space relevant to a theory is not fixed at the theory's formulation; new instruments, new regimes, and new experimental designs continually generate previously unavailable observations. A theory, being a finite statement, cannot represent this space exhaustively, and definition 2.2's projection structure therefore applies.

Proposition 19.2 (Revision Necessity). *Every finite theory admits possible future observations that would require its revision.*

Proof Sketch. This is proposition 4.7 applied to proposition 19.1's open observation space: a future observation τ^* can always be constructed that the theory, as a finite rendering, was not built to accommodate. This is essentially the falsifiability criterion, reframed within the admissibility apparatus of Chapter 4 rather than derived independently.

19.2 Revision, Not Augmentation

Unlike chess (chapter 14), where insufficiency is resolved by augmenting a closed hidden-state set, and unlike maps (chapter 16), where insufficiency is resolved by declaring the discard, a theory's response to proposition 19.2 is typically wholesale replacement of the theory itself (definition 8.6, section 8.6), not a patch to the existing substrate. This is the one domain treated in this monograph where global insufficiency, per principle 1.1, is not merely tolerated but functions as the intended mechanism of progress: a theory's eventual insufficiency is what licenses its replacement by a better one.

Example 19.3. A theory calibrated against observations within a certain regime – low relative velocities, say – can be sufficient (definition 4.5) for every task posed within that regime while being straightforwardly insufficient for tasks posed in a regime its observational basis never covered. The historical replacement of such a theory by one validated across a wider observational range is a canonical instance of definition 8.6: the newer theory is not derived from the older one by augmentation, in the sense proposition 4.9 would require, since the older theory’s observational basis was never closed to begin with; it is a new production built from an enlarged evidential base.

19.3 A Note on Adjacent Debates

This chapter’s claim is narrower than it might appear next to some well-known positions in philosophy of science, and it is worth marking the boundary rather than letting the two blur together. proposition 19.2 claims only that a finite theory cannot anticipate every future observation, which is a direct instance of proposition 4.7 and carries no commitment about whether successive theories are commensurable, whether observation itself is theory-laden in ways that complicate what counts as a “new” observation, or whether scientific change is better described by gradual accumulation or by more discontinuous conceptual shifts. These are substantive, contested questions in their own right, and this monograph does not adjudicate them; definition 8.6’s formal claim – that a superseded chain is replaced rather than repaired – is compatible with more than one position on how sharp or continuous that replacement is. Chapter 3 (proposition 3.33) draws the same boundary from the other direction, formally: relay revision corresponds to Distinguishability Geometry’s revision operation only when the underlying state space is held fixed and the ontology merely expands. The sharper case, where the state space itself changes, is

not left uncharacterized: proposition 3.34 shows it corresponds instead to Observability Is Restorability's Paradigm Shift Theorem (theorem 3.21), under which the two eras' restorable distinctions become genuinely incomparable rather than one refining the other. This monograph still does not claim to have resolved which of science's historical episodes are which; it claims only that both candidate cases now have a formal home.

19.4 Scientific Revolutions as Relay Replacement

It is worth drawing the last three sections together into a single picture, since they were developed at different points in this monograph and can otherwise read as three separate remarks rather than one account. A scientific tradition is a substrate chain in the sense of Chapter 7: successive theoretical and observational stages, each locally admissible with respect to the evidential standards and instruments available at the time. proposition 19.2 guarantees this chain eventually meets an observation it cannot accommodate. What happens next depends on which of two cases obtains. If the anomaly can be absorbed by extending the evidential ontology without disturbing what counts as a state of the system under study – a more precise instrument within the same conceptual framework – proposition 3.33 applies, Distinguishability Geometry's Revision Monotonicity theorem governs the transition, and the new theory's coding deficit is guaranteed no worse than the old one's: this is normal science, in something close to the textbook sense, absorbing anomalies by controlled extension. If instead the anomaly cannot be absorbed this way – if accommodating it requires changing what counts as a state at all – proposition 3.34 applies instead, and Observability Is Restorability's Paradigm Shift Theorem characterizes the result: the old and new theories' restorable distinctions become genuinely incom-

parable, with real losses alongside the real gains, rather than the new theory simply subsuming the old one. Both cases are relay replacement in definition 8.6's sense; what distinguishes them is not sharpness of the break in some vague, qualitative sense but a precise, checkable structural fact — whether the state space was held fixed. This monograph does not attempt to sort particular historical episodes into one case or the other, since doing so responsibly would require the kind of close historical argument this book's method does not supply; what it offers instead is the distinction itself, sharp enough that such sorting could in principle be attempted by those equipped to attempt it.

Chapter 20

Memory: Reconstitution and the Recursive Limit

Synopsis. This chapter returns to memory as a concrete illustration of reconstitution. Memories are analyzed as traces requiring reconstruction through schemas, priors, and contextual cues; eyewitness testimony and autobiographical recall are examined as continuation under severe informational constraint. The chapter shows that memory does not escape the framework's central limitations but reproduces them in concentrated form, making visible, at the scale of a single mind, dynamics that are otherwise easiest to see in large institutional relay systems.

This chapter proceeds under the narrow reading of MEM|8 identified in Chapter 12; wherever a claim below would be affected by the broad reading instead, this is noted explicitly rather than left implicit.

20.1 Memory as a Three-Stage Relay

Standard treatments of memory in cognitive psychology distinguish encoding, consolidation, and retrieval as separate processes, each with its own resource constraints. This structure maps directly onto definition 7.1.

Construction 20.1 (Memory as a Substrate Chain). Let A_0 be an experienced event, A_1 the encoded trace immediately following it, A_2 the consolidated trace some time later, and A_3 the recollection produced at retrieval. The transition $A_0 \rightarrow A_1$ is governed by encoding resources (attention, salience) as its budget β_0 ; $A_1 \rightarrow A_2$ by consolidation resources (rehearsal, sleep, interference from subsequent events) as β_1 ; and $A_2 \rightarrow A_3$ by retrieval resources (available cues, time pressure) as β_2 .

Each transition in construction 20.1 satisfies definition 7.6's independence condition in the ordinary case: an event's encoding is not typically decided with reference to what a much later retrieval attempt will need, consolidation proceeds according to its own resource constraints without reference to encoding's particular priorities, and a specific retrieval attempt draws on whatever cues are available at that moment without renegotiating what consolidation preserved. construction 20.1 therefore is not merely analogous to a budget relay, in the way Chapter 11 was careful to establish for reconstitution generally (proposition 11.10): it is a three-stage instance of one, with each stage's budget independently and plausibly set.

Proposition 20.2 (Memory Retrieval Is a Relay Transition). *The transition trace $\rightarrow$ recollection satisfies the conditions of a substrate-chain transition (definition 7.1).*

Proof Sketch. Retrieval takes a residual trace and a current schema and produces a usable recollection, incurring costs (effort, plausibility constraints supplied by the schema) and

losses (aspects of the original event the trace does not constrain) in the sense of definition 7.2. This satisfies definition 7.1 directly, and is the final stage of construction 20.1.

20.2 Schema-Driven Confident Error

Eyewitness-memory research documents confident, richly reconstructed testimony that is nonetheless factually wrong: the schema fills a gap the trace does not constrain, and the result is presented, to the rememberer and to others, as though it were retrieved rather than regenerated. This is a concrete instance of proposition 11.10's "priors patch" (section 8.4) operating at retrieval time on an already-insufficient trace, and is structurally the same failure mode as proposition 15.2's painting case, transposed from an external image to an internal reconstruction. In terms of construction 20.1, this is exactly the pattern theorem 10.2 predicts: a distinction lost or degraded at encoding or consolidation – because τ_0 or τ_1 , whatever was salient or well-rehearsed at the time, did not prioritize it – is unavailable to retrieval regardless of how confidently retrieval's own schema fills the resulting gap, since confidence at A_3 cannot restore what A_1 or A_2 never carried forward.

Proposition 20.3 (Memory Does Not Escape Composition Failure). *Memory's reconstitutive character does not constitute an exception to theorem 10.2.*

Proof Sketch. By proposition 20.2 and construction 20.1, memory retrieval is a relay transition, indeed the final stage of a three-stage relay; by proposition 11.10, this transition inherits the open-task-space vulnerability of any other stage in a budget relay. theorem 10.2 therefore applies to memory exactly as it applies to the media-pipeline case of Chapter 21, contrary to an informal suggestion, made earlier in the research program that produced this monograph, that reconstitution somehow

“changes the game” rather than relocating the same problem to a new substrate.

This chapter’s opening noted that any claim affected by Chapter 12’s broad reading would be flagged explicitly as it arose. None was: proposition 20.2 and proposition 20.3 both rest on proposition 11.10, which Chapter 12’s own table records as holding under either reading. The narrow reading was invoked at the start of this chapter for its independent empirical support, not because this chapter’s conclusions require it; memory turns out to be a case, unlike the painting and chess cases of remark 12.2, where the fork simply does not enter the argument at all.

Chapter 21

Media Pipelines: Budget Relays Made Concrete

Synopsis. This chapter applies the full framework to a real transformation chain: audio recording, signal processing, transcription, timestamp generation, summarization, and archiving, each analyzed as a stage in a budget relay preserving some distinctions while discarding others according to local priorities. The chapter provides a concrete, constructive witness for the Composition Failure Theorem by exhibiting a distinction lost early in the chain that becomes critical for a later task no stage anticipated, and closes with practical recommendations for provenance design, root-object preservation, and re-derivable archival systems.

21.1 The Full Worked Chain, Named and Numbered

The chain audio $\rightarrow$ transcript $\rightarrow$ summary $\rightarrow$ archive is mapped directly onto definition 7.1’s notation, but rather than leave the distinctions involved abstract, as Appendix A’s canonical witness does, this section fixes concrete, named ones, so that the constructive witness promised by this chapter’s synopsis is worked with real content rather than only with placeholders standing in for it.

Construction 21.1 (Named Pipeline Instance). Let A_0 carry four named distinctions from a recorded utterance: lexical content d_{lex} (cost 3), timestamp precision d_{time} (cost 1), sentence-level structure d_{struct} (cost 2), and emotional tone d_{tone} (cost 4). The transcription stage has task τ_0 requiring exactly $\{d_{\text{lex}}, d_{\text{time}}\}$, with budget $\beta_0 = 4$. The summarization stage has task τ_1 requiring exactly $\{d_{\text{lex}}\}$, with budget $\beta_1 = 3$.

Proposition 21.2 (The Named Instance Exhibits Composition Failure). *In construction 21.1, d_{tone} is dropped at stage 0 under any construction satisfying definition 7.2, and the resulting archive is insufficient for any task recovering d_{tone} .*

Proof. The required set at stage 0, $\{d_{\text{lex}}, d_{\text{time}}\}$, has cost $R_0 = 3 + 1 = 4 = \beta_0$ exactly. The non-required distinctions have costs $C(d_{\text{struct}}) = 2 > \beta_0 - R_0 = 0$ and $C(d_{\text{tone}}) = 4 > 0$; by lemma A.4, both are dropped under any construction, so $D_0 = \{d_{\text{struct}}, d_{\text{tone}}\}$. At stage 1, A_1 retains only $\{d_{\text{lex}}, d_{\text{time}}\}$; the required set $\{d_{\text{lex}}\}$ has cost $R_1 = 3 = \beta_1$ exactly, and the sole non-required distinction present, d_{time} , has cost $1 > \beta_1 - R_1 = 0$, so $D_1 = \{d_{\text{time}}\}$ by the same lemma. A_2 retains only $\{d_{\text{lex}}\}$. Let $\tau^* = \tau_{d_{\text{tone}}}$, the recovery task (definition A.1) for whether the speaker’s tone on a given claim was, for instance, sincere or sarcastic. Since $d_{\text{tone}} \in D_0 \subseteq D(A_0 \Rightarrow A_2)$, by lemma A.2 A_2 is insufficient for τ^* . Neither τ_0 nor τ_1 is τ^* : both are preservation

tasks over different, disjoint required sets not including d_{tone} 's recovery. $\square$

This is Appendix A's abstract construction re-run with distinctions a reader can recognize without translation: a transcription system tuned for lexical accuracy and rough timing, and a summarizer tuned for gist, jointly and irreversibly discard whether a speaker meant something sincerely or sarcastically – not through negligence at either stage, but because neither stage's task ever prioritized it, exactly as theorem 10.2 predicts.

21.2 Fan-Out, Concretized

construction 21.1 extends directly to the fan-out case of Chapter 9 by adding a second branch from A_0 : an acoustic-tone-analysis branch Y with task τ_Y requiring exactly $\{d_{\text{tone}}\}$, budget $\beta_Y = 4$ (funding d_{tone} exactly), dropping everything else. This is construction 9.5 instantiated with named distinctions: branch X (transcription, as above) and branch Y (tone analysis) are both locally admissible, both derived from the identical A_0 , and disagree silently in precisely definition 9.4's sense – a consumer relying on the transcript alone has no signal that tone was ever at issue, while branch Y , unconsulted, would have flagged it directly.

Proposition 21.3 (Pipeline Witness). *There exists an explicit audio $\rightarrow$ transcript $\rightarrow$ summary chain and a task τ^* , constructible without appeal to hypothetical or idealized stages, such that τ^* requires a distinction discarded upstream and no downstream stage's loss ledger records enough about that distinction for a consumer pursuing τ^* to detect the gap.*

Proof Sketch. construction 21.1 together with proposition 21.2 constitutes exactly this witness, with $\tau^* = \tau_{d_{\text{tone}}}$ constructed explicitly rather than posited. This result was labeled a Theorem

in an earlier draft; the label has been corrected to Proposition, consistent with the monograph’s epistemic-labeling discipline, since it is a single explicit instantiation of the general result already proved in Appendix A rather than an independent derivation.

21.3 Practical Recommendations

Preserving the raw audio as a root object (definition 7.8) and treating every downstream rendering as disposable and re-derivable directly operationalizes proposition 7.9: a discovered insufficiency at the summary or transcript stage can then be repaired by re-deriving a new rendering from the root under a revised discard policy, rather than requiring recovery of information that no longer exists anywhere in the chain. In terms of construction 21.1, this means that discovering $\tau_{d_{\text{tone}}}$ ’s relevance after the fact is recoverable, at the cost of reprocessing, if and only if the raw audio was retained; it is permanently unrecoverable if only the transcript or summary survived.

A budget-aware loss ledger (definition 7.13) attached to each stage should record not merely that d_{struct} and d_{tone} were dropped at transcription, but the budget $\beta_0 = 4$ and required set $\{d_{\text{lex}}, d_{\text{time}}\}$ that produced the drop, so a later archivist can distinguish a distinction dropped because it was judged irrelevant from one dropped purely because the transcription budget was small – exactly the distinction definition 7.13 was built to preserve. Where a fan-out structure exists, as in Section 8.3’s sense of provenance-sharing extended across branches rather than only along a single chain, cross-referencing each branch’s ledger against the others makes definition 9.4’s failure mode detectable after the fact, even though, as proposition 8.4 showed, detectability alone does not prevent the original loss.

Part V

Synthesis

Chapter 22

The Closing Statement

Synopsis. This closing chapter integrates the monograph's concepts into a single narrative: distinguishability explains what survives; admissibility explains what remains possible; repair economics explains why a distinction was or was not maintained; budget relays explain how losses accumulate across time and institutions; the fidelity continuum unifies preservation and reconstitution; and Composition Failure explains why locally rational decisions can still produce globally insufficient outcomes. The chapter closes by naming what remains unresolved, above all the MEM|8 dependency, confronting directly the strongest available objection to the whole enterprise – that its central theorem may be little more than task-unavailability restated in heavier notation – and restates the central claim that continuation is constrained not by malice or error but by the unavoidable limits of finite repair budgets operating in an open future.

22.1 Restating the Arc

Distinguishability (chapter 2) established what survives a projection. Admissibility (chapter 4) established what continuations remain reachable given what survives. Repair economics (chapter 5) established why a given distinction was or was not maintained, and, via corollary 5.8, established that this monograph could not simply inherit its central claim from *Conservation of Ambiguity*. The budget relay (chapter 7) formalized how discard compounds across uncoordinated stages, and the Composition Failure Theorem (theorem 10.2) proved the diachronic analogue the Redistribution Theorem does not cover. The fidelity continuum (chapter 11) collapsed preservation and reconstitution into a single variable, conditional on the unresolved question raised in Chapter 12.

Part IV then did more than illustrate this arc. Chess and maps showed the arc's positive case: sufficiency genuinely achieved, by augmentation or by declared discard, when hidden state is closed. Painting showed its negative case, and showed that the negative case is not a defect: ambiguity a task does not need to resolve is not a gap to be regretted. Proofs supplied the arc's sharpest internal check, proposition 17.3, which deserves to sit alongside theorem 10.2 in this summary rather than beneath it: where the Composition Failure Theorem shows how differing local tasks compose into a gap no stage intended, proposition 17.3 shows the exact condition under which they do not – a single task, adequately funded, held constant across every stage. The two results are converses of each other, and together they say more than either does alone: composition failure is not a property of chains in general, it is a property of chains whose stages do not agree, deliberately or by accident, on what matters. Contracts, science, memory, and pipelines each instantiated this same pair of results in a different institutional setting, and the pipeline chapter's numer-

ically explicit witness (proposition 21.3) closed the loop back to Appendix A's abstract construction, confirming the two are the same argument rather than merely similar ones.

22.2 The Closing Statement

Conjecture 22.1 (Closing Statement). Continuation depends on the substrate available to a consumer, whether preserved with high fidelity or regenerated from traces at low fidelity; in either case the substrate is bounded by the distinctions its production budget could afford to maintain. Because relays compose without guaranteed coordination (definition 7.6) and because future task spaces remain open (proposition 4.7), no finite chain of substrates can be guaranteed sufficient for arbitrary future continuation – not because information is destroyed, but because no stage can afford, or is positioned, to preserve everything a task it cannot yet foresee will require.

This statement is labeled a conjecture rather than a theorem deliberately: it synthesizes theorem 10.2 together with the fidelity continuum of Chapter 11, but the synthesis itself, as a single unified statement, has not been given an independent proof beyond what its components already establish.

22.3 What Remains Open

Appendix F consolidates every conjecture and open question raised in the body in one register, organized by which part of the framework each bears on rather than by chapter, and this section does not attempt to duplicate it. Three items are worth naming here because of their scope rather than their difficulty. The MEM|8 fork (open Question 12.1) is the largest: it bears specifically on whether a consumer's perceptual access to a given rendering can itself reach fidelity 1, not on

whether an already-open task's ambiguity can be resolved by perceptual directness alone – remark 12.2 states this boundary precisely, and it is worth a reader's attention, since conflating the two was an error this monograph made and corrected during its own drafting. The tightness of corollary 10.4's growth bound is conjectural: this monograph shows cumulative discard does not shrink, not that discard *risk* grows at any particular rate. Whether a general coordination protocol across budget relays could bound, though not eliminate, theorem 10.2's failure mode is left as a direction for future work; proposition C.1 shows one specific, more radical alternative to coordination succeeding in one instance, which is evidence that the broader question is not hopeless, not an answer to it.

22.4 What This Monograph Does Not Claim

This monograph does not claim that all cognition is reconstructive; it restricts that claim, via chapter 12's narrow reading, to cases independently supported by existing literature, and Chapter 20 found, on inspection, that none of its own conclusions actually required the fork to be resolved either way. It does not claim that *Conservation of Ambiguity's* Redistribution Theorem is superseded; theorem 5.7 is imported and used, not revised, and Appendix C shows concretely, with a specific allocation of resources that succeeds under pooling where the separated relay fails, that the two theorems answer different questions rather than one subsuming the other. It does not claim to have resolved the preservation/reconstitution question definitively, since that resolution remains contingent on open Question 12.1. And it does not claim that the closing statement above is the only way to unify its results; it claims only that this particular unification is not, on inspection, derivable from any single component alone, which is the most a conjecture honestly labeled as such should claim.

22.5 The Strongest Objection

The most serious challenge to this monograph is not a technical one, and it deserves a direct answer rather than a deflection into further formalism. The objection: theorem 10.2, once stated precisely, is a fairly direct consequence of proposition 4.7 – no substrate can anticipate an unforeseeable future task – dressed in the additional vocabulary of budgets, stages, and discard sets. If that is right, has this monograph built an elaborate apparatus to reprove something close to common sense: that you cannot plan for what you do not know?

The honest answer has three parts, and none of them is a denial that the objection identifies something real. First, the *existence* half of theorem 10.2 is indeed a close relative of proposition 4.7, and this monograph has not claimed otherwise; Chapter 10's own proof sketch is built directly on top of it. Second, and this is where the objection proves too much if pressed further, the theorem's genuine content is not the existence claim but two things the existence claim alone does not supply: the exact mechanism by which failure occurs (budget exhaustion after funding a stage's own required distinctions, not some unspecified misfortune), robust to the choice of local discard procedure (lemma A.4), and the precise converse identifying when failure provably does not occur (proposition 17.3), which is not a consequence of proposition 4.7 at all and could not have been stated without the budget-relay apparatus this monograph built to state it. A theorem that only says "the unforeseeable cannot be foreseen" does not tell you that a shared, adequately funded task throughout a chain is exactly what defends against it; this monograph's apparatus does. Third, and this is the part worth being fully candid about: whether that additional content justifies the apparatus's formal weight, as opposed to a shorter and less machinery-heavy statement of the same two facts, is a matter of taste this monograph cannot

settle for a skeptical reader by further argument. The worked domains of Part IV are this monograph’s actual case for the apparatus’s usefulness – chess, painting, proofs, and pipelines each get a sharper account of their own structure from having the vocabulary available – and a reader unconvinced by those chapters will not be talked into the apparatus’s worth by this section restating that it exists.

22.6 Future Research Directions

Appendix F is the complete register; this section does not duplicate it, only names the three items there most likely to reward attention first. The tightness of corollary 10.4’s growth bound is the most tractable: it asks a concrete quantitative question about a construction already fully specified in Appendix A. The measure of a persistence set is the most promising: remark 3.30’s reconstruction redundancy $\rho_W(r)$, though unverified, is concrete enough to check directly against this monograph’s root-recovery apparatus (proposition 7.9) without requiring new machinery. The MEM|8 fork is the most consequential and the least tractable by this monograph’s methods: resolving it requires empirical and philosophical work well outside what a framework built from worked examples and formal proof can settle on its own, and this monograph’s contribution is limited to having shown precisely which of its own results would move, and which would not, once someone else resolves it.

Appendix A

Full Proof of the Composition Failure Theorem

Synopsis. This appendix gives a complete proof of Theorem 10.2, together with the explicit construction on which it rests. The proof is built to satisfy the robustness requirement raised in remark 7.5: rather than exhibiting a witness that happens to work under one convenient local discard rule, lemma A.4 below shows that the witnessing relay’s key discard is forced under *any* construction obeying definition 7.2, closing that specific open question for the purposes of this theorem even though the broader question of whether every claim in this monograph enjoys the same robustness remains open (appendix F).

A.1 Recovery Tasks and the Witnessing Task Family

The theorem requires an explicit, non-enumerable-in-advance family of tasks T , together with a specific $\tau^* \in T$ for which insufficiency can be verified directly rather than merely asserted. Both are supplied by a single, simple family of tasks built directly from the minimal-pair machinery of Chapter 2.

Definition A.1 (Recovery Task). For a named distinction d with minimal pair $\{x_d, x_{\neg d}\}$ (definition 2.4), the recovery task τ_d is the task of correctly reporting, given a substrate, whether the source object e satisfies x_d or $x_{\neg d}$. Its correctness set (definition 4.4) is the singleton containing whichever of the two reports matches e 's actual state.

Lemma A.2 (Sufficiency for a Recovery Task Iff No Discard). *A rendering r is sufficient (definition 4.5) for τ_d if and only if $d \notin D(r)$.*

Proof. If $d \notin D(r)$, the pair $\{x_d, x_{\neg d}\}$ is not identified by r (definition 2.3, definition 2.5), so r distinguishes which of the two states e occupies, and the correct report is the unique continuation reachable: $\mathcal{A}_{\tau_d}(r) = \mathcal{C}_{\tau_d}(e)$, satisfying definition 4.5. If $d \in D(r)$, the pair is identified, so a consumer with access only to r cannot rule out either report: both the correct and the incorrect report are reachable, $\mathcal{A}_{\tau_d}(r) \supsetneq \mathcal{C}_{\tau_d}(e)$, and definition 4.5's containment fails. $\square$

Definition A.3 (Witnessing Task Family). $T_{\text{rec}} := \{\tau_d : d\}$, where d ranges over the open-ended space of possible named distinctions.

T_{rec} is open in the sense required by proposition 4.7: the argument given there — that a task requiring a distinction not represented in a fixed candidate list can always be constructed after the fact — applies verbatim to T_{rec} , since the space of possible minimal pairs (definition 2.4) that might later be named

as relevant is not fixed in advance by any finite specification of a substrate chain's producers.

A.2 A Construction-Independent Exhaustion Lemma

remark 7.5 showed that definition 7.2 is satisfied by more than one non-equivalent construction. The following lemma isolates a sufficient condition for a discard to be forced under every such construction, so that the theorem's witness does not depend on which one a given stage happens to use.

Lemma A.4 (Exhaustion Forces Discard). *Fix a stage with budget β_i and suppose the distinctions required by τ_i (in the sense of definition 2.7) have total cost $R_i = \sum_{d \text{ required}} C_i(d) \leq \beta_i$. If every non-required distinction d' satisfies $C_i(d') > \beta_i - R_i$, then every non-required distinction is dropped – $d' \in D_i$ for every such d' – under any construction satisfying definition 7.2 together with the requirement that required distinctions are ranked strictly above non-required ones.*

Proof. Any construction satisfying definition 7.2 must respect the budget constraint $\sum_{d \in S} C_i(d) \leq \beta_i$ for the funded set S , and, by hypothesis, must rank every required distinction above every non-required one. Since the required distinctions have total cost $R_i \leq \beta_i$, funding all of them is feasible and, given their priority, is favored by both construction 7.3 (which processes higher-priority items first and encounters no obstruction, since $R_i \leq \beta_i$) and construction 7.4 (for which funding every required item dominates any allocation that sacrifices one, provided required items are assigned value strictly exceeding any combination of non-required items, which the priority requirement above entails). After funding the required set, the remaining budget is exactly $\beta_i - R_i$. By hypothesis, $C_i(d') > \beta_i - R_i$ for every non-required d' , so no non-required

distinction can be funded in addition to the required set under either construction, nor under any construction obeying the same budget constraint, since funding any single such d' alongside the required set would require total expenditure exceeding β_i . Hence $d' \in D_i$ for every non-required d' , regardless of which construction produced the allocation. $\square$

lemma A.4 is the appendix's answer to the robustness question raised in Chapter 7: it does not show that every budget relay behaves identically under every construction, but it identifies a class of stages – those where the required set exactly, or near-exactly, exhausts the budget – whose discard behavior is construction-independent, and the witness built below is chosen to fall into exactly this class.

A.3 Recalling Local Admissibility

Chapter 10 makes precise, in definition 10.1, what Theorem 10.2 means by a transition being “locally admissible with respect to its own stage’s task τ_i .” That definition does not privilege construction 7.3 over construction 7.4, or vice versa; it constrains only the priority ordering’s relationship to τ_i ’s own stated requirements, consistent with remark 7.5’s observation that definition 7.2 constrains outcomes, not procedures. It is used throughout the remainder of this appendix exactly as stated there.

A.4 The Canonical Witness

Construction A.5 (Two-Stage Witness). Let A_0 carry five distinctions $d_1, \dots, d_5$ with costs $C(d_1) = 1, C(d_2) = 2, C(d_3) = 3, C(d_4) = 2, C(d_5) = 4$ – the same instance introduced in section 7.8 and section 5.5. Let τ_0 require exactly $\{d_1, d_2\}$, with $\beta_0 = 4$. Let τ_1 require exactly $\{d_1, d_2\}$ as well, with $\beta_1 = 3$.

Proposition A.6 (The Witness Is Locally Admissible and Forces the Same Discard Under Any Construction). *In construction A.5: (a) B_0 and B_1 are locally admissible (definition 10.1) with respect to τ_0 and τ_1 respectively, under any construction satisfying definition 7.2; and (b) $D_0 = \{d_3, d_4, d_5\}$ and $D_1 = \emptyset$ regardless of which such construction is used.*

Proof. For stage 0: the required set $\{d_1, d_2\}$ has total cost $R_0 = 1 + 2 = 3 \leq \beta_0 = 4$. Every non-required distinction has cost exceeding $\beta_0 - R_0 = 1$: $C(d_3) = 3 > 1$, $C(d_4) = 2 > 1$, $C(d_5) = 4 > 1$. lemma A.4 applies directly, giving $D_0 = \{d_3, d_4, d_5\}$ under any construction satisfying definition 7.2 with required distinctions ranked above non-required ones – which is exactly definition 10.1’s condition, so (a) and (b) both hold for stage 0.

For stage 1: A_1 retains only $\{d_1, d_2\}$ (nothing else survived stage 0), both required by τ_1 , with total cost $R_1 = 3 = \beta_1$ exactly. There are no non-required distinctions present at this stage to consider, so the exhaustion hypothesis of lemma A.4 is vacuously about an empty set, and any construction funds $\{d_1, d_2\}$ in full (the only feasible allocation using the entire budget), giving $D_1 = \emptyset$ regardless of construction. Local admissibility for stage 1 follows immediately, since there is nothing non-required present to rank incorrectly. $\square$

A.5 Extension to Arbitrary Relay Length

Construction A.7 (Pass-Through Stage). A stage j is a pass-through stage if τ_j requires exactly the distinctions present in A_j , and β_j equals their total cost exactly.

Proposition A.8 (Extension to Arbitrary Length). *For every $n \geq 2$, there is an n -stage budget relay, every transition of which is locally admissible, with $D(A_0 \Rightarrow A_n) = \{d_3, d_4, d_5\}$ exactly as in construction A.5.*

APPENDIX A. FULL PROOF OF THE COMPOSITION FAILURE THEOREM

Proof. Take construction A.5's two stages as stages 0 and 1, and insert $n - 2$ pass-through stages (construction A.7) at positions $2, \dots, n - 1$, each requiring exactly $\{d_1, d_2\}$ at cost 3, matching A_1 's (and every subsequent stage's) content exactly. Each pass-through stage is locally admissible by the same vacuous argument given for stage 1 in the proof of proposition A.6, and discards nothing, so $D_j = \emptyset$ for $j \geq 2$. By definition 7.11, $D(A_0 \Rightarrow A_n) = D_0 \cup D_1 \cup \dots \cup D_{n-1} = \{d_3, d_4, d_5\} \cup \emptyset \cup \dots \cup \emptyset = \{d_3, d_4, d_5\}$. $\square$

A.6 Proof of Theorem 10.2

Proof. Fix any $n \geq 2$ and take the n -stage relay of proposition A.8, with every transition locally admissible by construction. Let $\tau^* := \tau_{d_5} \in T_{\text{rec}}$ (definition A.1, definition A.3). By proposition A.6 and proposition A.8, $d_5 \in D(A_0 \Rightarrow A_n)$, so $d_5 \in D(A_n)$ relative to A_0 . By lemma A.2, A_n is insufficient for τ_{d_5} .

It remains to check $\tau^* \notin \{\tau_0, \dots, \tau_{n-1}\}$. Every τ_i in the construction is a preservation task, requiring exactly $\{d_1, d_2\}$, with correctness set consisting of continuations that retain both; $\tau^* = \tau_{d_5}$ is a recovery task in the sense of definition A.1, with correctness set consisting of a single report about d_5 alone. These are distinct formal objects with distinct correctness sets for any nontrivial choice of underlying object e , so $\tau^* \neq \tau_i$ for any i .

T_{rec} is an open task space containing τ^* , as argued following definition A.3. This exhibits, for every $n \geq 2$, a budget relay and an open task space $T = T_{\text{rec}}$ satisfying every clause of Theorem 10.2. $\square$

A.7 Verifying the Corollaries

Verification of corollary 10.3. Suppose β_1 is increased in the witness of construction A.5, holding $\beta_0 = 4$ fixed. Stage 1 receives only $A_1 = \{d_1, d_2\}$ regardless of β_1 's value, since d_3, d_4, d_5 were already absent from A_1 by the time stage 1 acts; no increase in β_1 can fund a distinction that is not present to be funded. $D(A_0 \Rightarrow A_n)$ is therefore unaffected by any change to β_1 (or to any β_j for $j \geq 1$), confirming that only an improvement to β_0 specifically – the stage where the loss actually occurred – could have prevented the eventual insufficiency, and even that requires τ_0 's producer to have known, at stage 0, that d_5 would later matter: precisely the knowledge proposition 4.7 shows is unavailable in general. $\square$

Partial verification of corollary 10.4. The existence half of corollary 10.4 – that cumulative discard is non-decreasing in relay length – is exactly proposition 7.12, and proposition A.8 shows this bound is not merely non-strict but can be realized with equality across arbitrarily many additional stages (the pass-through stages contribute no further discard). This appendix does not, however, establish the stronger, quantitative claim that cumulative discard *risk* grows across relays in general, only that it does not shrink in this particular family of constructions. The tightness question — whether a sharper growth bound holds under specific coordination mechanisms – remains open, exactly as flagged in Appendix F, and is not resolved here. $\square$

A.8 What This Proof Does Not Establish

This appendix proves Theorem 10.2 and corollary 10.3 in full, and establishes the existence half of corollary 10.4. It does not establish that every budget relay of practical interest falls into the construction-independent class identified by lemma A.4;

APPENDIX A. FULL PROOF OF THE COMPOSITION FAILURE THEOREM

relays in which the required set does not closely exhaust the local budget may genuinely behave differently under construction 7.3 than under construction 7.4, and nothing here rules that out for such cases. It also does not establish corollary 10.4's quantitative tightness, address the infinite-relay case flagged in Chapter 7, or extend to the branching relays of Chapter 9, whose fan-out structure introduces the further complication of proposition 9.7 and is not treated by the linear construction given here. Appendix F is updated to reflect exactly this scope.

Appendix B

Notation Reference

Synopsis. This appendix collects the symbols, definitions, and formal objects used throughout the monograph into a single reference table.

Symbol	Meaning
$(X, \sim_X)$	Distinction space with equivalence structure (Ch. 2)
r, A_i	A rendering / stage in a substrate chain
$D(r)$	Discarded distinction set of rendering r (definition 2.5)
$\mathcal{A}_\tau(r)$	Admissible continuation set for task τ given r (definition 4.1)
$C_R(d)$	Repair cost of distinction d (definition 5.3)
$A(d)$	Operational ambiguity of d (definition 5.4)
$I(d_i, d_j)$	Repair interference between d_i and d_j (definition 5.6)
Φ	Repair budget functional (theorem 5.7)
$B_i = (C_i, I_i, \beta_i)$	Local budget triple at relay stage i (definition 7.1)
$L_i = (D_i, B_i)$	Budget-aware loss ledger at stage i (definition 7.13)

APPENDIX B. NOTATION REFERENCE

Symbol	Meaning
Q_i	Confidence/quality profile of branch i (proposition 9.8)
$F(r, \tau)$	Fidelity of r for task τ (definition 11.2)
$\text{Rest}(\mathcal{R})$	Repair-stable domain (imported, <i>Observability Is Restorability</i>)
T	Task space; <i>open</i> if unbounded at production time
H	Enumerable hidden-state set (closed domains only, definition 4.8)
$M(c, \tau, t)$	Access-mode relation over consumer, task, time (proposition 13.3)
$D(A_i \Rightarrow A_j)$	Cumulative discard set between stages i and j (definition 7.11)
$C_i(d), I_i, \beta_i$	Stage-local repair cost, interference, and budget at relay stage i (definition 7.1)
$\{x_d, x_{-d}\}$	Minimal pair representing named distinction d (definition 2.4)
$\mathcal{C}_\tau(e)$	Correctness set of task τ evaluated against true object e (definition 4.4)
$r_{c,t}$	Substrate available to consumer c at time t (definition 13.1)
τ_d	Recovery task for distinction d (definition A.1)
T_{rec}	Witnessing family of recovery tasks (definition A.3)
$\theta(s)$	Resolution threshold for a map at scale $1:s$ (definition 16.1)
$(X, \sim, T)$	Ontological triple: states, indistinguishability, ontology (definition 3.1)
$\delta_T(x)$	Coding deficit of x under ontology T (definition 3.2)
$\Delta_T(x)$	Distinguishability deficit of x under ontology T (definition 3.3)
$(X, \mathcal{T}, \mathcal{D}, \mathcal{R})$	Structured domain: states, transformations, distinctions, reconstructions (definition 3.15)
$P(d)$	Persistence set of distinction d (definition 3.17)
E, S, O	Evaluation function, internal signals, external channel (definition 3.25)

Symbol	Meaning
$\rho = \partial E / \partial O$	Coupling coefficient between evaluation and external reality (definition 3.26)
$C(r), \rho_W(r)$	Reconstruction cover and redundancy, reported tier (§3.8)
A^{merge}	Reconciled rendering combining two branches (definition 9.11)

The symbol ρ_W above is a renaming, introduced in this monograph, of the reported $\rho(r)$ from The Geometry of Witnesses (§3.8), undertaken specifically to avoid collision with the unrelated coupling coefficient $\rho = \partial E / \partial O$ from The Geometry of Correction, two sources that apparently chose the same symbol independently (remark 3.29).

Appendix C

Relation to *Conservation of Ambiguity*: A Side-by-Side Comparison

Synopsis. This appendix compares the Composition Failure Theorem with the Redistribution Theorem of *Conservation of Ambiguity*, setting out their different assumptions, domains, and implications side by side, and arguing explicitly that neither is derivable from the other without the budget-relay machinery of Part II.

APPENDIX C. RELATION TO CONSERVATION OF AMBIGUITY: A SIDE-BY-SIDE COMPARISON

	Redistribution Theorem (theorem 5.7)	Theo- rem	Composition Theorem (theorem 10.2)	Failure
Structure	Synchronic		Diachronic	
Object	Single observer's budget Φ		Sequence of independent budgets B_i	
Distinctions involved	Many, competing simultaneously		One (or a bounded cluster), tracked over stages	
Conserved quantity	Total repair budget Φ_{tot}		None conserved; cumulative discard is monotone (proposition 7.12)	
Coordination assumed	Full, by construction (one observer)		None, by construction (definition 7.6)	
Failure mode	Local sharpening forces redistribution		Locally rational stages compose into global gaps	

C.1 Why One Does Not Reduce to the Other

definition 7.6's defining condition is that each stage's budget β_i is set without reference to any other stage's budget or ledger. It is worth checking directly what happens to Appendix A's witness if that condition is dropped and the stages' budgets are pooled into a single, simultaneously-allocated Φ_{tot} instead – not merely to confirm that theorem 5.7 then applies, but to see concretely what difference pooling makes to the actual outcome.

Recall the witness of construction A.5: five distinctions with costs 1, 2, 3, 2, 4, and two stage-local budgets $\beta_0 = 4$ and $\beta_1 = 3$, applied sequentially. Under definition 7.6, β_0 's unspent remainder (after funding $\{d_1, d_2\}$ at cost 3, one unit of β_0 goes unused, since no non-required distinction costs 1 or less) cannot be carried forward to supplement β_1 ; each stage's allocation is decided in isolation, and lemma A.4 shows d_5 is dropped at stage 0 regardless of construction.

Proposition C.1 (Pooling Can Restore Sufficiency). *Let $\Phi_{\text{tot}} = \beta_0 + \beta_1 = 7$ be made available to a single observer as one simultaneous allocation over all five distinctions of construction A.5, rather than as two separate stage-local budgets. Then a feasible allocation exists that funds $\{d_1, d_2, d_5\}$ in full, avoiding the insufficiency for τ_{d_5} that Appendix A shows is unavoidable under the separated relay.*

Proof. $C(d_1) + C(d_2) + C(d_5) = 1 + 2 + 4 = 7 = \Phi_{\text{tot}}$, so funding exactly $\{d_1, d_2, d_5\}$ is feasible under a single, simultaneous budget constraint of 7, using the full amount with nothing left over. Under this allocation, $d_5 \notin D(A_1)$, so by lemma A.2, sufficiency for τ_{d_5} is preserved. This allocation is unavailable to the separated relay of construction A.5 because $\beta_0 = 4$ alone is insufficient to fund $\{d_1, d_2, d_5\}$ (cost $7 > 4$), and proposition 7.12 forbids recovering d_5 once stage 0 has discarded it, regardless of how much budget β_1 later provides. $\square$

proposition C.1 is the concrete content behind the claim, made in prose in Chapter 10, that assuming a shared budget across all stages of a relay is precisely the condition under which theorem 10.2's construction becomes unavailable. It is stronger than merely showing the construction *fails to apply* under pooling: it exhibits an allocation that actively *succeeds* where the separated relay fails, using the identical total nominal resource. The insufficiency exhibited in Appendix A is therefore not a consequence of scarcity in any absolute sense – $\Phi_{\text{tot}} = 7$ was always, in total, enough to have funded $\{d_1, d_2, d_5\}$ – but a consequence specifically of definition 7.6's prohibition on transferring unspent capacity from one stage to a later one. This is the precise sense in which the two theorems occupy disjoint settings rather than one subsuming the other: theorem 5.7 presupposes exactly the transferability that theorem 10.2's witness depends on the absence of.

Appendix D

Extended Worked Examples

Synopsis. This appendix sketches additional domains — textual transmission, oral tradition, and biological inheritance — that further test and illustrate the framework beyond the chapters given full treatment in Part IV. Treatment here is at appendix depth: each example is developed far enough to show the fit is real, not far enough to constitute a chapter.

D.1 Manuscript Transmission

A chain of scribal copies A_0 (the autograph) $\rightarrow A_1 \rightarrow \cdots \rightarrow A_n$ (an extant manuscript) is a substrate chain in the sense of definition 7.1, with each transition governed by a copyist's own budget: time, eyesight, familiarity with the source language, and attentiveness all bound β_i . Unlike Chapter 16's scale bar, a scribal error is essentially never declared: a miscopied word is presented exactly as confidently as a correctly copied one, making most manuscript corruption an instance of definition 2.8

rather than definition 2.7.

The chain structure is not merely formally convenient here; it is the literal basis of the scholarly discipline of textual criticism, which reconstructs a stemma — a family tree of copies — from surviving manuscripts and infers readings of the lost autograph by comparing independent branches.

Proposition D.1 (Branch Agreement as Evidence for the Archetype). *If two branches of a manuscript stemma, descended independently from a common ancestor, agree on a reading, and copying errors are independent across branches, that reading is more likely to reflect the ancestor than a reading found in only one branch.*

Proof Sketch. This is definition 9.3 applied to the manuscript domain: if errors in branch X and branch Y are independent, the probability that both branches independently introduce the *same* erroneous reading is far lower than the probability that both correctly preserve a shared true one. Agreement is therefore stronger evidence for the ancestor's reading than either branch's testimony alone. This is the formal content behind the standard editorial practice of preferring readings attested across independent branches of a stemma.

Where branches disagree, an editor constructing a critical text must choose a reading without being able to fully recover which branch, if either, preserved the ancestor correctly – a decision structurally similar to section 8.5's treatment of adjudication: the editor's judgment substitutes an operative reading for an underlying fact that may never be recoverable, though editorial judgment aims at recovering the truth in a way judicial adjudication does not always claim to.

D.2 Oral Tradition

Each retelling of an oral epic is a transition in the sense of proposition 20.2: a performer's schema, operating on whatever

residual trace of prior tellings they carry, produces a new performance. Oral-formulaic composition — fixed epithets, metrical formulae, repeated type-scenes, of the kind documented in the study of oral epic tradition — functions as a strong negative-interference device in the sense of definition 5.6: because performer and audience share the same formulaic expectations, maintaining a formula costs the performer far less than reconstructing an equivalent passage from scratch would, exactly as remark 15.1 showed for a painting's viewer.

This case surfaces a complication genuinely sharper than anything Part IV's chapters needed to confront. Chapter 7's root object (definition 7.8) presupposes some A_0 existing prior to the chain of transitions that follow it. An oral tradition without a fixed textual original may have no such A_0 at all: each performance is arguably a new production rather than a rendering of a prior fixed telling, in which case the entire preservation/reconstitution question of Part III may not have a privileged answer to converge toward, since there may be no single original for any performance's fidelity (definition 11.2) to be measured against. This monograph does not resolve this; it is flagged here as a further, distinct complication from the MEM|8 fork of Chapter 12, arising even under the narrow reading, since it concerns whether a root object exists at all rather than how faithfully a consumer accesses one. Where a tradition does have something root-object-like – a single elder who knows a telling no one else does, as opposed to a story many performers independently carry – remark 3.30 (Chapter 3) offers unverified but concrete vocabulary for exactly this distinction: the elder's version has reconstruction redundancy $\rho_W = 1$, a single point of failure, while the widely-carried story has $\rho_W \gg 1$.

D.3 Biological Inheritance

A caveat before this section. Conservation of Ambiguity reportedly applies repair-budget theory to biological evolution directly. The sketch below has not been checked against that treatment, and it is stated as an independent, appendix-level observation rather than as an extension building on it; a careful development of this case should verify the relationship before treating either as complete.

Reproduction with recombination and mutation can be modeled as a substrate chain: a parental genome A_0 , an offspring genome A_1 produced by a transition governed by the fidelity of DNA repair and replication machinery — a budget in a sense closer to literal than metaphorical, since the relevant biological machinery is itself called repair machinery. Traits neutral under current selective pressure are exactly the non-required distinctions of definition 7.2: they are not funded preferentially, and under relaxed selection or genetic drift they can be lost across generations, exactly as proposition 7.12 describes cumulative discard accumulating along an uncoordinated chain.

Proposition D.2 (Environmental Change as Task Discovery). *If a trait is not required under the selective pressure τ_i operative during generations $0, \dots, n - 1$, and a later environmental change constitutes a new selective task τ^* requiring that trait, the trait's loss under $\tau_0, \dots, \tau_{n-1}$ is unrecoverable by proposition 7.12, regardless of how strongly τ^* would now favor it.*

Proof Sketch. Immediate from treating successive generations as relay stages and environmental change as introducing an unanticipated task, exactly in the sense of proposition 4.7. This is the formal counterpart of a familiar evolutionary observation: a lineage cannot retroactively retain a trait it has already lost to

drift or relaxed selection merely because a later environment would have favored it.

This proposition is offered cautiously, as a structural observation about the fit between the two frameworks rather than as a contribution to evolutionary theory, and its relationship to *Conservation of Ambiguity's* own treatment of the same domain remains, per the caveat above, to be checked.

Appendix E

Toward a Simulation Framework

Synopsis. This appendix outlines, without implementing, how budget relays could be represented computationally and investigated experimentally through simulation.

This appendix sketches, without implementing, how a budget relay (definition 7.6) could be simulated computationally. It is explicitly scoped as a direction for future computational work, not as part of the monograph’s formal content.

Each stage of a relay would be represented as a process holding a declared triple (C_i, I_i, β_i) , applying definition 7.2 to a set of input distinctions and producing an output rendering together with a loss ledger L_i per definition 7.13. A downstream query interface would then test theorem 10.2’s failure mode directly, by constructing task instances τ^* outside the set of tasks any stage was configured to anticipate, and checking whether the final rendering’s admissible continuation set $\mathcal{A}_{\tau^*}(A_n)$ contains the correct continuation.

Such a framework would allow empirical (rather than purely constructive) exploration of corollary 10.4’s conjectured growth

bound across relays of varying length, branching factor (chapter 9), and coordination mechanism (chapter 8), which is currently stated only as a conjecture in this monograph.

The monograph's own worked constructions supply natural first test cases rather than requiring new ones to be invented: construction A.5 and its extension (proposition A.8) give an exact, small instance against which a simulator's discard behavior could be checked directly, construction 21.1 gives a named, domain-flavored instance of the same size, and construction 9.5 gives the minimal fan-out case. A useful early milestone for any implementation would be reproducing these four constructions' discard sets exactly, before attempting the larger, randomized relays needed to explore corollary 10.4's growth-rate question empirically.

Appendix F

Open Questions Register

Synopsis. This appendix maintains a single consolidated record of every conjecture and open question raised in the body, keeping a clear separation between established results and speculative extensions.

Per the epistemic-labeling discipline stated in the front matter, this appendix consolidates every claim in the body labeled Conjecture or Open Question, together with every substantive open item flagged informally in a Remark rather than housed in a formal environment, kept separate from proved content so that the monograph's unproved commitments are never left implicit. Entries are grouped by the part of the framework they bear on, rather than by chapter order, since several recur across chapters far apart in the text.

- **The MEM|8 fork** (open Question 12.1, Ch. 12). The single largest open dependency of the monograph: whether MEM|8's claim generalizes from biological recall specifically (narrow reading) to representation as such (broad reading), and, if the latter, whether Part I requires revision from the ground up. open Question 11.9 (Ch. 11) is

the same fork applied to the specific question of whether any instance of “direct access” involves strictly zero mediation.

- **Discard-rule robustness — partially resolved** (Ch. 7, remark 7.5; Appendix A). definition 7.2 is satisfied by more than one non-equivalent construction — at minimum construction 7.3 and construction 7.4 — and the two can select different discard sets for the same budget and priorities in general. lemma A.4 resolves this specifically for theorem 10.2’s own proof, by exhibiting a witness relay whose key discard is forced under any construction satisfying definition 7.2, but only for stages where the required distinctions’ cost closely exhausts the local budget (**non-exhausting relays**, below). What remains open is whether every *other* claim in this monograph stated to hold for “every” budget relay enjoys the same construction-independence, or whether some depend, unexamined, on a specific choice of local discard rule.
- **Non-exhausting relays** (Appendix A). lemma A.4 guarantees construction-independent discard only for stages where the required distinctions’ cost closely exhausts the local budget. Whether theorem 10.2’s conclusion is robust to choice of construction for relays *without* this near-exhaustion property — where genuine slack remains after funding every required distinction — is open; construction 7.3 and construction 7.4 may disagree on which non-required distinctions such slack goes to, with unexamined consequences for which future tasks such a relay turns out to support.
- **Root-preservation bootstrapping** (Ch. 7, remark 7.10). Preserving a root object (definition 7.8) so that proposition 7.9 remains available is itself a repair decision that must be funded before the future task space justifying

it is known. Whether this can be given a more precise decision-theoretic treatment — for instance, as an expected-value calculation over a distribution of possible future tasks — rather than being left at the level of a qualitative design recommendation, is open.

- **Infinite relays** (Ch. 7, definition 7.1). This monograph treats only substrate chains with finitely many stages. Whether an infinite chain behaves differently in kind, rather than merely in degree, from the finite case — for instance, whether proposition 7.12 or theorem 10.2 continue to hold, require additional hypotheses, or fail outright in the infinite setting — is unaddressed.
- **Branching relays** (Appendix A, Ch. 9). Appendix A's construction is linear (definition 7.1) throughout. Whether an analogous construction-independent witness exists for the branching relays of Chapter 9, given the additional complication of proposition 9.7, has not been checked.
- **Quantitative tightness of corollary 10.4** (Ch. 10, Appendix A). Appendix A establishes only the existence half of this corollary — that cumulative discard is non-decreasing along the witnessing construction, realized with equality across arbitrarily many pass-through stages — and explicitly does not establish the stronger claim that cumulative discard *risk* grows across relays in general, nor whether a sharper bound holds under specific coordination mechanisms.
- **General coordination protocols** (Ch. 8, Ch. 22). Whether a coordination protocol could be designed to bound, though not eliminate, theorem 10.2's failure mode across an entire class of relays, rather than only via the domain-specific mechanisms of Chapter 8. proposition C.1 (Appendix C) shows that one particular, more radical move

- pooling stage budgets rather than coordinating separate ones, per Section 8.7’s distinction — can restore sufficiency in a specific witness; it does not answer this broader question, since pooling eliminates the stage-separation the question is asking about coordinating *within*, rather than supplying a coordination protocol in the sense of Chapter 8’s taxonomy.
- **The closing synthesis** (conjecture 22.1, Ch. 22). Labeled a conjecture because the unification of theorem 10.2 with the fidelity continuum has not itself been given an independent proof beyond what its components separately establish.
- **Fidelity and Transport Stability** (remark 3.36, Ch. 3; Ch. 11). Whether $F(r, \tau)$ (definition 11.2) can be derived from, bounded by, or otherwise shown compatible with Distinguishability Geometry’s Transport Stability theorem (theorem 3.13) is unexamined; the two are currently related only by analogy.
- **Revision under a changing state space — resolved** (proposition 3.34, Ch. 3; Ch. 19). Previously open: whether relay revision under a changing state space could be expressed formally at all, given that Distinguishability Geometry’s revision operation requires a fixed state space. Resolved by Observability Is Restorability’s Paradigm Shift Theorem (theorem 3.21), which characterizes exactly this case as repair-repertoire discontinuity producing incomparable restorable-distinction sets. What remains open is only the interpretive question of which of science’s historical episodes fall under the fixed-state-space case (proposition 3.33) versus this one, which this monograph does not attempt to adjudicate.
- **Quantitative deficit bounds for this monograph’s dis-**

card sets (proposition 3.31, Ch. 3). Every discard set in this monograph (definition 2.5) could in principle carry a quantitative lower bound via Distinguishability Geometry’s Projection Deficit Bound (theorem 3.6), $\Delta\delta(r) \geq H(X \mid Y)$. No chapter has computed this bound for any of its worked instances; the discard-set apparatus throughout this book remains qualitative (present/absent) rather than quantitative.

- **The measure of a persistence set** (remark 3.35, Ch. 3; Ch. 7). Root recovery (proposition 7.9) establishes that a specific transformation lies in a persistence set $P(d)$ (definition 3.17) when a root object is preserved. What $\mu(P(d))$ is in general — what fraction of the relevant transformation family admits recovery, and at what cost it can be enlarged — has not been computed, and would give the root-preservation bootstrapping problem above a quantitative rather than qualitative target. remark 3.30 records a specific candidate from a weaker-tier source, The Geometry of Witnesses’s reconstruction redundancy $\rho_W(r) = |C(r)|$, which is unverified but concrete enough to be worth checking directly rather than left as a placeholder for “some measure or other.”
- **One remaining unverified import, one retraction** (Front matter, Relation to Prior Work). *The Geometry of Reconstruction* was cited in an earlier draft and has been retracted: no such text has been located, and the idea it was cited for (second-order repair operators maintained by witnesses) is now attributed instead to *The Geometry of Witnesses*, a text the author reports as forthcoming or separately authored. This replacement citation has itself not been independently read or confirmed as existing in completed form, and should be treated as no more reliable than the citation it replaced until it is checked

directly.

- **The transcription tier** (Front matter, Relation to Prior Work; §§3.6–3.7). *Admissibility Before Truth* and *The Geometry of Correction* are integrated on the strength of formulas and theorem statements the author transcribed directly from source PDFs, marked (*Transcribed from by the author; not independently verified against primary source text.*) throughout rather than (*Imported from .*). This is stronger evidence than the paraphrase-level citations these two sources previously carried, but it is not the same as an independent reading of the primary text, which remains the standard met only by *Conservation of Ambiguity*, *Distinguishability Geometry*, *Persistence Before Truth*, and *Observability Is Restorability*. Whether the transcribed formulas are complete and correctly contextualized relative to their source documents has not been independently checked.
- **The admissibility collision, sharpened but not resolved** (§3.6, Ch. 3). Four distinct formal notions bearing the name “admissibility” are now on record: this monograph’s epistemic $\mathcal{A}_\tau(r)$, the activation-manifolds application paper’s $A : M \rightarrow \mathbb{R}_{\geq 0}$, *Observability Is Restorability*’s observer-relative $\mathcal{A}_O$, and *Admissibility Before Truth*’s domain restriction $\text{dom}(\mathcal{T}) \subseteq \text{Rest}(\mathcal{R})$. proposition 3.24 connects the last of these directly to this monograph’s own apparatus by a stated proof, which is more traction than the other two comparisons have yielded, but reconciling all four, or the primary *Admissibility Field* text none of them may fully represent, remains undone.
- **Repair-interference formula discrepancy, internal to the corpus** (remark 3.20, Ch. 3). *Conservation of Ambiguity* attributes its derivative-form repair interference functional to *Observability Is Restorability*; *Observability*

Is Restorability’s own verified text defines interference algebraically instead ($I(d_i, d_j) = C_R(d_i \wedge d_j) - C_R(d_i) - C_R(d_j)$). This monograph uses the derivative form consistently and correctly attributes it to *Conservation of Ambiguity*, but the discrepancy between the two source texts themselves is unresolved and not this monograph’s to settle.

- **When reconciliation resolves automatically versus requires adjudication** (remark 9.13, Ch. 9). Reconciliation (definition 9.11) recovers what non-overlapping branch discard destroyed, but a genuine conflict – both branches preserving the same distinction with incompatible values – is not resolved by the intersection rule and falls back on Chapter 8’s mechanisms instead. Whether the boundary between these two cases can be characterized precisely in advance, rather than discovered only by attempting a merge and finding a conflict, is open.
- **Coordinating the coordinators** (§3.8, Ch. 8). Chapter 8’s taxonomy coordinates distinctions across separated stages but is silent about what maintains the coordinating mechanism itself – the shared ledger, the standing court, the augmentation policy. This is precisely the question §3.8 reports *The Geometry of Witnesses* as raising one level up, at that source’s weaker, unverified tier; this monograph does not develop a coordinated-coordinator layer of its own, and the gap between coordinating distinctions and coordinating the capacity to coordinate them remains open.

Bibliography Note

This monograph uses an inline bibliography rather than a separate reference list, citing sources by name at the point of use. Seven prior works from the same research program are engaged at three explicitly distinguished tiers of confidence, set out in full in the front matter's Relation to Prior Work and in Chapter 3: *Conservation of Ambiguity*, Distinguishability Geometry, Persistence Before Truth, and Observability Is Restorability are checked against primary source text directly; *Admissibility Before Truth* and *The Geometry of Correction* are integrated from formulas the author transcribed from source PDFs, marked (*Transcribed from by the author; not independently verified against primary source text.*) throughout; and *The Geometry of Witnesses* is reported only from a prose summary, marked (*Reported from a prose summary of by the author; neither transcribed formulas nor primary source text independently available.*). One further citation, to *The Geometry of Reconstruction*, was retracted during drafting; no such text has been located, and Appendix F records the retraction rather than silently removing the trace of it.

External sources outside this research program are cited inline at first mention rather than deferred here: the eyewitness-memory literature referenced in Chapters 12 and 20, the oral-formulaic and stemmatic-philology literature referenced in Appendix D, and Arrow's impossibility theorem referenced in Chapter 10 are each named at the point they are used rather

than collected into a separate list. This monograph does not otherwise self-cite beyond the six works named above and *Conservation of Ambiguity*, from which the Redistribution Theorem is imported in full in Chapter 5.